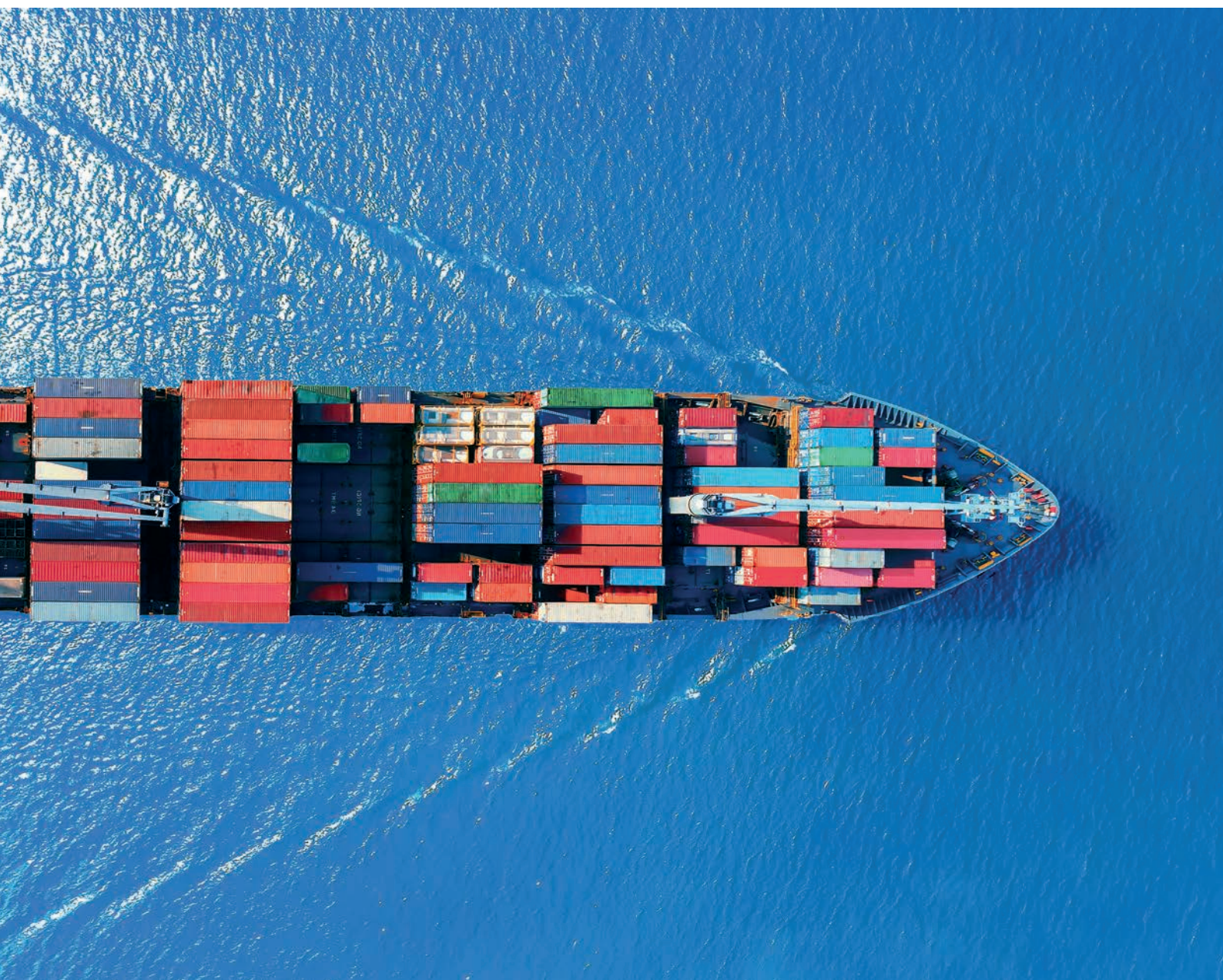


The Role of Shipbuilding in Maritime Decarbonisation

Impacts of Technology Developments and Policy Measures



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IMPACTS OF TECHNOLOGY DEVELOPMENTS
AND POLICY MEASURES

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Preface

The global maritime sector is at a turning point. While shipping remains the most carbon-efficient mode of transport on a per-tonne basis, it is also responsible for nearly 3% of global greenhouse gas emissions—a figure that will continue to rise without significant intervention. The challenge ahead is clear: if the shipping industry is to meet its decarbonisation targets, the fleet must evolve. Shipbuilding plays a central role in this transformation, not only by developing new vessels equipped for alternative fuels but also by ensuring that existing ships can be retrofitted with energy-saving and low-emission technologies.

The transition to a lower-emission fleet is already underway, but progress is uneven. More than half of the current orderbook consists of vessels designed for alternative fuels, yet only 7% of the global fleet is capable of running on them. Retrofitting efforts remain limited, and the widespread adoption of advanced technologies is constrained by high costs, supply chain challenges, and uncertainty around future fuel availability. Meanwhile, global shipbuilding capacity remains concentrated in a few regions, and the sector faces growing competitive pressures as it adapts to new demands. Addressing these challenges requires a balanced and coordinated approach. Industrial strategies must be designed to support innovation without distorting markets, ensuring that investments in alternative fuels, digitalisation, and energy efficiency technologies remain both commercially viable and globally competitive. Expanding access to ship finance can help shipowners, shipbuilders and fuel suppliers accelerate the adoption of low-emission solutions.

The OECD Shipbuilding Committee provides a unique platform which aims to contribute to establishing normal competitive conditions in the shipbuilding sector and to address other key issues. With its renewed mandate for 2024-2028, the Committee is indeed deepening its focus on the economic implications of maritime decarbonisation and ensuring that policy measures encourage innovation while maintaining fair competition in the global shipbuilding sector. One key area of work is the ongoing discussion on possible more liberal export credit terms for low- and zero-emission vessels, which has the potential to reduce financial barriers and incentivise investment across the sector.

This report provides a timely assessment of the shipbuilding industry's role in maritime decarbonisation, exploring the impact of technology developments and policy measures on the sector's future. Ensuring a successful transition will require close cooperation between actors across the maritime ecosystem and supply chain—from shipowners, shipbuilders, marine technology designers and fuel suppliers, to policymakers and financial institutions. The choices made today will not only determine the carbon impact of shipping but also shape the long-term competitiveness and resilience of the global shipbuilding industry.



H.E. Ambassador Halvor Hvideberg,
Permanent Representative of Norway to the OECD,
Chair of the OECD Shipbuilding Committee

Foreword

The shipping industry accounts for approximately 3% of global greenhouse gas (GHG) emissions, and without intervention, its share is expected to rise due to increasing seaborne trade. As decarbonisation efforts gain momentum, policies are being implemented to curb emissions, improve vessel efficiency, integrate digital and energy-saving technologies (ESTs), and accelerate the adoption of alternative fuels. Recognising the crucial role of shipbuilding in these efforts, the OECD Shipbuilding Committee (SBC) works to promote fair competition and transparency in the global shipbuilding market.

In 2023, the OECD Council renewed the SBC's mandate for 2024–2028, strengthening its focus on decarbonisation in line with the International Maritime Organization's (IMO) GHG strategy, which aims for net-zero emissions from ships 'by or around, i.e. close to,' 2050. Achieving this goal will require the replacement or retrofitting of around 50,000 ocean-going vessels, necessitating significant investments in shipbuilding and marine equipment industries. The SBC's Program of Work and Budget (PWB) for the 2023–24 biennium reflected this shift, introducing a dedicated Intermediate Output Result (IOR) on decarbonisation and digitalisation in shipbuilding and shipping. The output, titled "Analysis of the impact of new (digital) technologies and comparative analysis of maritime decarbonisation policies and support measures in SBC and non-SBC members" addressed two key areas: (1) the impact of emerging technologies, including energy-saving solutions, digitalisation, and alternative fuel systems, on shipbuilding capacity and costs, and (2) a comparative analysis of maritime decarbonisation policies, focusing on policy transparency, the design of regulatory and support measures, and the role of international cooperation in ensuring regulatory stability.

This report is the outcome of research over the 2023–24 biennium. It discusses the critical role of shipbuilding in decarbonising the maritime sector, both by reducing emissions from ship production and by developing energy-efficient ships, integrating zero/low-carbon propulsion technologies, and fostering innovation in ship design and equipment. The OECD's work in this area aligns with broader industrial decarbonisation efforts, providing a platform for international policy exchange, knowledge sharing, and mutual learning to support a well-managed transition towards a zero-emission fleet. By facilitating cross-country dialogue and improving transparency in policy frameworks, the OECD contributes to shaping more resilient and competitive shipbuilding in the context of industry transformations.

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Acronyms and abbreviations

CCS	carbon capture and storage
CGT	compensated gross tonnage
CO ₂	carbon dioxide
EPO	European Patent Office
EU	European Union
EU ETS	EU Emissions Trading Scheme
EST	energy saving technology
GHG	greenhouse gas
GT	gross tonnage
IEA	International Energy Agency
IMO	International Maritime Organization
ITF	International Transport Forum
LCA	lifecycle assessment
LNG	liquified natural gas
LPG	liquified petroleum gas
MARPOL	International Convention for the Prevention of Pollution from Ships
MEPC	Marine Environment Protection Committee
MtCO ₂	million tonnes of CO ₂ equivalent
ROW	Rest of World
UNCTAD	UN Trade and Development
WtW	well-to-wake

Executive Summary

Maritime transport is central to global trade, facilitating around 80% of the world's exchange of goods. Despite being the most carbon-efficient mode of transport on a per tonne basis, the shipping industry is responsible for almost 3% of global greenhouse gas (GHG) emissions, and without decisive action, the share of emissions from shipping could rise significantly by 2050. The shipbuilding industry plays a crucial role in enabling this transformation, as it is responsible for designing and constructing vessels that are compatible with alternative fuels and advanced energy-efficient technologies.

Recent regulatory measures, notably the 2023 GHG strategy by the International Maritime Organisation (IMO), signal an intensified commitment to decarbonise maritime transport, with targets set for net-zero greenhouse gas emissions 'by or around, i.e. close to,' 2050. This global effort is complemented by national and regional policies, such as the European Union's inclusion of maritime transport into its Emissions Trading System (ETS) and the FuelEU Maritime Regulation. Spurred by this wave of net-zero measures, the maritime sector is undergoing a significant transformation, marked by the development and uptake of low/zero-emission and digital solutions.

Despite this progress, key challenges to achieving net-zero shipping remain, including an aging global fleet and the heterogeneity of the maritime industry rendering a "one-size-fits-all" decarbonisation approach unfeasible. Key issues include the uneven pace of transition across ship segments and regions, the need to sustain innovation for ongoing advancement and cost reduction of low- and (near) zero-emission solutions, the limited infrastructure and high cost for alternative fuels, and the critical challenge of financing the transition.

Technology trends for alternative fuels and propulsion

While the construction of alternative fuel capable ships is increasing, fleet uptake remains low. The maritime sector is shifting toward low/zero-emission solutions, with ships capable of using alternative fuels making up over 52% gross tonnage in the global orderbook. Yet only around 7% of the global fleet can currently operate with these technologies, and maritime GHG emissions continue to rise.

Fuel optionality in ships is preferred to one specific fuel choice. Fuel options in the orderbook point to increased diversification, with around 37% of vessels (by gross tonnage) being LNG capable, 9.7% methanol capable and 0.55% ammonia capable. Emerging technologies to reduce alternative fuel demand, such as battery, hybrid or nuclear propulsion, are also gaining traction for certain ship types but their uptake in the fleet remains relatively low (<1%).

Design and construction capabilities are regionally concentrated. Chinese yards make up 47% of the orderbook in alternative fuel capable vessels in CGT, followed by Korea (42%). The building of methanol and ammonia capable ships is mainly concentrated in Korea, China and Japan, while ships capable of running on biofuels show a more diverse set of producer countries and manufacturers. European engine designers supply over 65% of the alternative fuel capable fleet and orderbook.

Retrofits in selected energy saving technologies and fuel conversions more than doubled since 2020 but account for less than 1% of retrofits and repairs. European and Chinese yards are leading alternative fuel conversions. Chinese yards also primarily carry out propeller and hull EST retrofits. Alternative fuel conversions focus on passenger vessels, EST retrofits on bulk carriers and containerships. Fuel conversions take significantly longer to complete than other retrofits (+45 days), hinting at potential capacity bottlenecks as demand grows.

Shipbuilding capacity for low/zero-emission vessels

In 2024, 82 builders constructed alternative fuel capable vessels. The past ten years have seen growing capacity among shipyards to construct vessels capable of using alternative fuels, with their share of total active yards increasing to between 7 and 36% for different ship types in 2023. This trend is uneven across ship types and fuel options: bulk carriers and tankers lag in adoption, while a significant portion of yards build alternative fuel capable containerships and cruise ships. LNG currently leads in construction, with methanol also seeing increased growth.

The yearly growth in patenting activity in low-carbon technologies is slowing. Despite the growing pressure to decarbonise the maritime sector, the yearly share of low-carbon patenting activity in maritime technologies peaked between 2010 and 2015. Yearly patents have also decreased in total numbers. The EU and Japan have consistently shown the most patent filings in this field, but China has surpassed other key innovating countries in patenting activity in the past five years.

Main shipbuilding companies are engaging in decarbonisation efforts, but regional differences persist. Seven out of 15 studied major shipbuilding companies do not have a 2050 net-zero emissions target, indicating that commitments vary by region. Shipbuilding companies promote low/ zero-emission vessel development through technology demonstration projects, emphasising digitalisation and ‘Just Transition’ strategies.

Global plans for alternative fuel storage and bunkering are progressing, but implementation remains limited. Methanol and ammonia infrastructure for shipping is mostly at the planning stage, with major uncertainties around fuel production, supply chains, port readiness, and safety standards.

Policy measures and financing mechanisms

Government policies focus mainly on shipping, less on shipbuilding. While over 80% of assessed jurisdictions are implementing maritime decarbonisation strategies, most of these focus on the shipping sector, and less on the shipbuilding industry. Moreover, 5 out of 25 countries do not have a dedicated decarbonisation strategy for the maritime sector, potentially causing an inconsistent policy mix and enforcement of measures.

Governments emphasise support programmes, not mandatory measures. Most countries focus their policy measures on strategies, roadmaps and a spectrum of government-backed support programmes to support decarbonisation. Most common policy target areas are research and development (R&D) and alternative fuel uptake and supply. Binding measures, such as fuel standards or carbon pricing mechanisms are limited.

Increased government guidance can support green finance practices. The use of green finance instruments is largely limited to major shipbuilders, pointing to gaps in companies’ decarbonisation efforts. While some jurisdictions have implemented green finance taxonomies, there are few measures specifically focused on access to finance and investment, highlighting the potential for increased government guidance to strengthen the business case for carbon-neutral fuels and mitigate investment risks.

1 Overview of shipbuilding and maritime decarbonisation

Shipbuilding plays a critical role in maritime decarbonisation, shaping the industry's ability to reduce greenhouse gas (GHG) emissions and adopt new low/zero-emission technologies and digital solutions. The transition to net-zero shipping is progressing, but key challenges persist, including financing and generating demand for alternative fuels, scaling up infrastructure, and addressing regulatory fragmentation. This chapter gives an overview of the analysis on technology developments and policy measures in the report and provides concluding recommendations on ensuring a holistic and effective net-zero transition. It highlights the need to coordinate efforts between first movers and (fast) followers, to sustain innovation for technology development, and to secure financing while establishing clear demand signals for alternative fuels and low/zero-emission vessels.

1.1. Decarbonising maritime transport— what’s the issue?

Maritime transport is central to global trade, facilitating around 80% of the world’s trade of goods (UNCTAD, 2024^[1]). Despite being the most carbon-efficient mode of transport on a per tonne basis, the shipping industry is responsible for almost 3% of global greenhouse gas (GHG) emissions, necessitating a response to mitigate its impact on climate change (International Maritime Organization, 2020^[2]). Further, the rebound in CO₂ emissions following a dip in 2020 due to the COVID-19 pandemic underscores the urgent need to decouple seaborne trade growth from emissions. Unlike many other sectors of the economy, CO₂ emissions from the world’s maritime fleet increased by 3.7% from 2020 to 2024 (Clarksons Research, 2024^[3]), highlighting the sector’s ongoing reliance on fossil fuels and the scale of the decarbonisation challenge.

With the increasing global emphasis on attaining carbon neutrality, the maritime sector is facing growing demands to diminish its environmental impact. These pressures come not only from regulators but also from stakeholders throughout the supply chain, including cargo owners, financiers, and port authorities, who are seeking tangible commitments to emissions reduction (COZEV, 2023^[4]). However, achieving net-zero emissions will require overcoming significant challenges, including secondary emissions, safety risks, crew welfare considerations, and economic feasibility. Coordinated industry-wide collaboration is essential to ensuring that decarbonisation efforts are effective, economically viable and resilient (Deng and Mi, 2023^[5]).

An ageing fleet and slow turnover highlight the urgency of future-proofed shipbuilding. The increasing age of the global fleet, averaging around 13 years in 2024 (on a tonnage weighted basis), underscores the magnitude of the tasks (UNCTAD, 2024^[1]). As vessels age, the risk of reduced energy efficiency increases. Further, ships have a slow turnover rate, with a typical lifespan of around 25 years, meaning that newbuild vessels and vessels in the orderbook today are likely to remain in service until 2050. This underscores the importance for newbuilds to be designed in alignment with future regulatory and technological requirements or constructed to allow for retrofitting. By the end of 2024, around 7% of the global fleet and 52% of the orderbook (in gross tonnage) are alternative fuel capable (Clarksons Research, 2024^[3]). This means that just over half of newbuilds can operate on alternative fuels or propulsion systems— many of which include fuels, which may not be fully aligned with net-zero objectives (Fricaudet et al., 2022^[6]).

Understanding the heterogeneities of the maritime sector is key. Given the heterogeneity of the maritime industry, characterised by diverse ship sizes and operating routes, implementing a uniform “one-size-fits-all” solution or technology for decarbonisation is not feasible. Instead, a portfolio approach is required, incorporating energy efficiency improvements, operational changes, and the adoption of new fuels. Further, each phase of a ship’s lifecycle— design, material sourcing, construction, operation, and end-of-life recycling— has emissions implications (Chatzinikolaou and Ventikos, 2015^[7]). This presents both challenges and business opportunities, particularly in the shipbuilding sector, which is seeing increased demand for dual-fuel and fuel-flexible vessels.

Financing and infrastructure challenges to scaling up alternative fuels remain. Amidst the accelerating shift in fuel technology within the maritime sector, the pursuit of scalable solutions is a prevailing concern. Energy efficiency and cost minimisation stand as key objectives in this transformative landscape. While pioneering initiatives within the industry have underscored the commercial viability of such transitions, key challenges for financing and scaling up alternative fuel production and supply remain. Research estimates suggest that decarbonising shipping with ammonia by 2050 would require USD 1.2–1.6 trillion, or USD 60–80 billion per year from 2030 to 2050. 87% of the investment would go toward fuel production, storage, and distribution, while 13% would be spent on ship upgrades, including engines, storage, and energy efficiency (Global Maritime Forum, 2020^[8]). Financing challenges are strongly rooted in market uncertainties, encompassing the anticipation of future technological advancements, the cost-

effectiveness of alternative fuels, safety considerations and the readiness of fuel supply and bunkering infrastructure at ports (DNV, 2023^[9]).

Pioneering projects demonstrate that the transition is commercially viable. Early adopters are already investing in alternative solutions, such as methanol capable ships, battery-electric short-sea vessels, and wind-assisted propulsion technologies. With the right policies and financial mechanisms, emerging technologies and fuels can be scaled. Strategic investment in port infrastructure and global bunkering networks for alternative fuels is critical to ensuring that new vessels can access the necessary refuelling options (DNV, 2024^[10]). Moreover, innovative financing tools—such as green bonds and carbon contracts for difference—are being developed to bridge the cost gap and de-risk investment in green shipping.

Regulatory momentum is increasing to drive new technologies and practices. Regulatory developments, such as the strengthened International Maritime Organization (IMO) Strategy on Reduction of GHG Emissions from Ships in July 2023 (the 2023 IMO GHG Strategy), reflect a commitment to decarbonise the sector, with targets for achieving net-zero greenhouse gas emissions “by and around, i.e. close to, 2050” (International Maritime Organization, 2023^[11]). Ongoing negotiations at the IMO are also focused on developing and implementing a “basket of measures” in support of the 2023 GHG Strategy (International Maritime Organization, n.d.^[12]). If designed and implemented effectively, such measures could support market certainty to help unlock large-scale investment in low/zero-emission shipping. Further, the European Union (EU) incorporated maritime transport in its Emissions Trading System (ETS) in 2024 and introduce the FuelEU Maritime Regulation in 2025, requiring for a 2% annual increase in ships’ energy efficiency (European Commission, 2023^[13]).

Beyond compliance, policy measures can create strong incentives for first movers and offer new revenue streams through innovative finance mechanisms. Shipbuilders that develop and foster expertise in constructing low/zero-emission, energy efficient vessels, along with shipowners that invest early in transitioning their fleet to low-carbon vessels through newbuilds and retrofitting and ports that position themselves as hubs for alternative fuel bunkering, can gain a competitive edge in the evolving maritime landscape. **As such, the transition to net-zero shipping represents not only a regulatory challenge but an opportunity to reshape the industry for long-term development and resilience.**

1.2. The transition to net-zero shipping is underway...

This report provides an assessment of the role of shipbuilding for maritime decarbonisation, evaluating both industry capacity to deliver and retrofit low/zero-emission ships and governments’ maritime decarbonisation policy measures relevant to shipbuilding. The analysis across chapters reveals several key developments in the shipping sector, particularly in the adoption of alternative fuels and the initiatives taken by both industry and governments to facilitate this transition.

52% of the orderbook can use alternative fuels and propulsion, but only 7% in the fleet. Shipowners are placing significant emphasis on integrating alternative fuels into both fleets and fuel supply chains. The post-2020 period has seen a notable rise in the production of alternative fuel capable ships, with fuel optionality increasingly overtaking the selection of a single fuel type. By October 2024, ships capable of using alternative fuels accounted for 52% of the orderbook in GT. This diversification of fuel types includes LNG, which makes up 37% of the alternative-fuel capable orderbook, as well as emerging fuel options, such as methanol at 9.7% and ammonia at 0.55%. Additionally, technologies aimed at reducing overall fuel demand, such as battery, hybrid and nuclear propulsion, are gaining traction for certain ship types but currently represent less than 1% of the existing fleet.

Alternative fuel retrofits have doubled since 2020 but remain limited. The retrofit market for alternative fuels and ESTs has expanded considerably since 2020, with conversions of alternative fuels more than

doubling. However, these retrofits still represent less than 1% of the total retrofit and repair market. European and Chinese shipyards are leading in alternative fuel conversions, with Chinese yards also showing high capacity for propeller and hull EST retrofits. Passenger vessels account for 45% of all alternative fuel conversions, while EST retrofits are most common among bulk carriers (32%) and containerships (29%). It is worth noting that fuel conversions take significantly longer to complete than other types of retrofits, averaging an additional 45 days, indicating potential capacity constraints as demand for conversions continues to grow.

In 2024, 82 shipyards globally are constructing alternative fuel capable vessels, reflecting the industry's increasing capacity to meet decarbonisation goals. Over the past decade, the number of shipyards capable of building such vessels has grown substantially, representing between 7% and 36% of all active yards across different ship types by 2024. However, this trend is uneven across ship types: alternative fuel newbuilding in bulk carriers and tankers are slower to adopt alternative fuels, while shipyards constructing containerships and cruise ships are at the forefront of this transition. LNG currently dominates new vessel construction, but methanol is quickly emerging as a key alternative fuel for the future.

Government policies are closely aligned with the industry's push toward alternative fuels, focusing heavily on the uptake of new alternative fuel technologies and low- and (near) zero-emission fuel supply, supporting carbon-neutral energy production and development of onshore infrastructure, particularly for fuel storage and bunkering. A key policy target area has been the creation of supportive infrastructure to enable widespread use of alternative fuels, highlighting the interdependence between technological advancements onboard ships and the necessary logistical systems on land.

1.3. ... but key challenges remain to meeting net-zero targets

While the transition to net-zero shipping is advancing, the report underscores several persistent challenges, from a shipbuilding and marine technology perspective, that must be addressed to achieve comprehensive and effective decarbonisation across the sector.

First movers and quick followers (?)

One key observation from the report's analysis is the significant variation in the speed of transition across different regions, stakeholders, and vessel types. This disparity raises questions about how to balance the role of first movers in advancing low/zero-emission solutions while ensuring that others—quick followers—can catch up without hindering the overall progress toward global net-zero goals.

The adoption of low/zero-emission technologies is advancing at uneven pace, with vessel segments, such as LNG and LPG carriers, containerships, and cruise ships, seeing higher uptake rates of energy-efficient technologies and alternative fuels. As of October 2024, around 7% of the global fleet by tonnage can use alternative fuels or propulsion systems. However, this share drops to approximately 1.8% when considering the number of vessels, suggesting that larger ships are being prioritised for alternative fuel capabilities.

There is considerable regional variation in capacities for the design, construction, and supply of alternative fuel capable vessels. Chinese and Korean shipbuilding dominates the orderbook for alternative fuels and propulsion, accounting for 47% and 42%, respectively, of vessels measured in CGT¹. The construction of ships capable of running on emerging alternative fuels, such as methanol and ammonia, is concentrated in these regions, with Japan also playing a significant role. In terms of engine design for alternative fuel or propulsion, Europe is the market leader, accounting for around 65% of vessels in the global fleet and orderbook.

Differences remain in major shipbuilding companies' decarbonisation targets. Of the 15 major shipbuilding companies studied, seven do not have a 2050 net-zero emissions target, reflecting varying levels of regional ambition and policy alignment. While many shipbuilders are promoting low- and (near) zero-emission vessel development through technology demonstration projects and 'Just Transition' strategies, the lack of uniform commitment can pose a challenge to global decarbonisation efforts.

There are significant regional disparities in alternative fuel supply chains. Global efforts are underway to develop storage and bunkering infrastructure for fuels like methanol and ammonia, but progress remains slow. Methanol bunkering capacity is currently most developed in East Asia, while ammonia storage and bunkering projects are largely concentrated in Europe and the Middle East and North Africa, though they are still in the planning stage. The development of Green Shipping Corridors also shows that first mover ports are taking the initiative to build up routes for low/zero-emission fuels, but gaps in fuel availability could slow the transition for lagging regions.

Addressing the slow-down in the growth of innovation activity

The yearly growth in patenting activity in low/zero-emission maritime technologies has slowed, despite strong commitments to research and development (R&D) by both industry and government. This disconnect between increased R&D focus expressed in companies' strategies and governments' policy measures and the slower pace of growth in innovation poses an obstacle to achieving sector-wide decarbonisation goals.

R&D is prioritised across policy measures and company strategies. Across the jurisdictions studied, R&D is highlighted as a cornerstone of maritime decarbonisation strategies. Significant resources are being allocated to advance technologies that address emissions reductions, with shipbuilders increasingly integrating decarbonisation into their business models.

The pace of patenting activity has decelerated in recent years. The share of low-carbon innovations within maritime technologies reached its peak between 2010 and 2015, and since then, the rate of patent filings has declined, including in OECD economies. The main exception to this trend is China, which has significantly increased its patent activity in the past five years, now surpassing other key innovating countries. Nevertheless, this increased activity in one region has not been sufficient to offset the global slowdown, underscoring a broader issue in translating R&D efforts into tangible innovations.

Cost remains a key barrier to the adoption of low/zero-emission solutions in shipping. Alternative fuel capable vessels typically carry a price premium— around 10–15% for ships designed for LNG or methanol— due to more complex engine systems and fuel storage requirements. There has been no clear downward trend in these premia, indicating limited cost reductions over time. However, the greater financial challenge lies in the high prices of alternative fuels, which remain significantly above those of conventional marine fuels. Without long-term price certainty, shipowners face elevated operational risks that may discourage investment, particularly in regions lacking regulatory or financial incentives.

Emerging technologies face technical and operation challenges. Emerging technologies, such as carbon capture, advanced nuclear propulsion, battery hybrid solutions, and wind-assisted propulsion, present important alternative solutions to decarbonisation. However, the development and deployment of these technologies face substantial challenges, including technical complexity and high upfront costs. Furthermore, the exploration of alternative fuels like ammonia and hydrogen presents additional safety and handling challenges, particularly in ensuring these fuels can be integrated safely into existing maritime operations.

Financing the transition and supporting clear demand signals

Financing the transition to low/zero-emission vessels, alongside securing a stable alternative fuel supply, remains a critical barrier to maritime decarbonisation. The report emphasises the need for targeted financial mechanisms and strong demand signals to drive investments in the necessary technologies and infrastructure.

The use of green financing instruments for low/zero-emission ships remains limited. While green finance instruments such as green bonds and loans are emerging, their uptake remains limited and is predominantly seen in globally leading shipbuilders. Smaller and less technologically advanced companies face substantial barriers to accessing capital, particularly for high-cost retrofitting projects and investments in alternative fuel capable vessels. Targeted government interventions—such as the inclusion of shipping-specific provision in green finance taxonomies, expanded access to export credit facilities, and risk-mitigation mechanisms—could help incentivise the use of green finance instruments and enable broader industry participation in the transition.

Inter-industry collaboration is key for alternative fuel supply. The development of robust supply chains relies on coordinated international efforts across sectors, including energy, shipping, and port infrastructure. Initiatives such as Green Shipping Corridors are promising but require significant scaling to deliver sector wide impact. Governments can play an important role by supporting partnerships between shipowners, fuel producers, and ports to secure fuel production and infrastructure deployment. Further, offtake agreements between shipping companies and energy providers are a useful tool for generating clear demand signals and mobilising investment, thereby reinforcing the resilience and scalability of supply chains. Expanding the concept of shipping corridors to include emerging technologies—such as carbon capture and storage (CCS)—could also accelerate adoption at sea and contribute to the development of integrated CCS value chains.

Addressing financial barriers requires a collaborative approach between governments, industry, and financial institutions. Cooperation across sectors—shipping, shipbuilding, energy, and finance—is crucial for developing financing tools that match decarbonisation needs. Existing instruments, like the Ship Sector Understanding of the Arrangement on Officially Supported Export Credits, should be adapted to support low/zero-emission technologies. The OECD can play an important role in convening stakeholders and aligning financing mechanisms with decarbonisation targets, ensuring investments flow where needed to meet net-zero goals.

1.4. Structure of the report

This report assesses the role of shipbuilding in maritime decarbonisation through several analytical perspectives:

- **Chapters 2 and 3** focus on shipbuilding capacity to deliver and retrofit low/zero-emission ships and low-carbon (digital) technologies. These chapters explore key technological developments and innovation activities in areas such as digital technologies, energy-saving measures, and alternative fuels, and their impact on low/zero-carbon shipping.
- **Chapter 4** evaluates the alignment of major shipbuilding companies' decarbonisation and digitalisation objectives and strategies with international and national net-zero targets.
- **Chapter 5** provides complementary analysis on onshore technological developments, specifically examining the storage and bunkering of alternative fuels within port infrastructure.
- **Chapter 6** presents a comparative analysis of maritime decarbonisation policy measures relevant to shipbuilding. This chapter includes a policy mapping exercise and assessments policy measures in both OECD Shipbuilding Committee members and non-member jurisdictions.

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Notes

¹ In Q4 2024, the share of Chinese shipbuilders in the total orderbook (in CGT) was 55% (vs. 47% in the alternative fuel capable orderbook). Korean shipbuilders' share of the total orderbook was 25% (vs. 42% in the alternative fuel capable orderbook).

2 Estimating shipbuilding capacity for the alternative fuel capable fleet

The adoption of low-carbon technologies in the maritime sector and the construction of low/zero-emission vessels provide shipbuilders with a crucial role in meeting the evolving demands of this rapidly transforming industry. This chapter discusses how the shipbuilding sector can contribute to maritime net-zero efforts across a ship's life cycle and assesses the progress in the adoption of alternative fuels and low/zero-emission technologies within the fleet and orderbook. It evaluates shipbuilders' capacity for decarbonisation efforts by examining the number of active shipbuilders producing vessels capable of running on alternative fuels as well as the price differential between alternative fuel capable vessels and their conventional counterparts. Analysis on newbuilding is complemented by looking into retrofits and an assessment of key trends in alternative fuel conversions.

Key messages

This chapter discusses alternative fuel uptake in both newbuilding and retrofitting and capacity indicators of shipyards building alternative fuel vessels. The main findings are:

- **Reducing maritime emissions requires a lifecycle approach**, as emissions from shipbuilding, operation, and end-of-life disposal are interconnected—choices made in design and material sourcing impact operational efficiency and recyclability, influencing the industry's decarbonisation potential.
- **Around 7% of the global fleet and 52% of the orderbook are alternative fuel capable**: the maritime sector has increased the commission and delivery of vessels capable of using alternative fuels, corresponding to the growing incentives for decarbonisation driven by regulatory measures and industry efforts. At the same time, only 7% of the global fleet can currently operate with these technologies and maritime GHG emissions continue to rise.
- **Capacity for alternative-fuel capable vessels has increased but strong regional differences exist**: there is a discernible uptrend in the number of shipyards that are active in the construction of vessels that can run on alternative fuels. This increase is observed both in absolute numbers and as a proportion of the total yards constructing vessels. The construction landscape for vessels using alternative fuels is heavily influenced by the builder country. LPG, methanol and ethane vessel construction is primarily dominated by Korea, China and Japan. In contrast, vessels designed for LNG or biofuels see a wider range of countries, in which these vessels are constructed, and in the number of builders.
- **Alternative fuel capability differs across ship types**: large differences exist in the construction of alternative fuel capable vessels across different ship types. Specifically, bulk carriers and tankers have been relatively slow to adopt, with only 7% and 14% of active yards, respectively, engaged in constructing alternative fuel capable vessels in the past year. In contrast, a more substantial portion of shipyards building containerships (1/5 of yards) and cruise ships (over 1/3) are moving towards alternative fuel vessel capabilities.
- **LNG is the dominant fuel choice, but other alternative fuels are increasing**: The transition to alternative fuel capable ships is not uniform, with significant differences between fuel types. LNG leads in terms of both the global fleet uptake and the number of active yards having previously constructed LNG capable vessels (37 yards in 2024). The rise of methanol as an alternative fuel is notable, with an increasing number of shipyards starting to offer methanol capable vessel construction (8 yards in 2024).
- **Price premium for alternative fuel capable ships varies across ship types**: A price premium exists for LNG capable and methanol capable ships across different ship types, averaging at around 12%. The extent of this premium varies significantly by ship type, with containerships, pure car carriers and cruise ships showing lower price premia than bulkers and different tanker types.
- **Alternative fuel conversions are increasing, especially in China and Europe**: primarily for LNG and biofuels, which are mostly undertaken on passenger ships (45% of total conversions). China and Norway are leading alternative fuel conversions, with retrofits by Chinese yards focusing on LPG, alongside some LNG projects. Norway demonstrates the most comprehensive range of fuel conversions, including biofuels, LNG and hydrogen.

2.1. The role of shipbuilding in maritime decarbonisation

Maritime transport is central to global trade, facilitating around 80% of the world's exchange of goods (UNCTAD, 2024^[1]). Despite being the most carbon-efficient mode of transport on a per tonne basis, the shipping industry is responsible for almost 3% of global greenhouse gas (GHG) emissions, necessitating a response to mitigate its impact on climate change (International Maritime Organization, 2020^[2]).

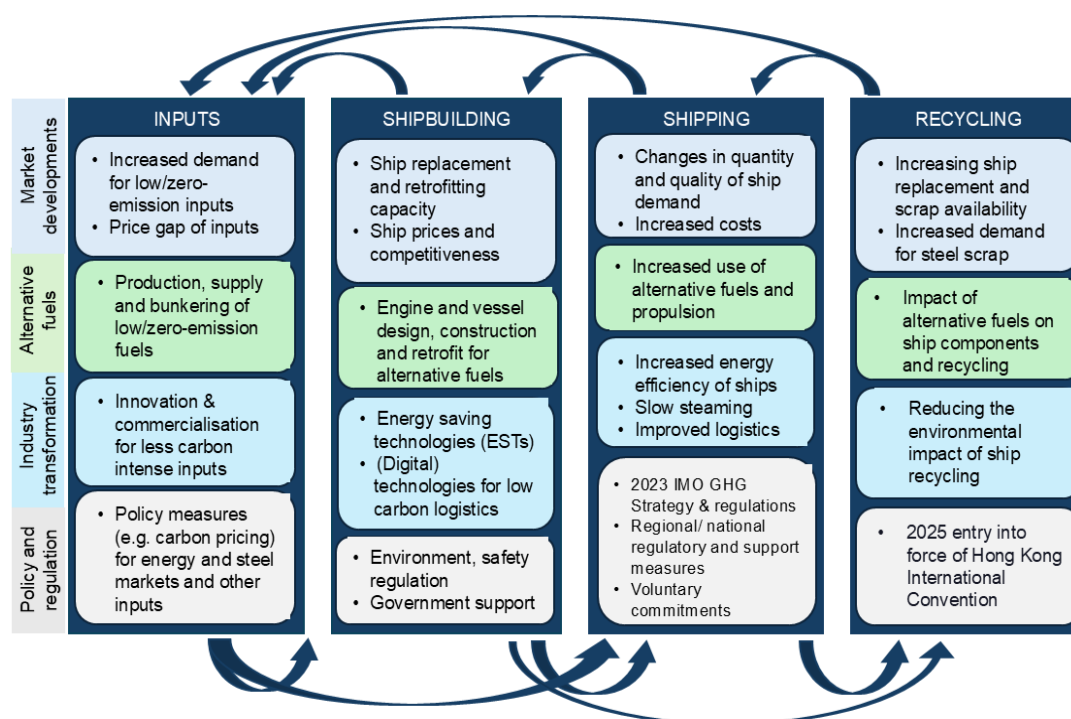
As the maritime sector becomes increasingly focused on reducing GHG emissions, the shipbuilding sector is undergoing a significant transformation to support climate change mitigation efforts and the advancement of low/zero-emission technologies, especially in the areas of alternative fuel and propulsion solutions, ship energy efficiency and digitalisation. To effectively reduce maritime emissions and identify potential areas for improvement or bottlenecks in decarbonisation efforts, it is important to consider shipbuilding within the broader lifecycle of the maritime industry.

A life-cycle assessment includes four phases of relevance to maritime decarbonisation: pre-production (upstream), production (the shipbuilding process itself), and post-production, including the operational and end-of-life phase (downstream) (Chatzinikolaou and Ventikos, 2015^[3]).

- In the **pre-production phase**, raw materials and components required for vessel construction are evaluated, including the extraction of raw materials, transportation and manufacturing of parts.
- The **production phase** involves the construction of the vessel, including the assembly of parts, painting and other finishing processes. For this scoping paper, the shipbuilding process is defined as the construction of ships and assembly of key marine equipment and technologies in shipbuilding.
- The **operational phase** includes fuel consumption, maintenance and retrofitting/ repair for maritime transportation.
- Finally, the **end-of-life phase** involves the disposal, recycling and potential reuse of the vessel's components.

Infographic 2.1. illustrates the key phases in the maritime life cycle as well as their relevant market trends, technology developments and recent policies and voluntary industry commitments. The diagram shows the critical role that shipbuilding plays in the delivery of decarbonisation in each of these phases and vice-versa how each of these phases is of relevance to shipbuilding decarbonisation itself.

Infographic 2.1. A life-cycle approach to the maritime industry



Shipbuilding and the costs of ship construction are strongly impacted by market dynamics and regulations in upstream sectors, such as steel, equipment parts, energy or raw materials (Gilbert et al., 2017^[4]). Many of these sectors are major contributors to global GHG emissions, for example, global steel production accounts for an estimated 8% of GHG emissions worldwide (International Energy Agency, 2022^[5]). Research by UMAS and the LR Maritime Decarbonisation Hub suggests that international shipping could reduce cumulative CO₂ emissions by around 776 million tonnes between 2024 and 2050 by progressively adopting hot rolled steel with lower embodied carbon— a reduction comparable to one year of international shipping’s operational emissions (UMAS, 2023^[6]). The transition towards low carbon energy and steel in global markets is also expected to affect volumes, cargo composition and the geography of supply and demand, and hence maritime transport, which subsequently is of relevance to shipbuilding.

Shipbuilding also plays a critical role in enabling downstream efforts towards the transition to net-zero maritime transportation, which is where the strongest regulatory focus currently lies. Decarbonisation of maritime transportation and regulatory uncertainty over alternative fuel pathways is expected to affect ship replacement rates, the types of ships replaced and owner’s retrofitting decisions.

At the end-of-life phase, ship recycling has the potential to provide an important source of scrap to enhance capacities for low carbon secondary steel production, directly impacting the Scope 1 GHG emissions for steel (Sustainable Shipping Initiative, 2023^[7]). However, challenges to the environmental safety of the process remain, as well as to the implementation of a cradle-to-cradle approach, i.e., the design and production of ships in such a way that at the end of their life, they can be recycled in a safe manner (Ismail et al., 2019^[8]). To help address these concerns, the Hong Kong International Convention for the Safe and Environmentally Sound Recycling of Ships, adopted in 2009 and set to enter into force in June 2025, establishes regulations to minimise environmental, health and safety risks associated with ship recycling. The Convention covers regulations 1) for the design, construction, operation and preparation of ships to support safe and environmentally friendly recycling practices while maintaining operational performance, 2) for ship recycling facilities to operate in a secure and environmentally sound manner, and

3) for an enforcement framework to report certification and reporting obligations (International Maritime Organization, n.d.^[9])

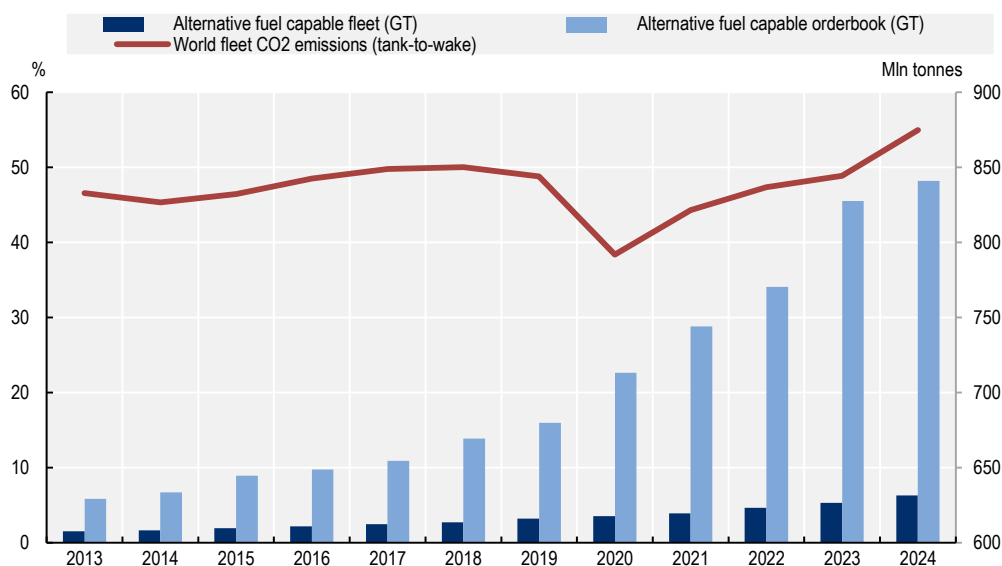
Maritime decarbonisation is strongly linked to wider changes in energy and raw material use in response to decarbonisation efforts, which may also impact shipbuilding and the quantity and quality of ship demand. The viability of low/zero-emission ships remains a significant challenge for maritime decarbonisation, and a price gap between conventional and alternative fuels risks limiting shipping demand for available onboard technological solutions. Consequently, bottlenecks for the uptake of low/zero-emission energy sources, both in terms of production cost and alternative energy supply chains, can pose challenges to shipbuilders (International Transport Forum, 2022^[10]). Further technological and commercial innovation is required to unlock the necessary investment decisions for the global fleet ahead of 2030, with shipbuilders and marine technology suppliers being tasked to ensure that engines and ship design are fit to support this transition.

2.2. Newbuilding for alternative fuel capability

Alternative fuel capable vessels in global ship orders are increasing. By the end of 2024, over half (52%) of the global ship orderbook in gross tonnage (GT) terms could use alternative fuels or battery hybrid propulsion. This marks a significant increase compared to previous years (up from 33% in 2021 and only 10.9% in 2017) and reflects a growing commitment by shipowners to alternative fuels.

The pace of adoption within the global fleet currently in operation has been considerably slower. Only 7.2% of the fleet by tonnage can use alternative fuels or propulsion as of 2024, a rise from 4.6% in early 2022 and 2.5% in 2017. In terms of the number of vessels, the share is smaller, at approximately 1.8%, indicating that larger vessels are being prioritised for alternative fuel capabilities (Clarksons Research, 2024^[11]). This disparity between tonnage and vessel count raises important questions about how to encourage smaller ships to adopt these technologies and ensure that the entire fleet transitions toward “greener” propulsion systems. Despite the growing uptake of alternative fuels, the maritime sector still faces significant challenges. Each fuel option comes with its own technical and economic hurdles. Additionally, uncertainty remains over which fuel or combination of fuels will become the long-term solution for maritime decarbonisation, adding complexity to the decision-making process for shipowners, shipbuilders, and policymakers.

Figure 2.1. Alternative fuel capable fleet and orderbook vs. maritime CO2 emissions (2013-2024)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

The global shipping fleet is aging, with the average age now at 13 years (on a GT weighted basis), up from a low of 9.7 years in 2013. This trend is consistent across several key sectors. Bulk carriers have an average age of 12.5 years, tankers average 13.5 years, and although the container fleet's age has started to decline slightly, it remains relatively high at 13.9 years (Clarksons Research, 2024^[12]). Currently, 34% of the global fleet's tonnage is over 15 years old. This increasing age, combined with stricter environmental regulations under the guise of the IMO, suggests growing potential for fleet renewal through replacement or retrofitting to meet carbon intensity targets.

LNG dual-fuel technology continues to be the primary choice for new vessel orders. As of October 2024, LNG fuel capability accounts for approximately 6% of the global fleet in GT. With over 1,018 vessels, LNG capable ships also represent 37.0% of the orderbook tonnage. Furthermore, as of October 2024, there were over 532 LNG ready vessels in the fleet, with an additional 134 on order.¹ The increased adoption of LNG fuel capability extends beyond LNG carriers, with around 597 non-LNG carrier vessels in the existing fleet and 628 more in the orderbook that are designed for LNG fuel use (Clarksons Research, 2024_[11]).

Box 2.1. The role of LNG as a transition fuel

LNG has been considered a promising alternative fuel for shipping due to its potential to reduce onboard GHG, sulfur oxide (SOx), nitrogen oxide (NOx) and particulate matter emissions compared to conventional marine fuels. However, its long-term suitability for achieving net-zero emissions in shipping is uncertain.

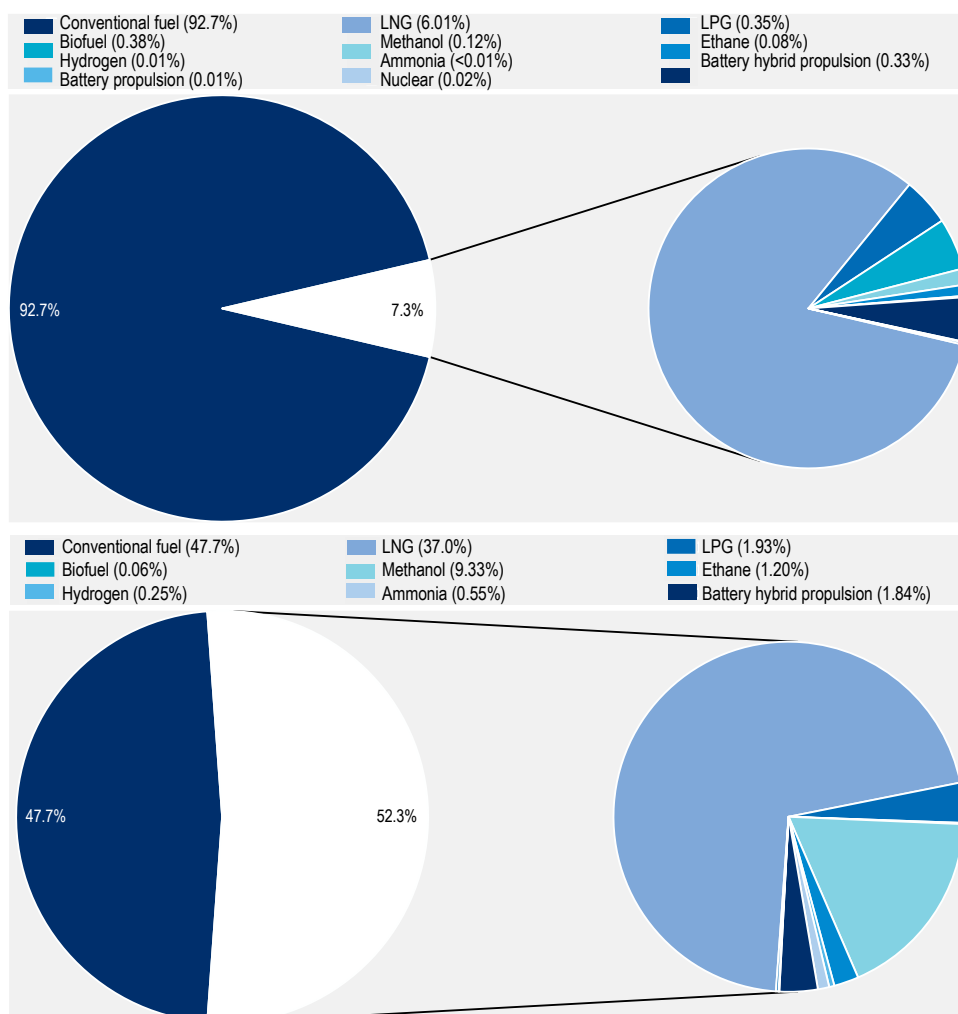
Challenges to LNG as an alternative fuel: A key concern about the usage of LNG as an alternative fuel is methane slip—the unintentional release of methane during extraction, transport, and combustion—which significantly undermines LNG’s climate benefits. Methane has a global warming potential approximately 28–34 times higher than CO₂ over a 100-year period. Additionally, there is a risk of stranded assets, as investments in LNG capable ships may become economically unviable under increasingly stringent decarbonisation regulations. Projections indicate a risk of future oversupply of LNG shipping capacity. Under the IEA’s Net Zero Emissions (NZE) scenario, no additional LNG carriers are needed, yet current orders could push oversupply beyond 400 vessels by 2030. Even under the more conservative Stated Policies Scenario (STEPS), excess capacity is expected to reach 40% of the 2023 fleet.

Long-term fuel pathways: LNG capable vessels have the technical potential to transition to low-emission fuels, such as e-methane (synthetic methane produced using renewable energy) or bio-methane, enabling them to remain relevant in a future net-zero ecosystem if appropriate retrofitting and infrastructure investments are made. Further, the development of engine designs focused on reducing methane emissions can support the attractiveness of LNG as a transitional fuel in the maritime sector. This adaptability could help mitigate some of the financial and environmental risks associated with LNG but requires clear policy frameworks and technological advancements to minimise methane leakage across the fuel’s lifecycle.

Source: (Fricaudet et al., 2022_[13]), (Climate Analytics, 2024_[14]), (International Energy Agency, 2024_[15])

Figure 2.2. Share of alternative fuels and propulsion in the fleet and the orderbook, as of October 2024

Share in the fleet (top) and orderbook (bottom) in terms of GT



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

There is a growing uptake of alternative fuel and propulsion solutions (other than LNG), including methanol, LPG, biofuels, ethane, hydrogen, and increasing interest in ammonia, synthetic methane, and nuclear propulsion options (as seen in Figure 2.2). In particular, the construction of methanol and LPG capable vessels is accelerating, alongside the emergence of newbuilds capable of running on ammonia and hydrogen. By Q4 2024, around 9.3% of the orderbook (272 vessels) was dedicated to methanol capable ships, while 1.9% (125 vessels) were set to use LPG. Additionally, approximately 3.0% of the orderbook was allocated to vessels using other alternative fuels, making up around 448 vessels. This includes 31 hydrogen capable vessels, 61 ethane capable vessels, 28 ammonia capable vessels, 22 biofuel capable vessels, and approximately 410 vessels equipped with battery or hybrid propulsion systems (Clarksons Research, 2024^[11]). Additionally, onboard carbon capture and storage (CCS) is considered a viable alternative to zero-emission fuels, and, by September 2024, carbon capture scrubbers were installed on 28 vessels. While onboard CCS technology is still in its early stages and requires

significant advancements, first movers are driving adoption— as seen with Singapore's Seatrion retrofitting a Norwegian-owned ethylene carrier with a full-scale carbon capture facility, with pilot testing poised to deliver key insights into feasibility and scalability (Offshore Energy, 2025^[16]).

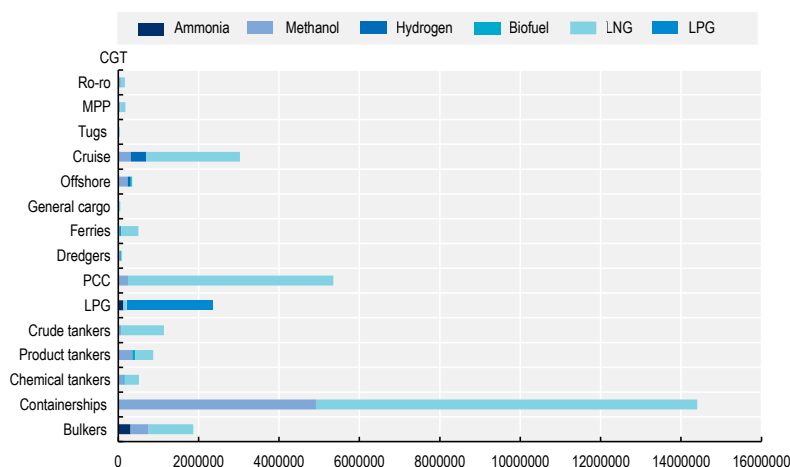
There are no ocean-going ammonia capable vessels in the fleet, as of October 2024, despite the orderbook showing increased ammonia readiness in future vessels (with 298 ammonia ready ships slated for construction). To ensure the uptake of ammonia, port-side investment projects, such as ammonia and hydrogen bunkering facilities, are also required. A significant milestone in advancing ammonia as a shipping fuel was achieved in 2024 with the world's first ship-to-ship transfer of ammonia in a working port environment (Yara, 2024^[17]). Various pilot initiatives for ships powered by clean ammonia are currently in progress, such as the West Australia-East Asia iron ore green corridor, which aims to have ships fuelled by clean ammonia servicing its iron ore shipping lanes by 2028, with an adoption target of 5% by 2030.

Fuel optionality continues to be preferred over fuel choice. Currently, there are 532 LNG ready ships in the global fleet and 134 more on order. In addition, 298 ammonia ready, 507 methanol ready, and 12 hydrogen ready vessels are on order. Alternative fuel ready vessels now account for approximately 20% of the tonnage ordered between January and September 2024, up from around 17% for the full year in 2023, signalling a clear shift toward greater flexibility and future adaptability in fuel choices across the industry.

(Hybrid) battery propulsion is gaining traction, while nuclear propulsion is being explored, as alternative methods of decarbonising the maritime sector. Battery and hybrid battery propulsion accounted for up to 1.88% of the orderbook in CGT by September 2024, up from just 0.34% in the global fleet. Battery propulsion and battery/diesel combinations are particularly gaining traction in smaller and coastal vessels, such as ferries, tugs, and offshore vessels, reflecting their suitability for short-range, lower-power maritime applications. Nuclear propulsion, while still a niche technology, also saw a slight increase from 0.02% of the fleet to 0.12% of the orderbook. While nuclear propulsion is primarily used for defence vessels, it is also being reconsidered for commercial ships, given its advantages of zero emissions, no need for bunkering, low weight, and the potential for high design speeds. These benefits must be weighed against significant challenges, including security concerns, complex monitoring requirements, high capital expenditure (CAPEX) and wider social and political risks (DNV, 2024^[18]). New small modular reactors (SMRs) have the potential to introduce low-maintenance reactor technology that could meet the propulsion and energy needs of commercial vessels (Lloyd's Register, 2024^[19]). In July 2024, Mokpo National University in Korea established the world's first SMR Ship Research Institute, with the goal of becoming a leading global centre for research and education on SMR-powered ships (world nuclear news, 2024^[20]).

Vessel types are emerging to seize opportunities of the net-zero transition, notably to support carbon capture and hydrogen transport (UNCTAD, 2024^[1]). Liquefied CO₂ (LCO₂) carriers are being developed to facilitate large-scale CO₂ transport, with Chinese shipbuilder, Dalian Shipbuilding Industry Co. (DSIC), constructing the first LNG capable LCO₂ carrier for Northern Lights in 2024 (Northern Lights, 2024^[21]). Meanwhile, Japanese NYK Line, K Line, MOL, Imabari Shipbuilding, JMU Corp, and Mitsubishi Heavy Industries secured approvals for standardised LCO₂ carrier designs, aiming for deployment by 2028 (Mitsubishi Corporation, 2024^[22]). Similarly, liquefied hydrogen (LH₂) carriers are advancing. In September 2024, DNV granted Approval in Principle (AiP) to HD Korea Shipbuilding & Offshore Engineering (HD KSOE) for its electric propulsion liquefied hydrogen (LH₂) carrier concept, designed to store and transport up to 80,000 cubic meters of LH₂ (DNV, 2024^[23]).

Figure 2.3. Breakdown of selected fuels by vessel type in the global orderbook (2024-2026)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

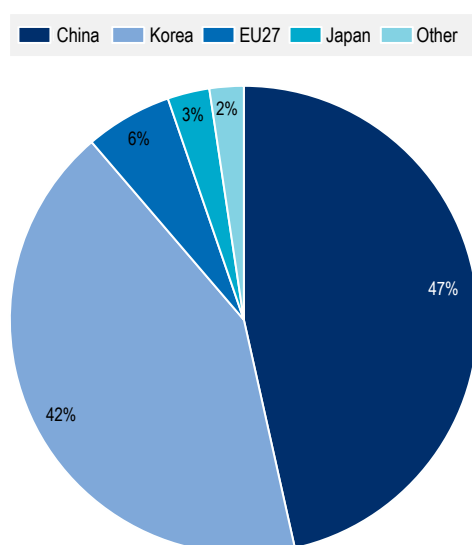
Across different vessel types, alternative fuel capability is increasingly adopted, with methanol featuring prominently in the containership orderbook, while LNG remaining the leading fuel choice in multiple segments. Bulk carriers show significant LNG adoption as well, with a smaller number of ammonia and hydrogen capable vessels. In the cruise ship sector, LNG is the predominant alternative fuel, though there are also some hydrogen capable newbuilds. For tankers, there is a diverse fuel mix, with increasing adoption of methanol and biofuels, particularly in product tankers. For vessels other than LNG carriers, alternative fuel capable tonnage accounted for 44% of total contracting in Q1-Q3 2024, up from 34% in 2023 (Clarksons Research, 2024_[11]).

2.2.1. Differences in regional production capacities

There are notable differences in production capacities for alternative fuel capable ships across regions. As of September 2024, China accounted for 47% of the CGT in the orderbook for alternative fuel capable vessels. This is slightly below its 55% share of the overall orderbook, including both conventional and alternative fuel capable ships. Korea follows closely, making up 42% of the alternative fuel capable orderbook—substantially higher than its 25% share of the total orderbook—demonstrating a stronger specialisation towards alternative fuel capable vessels, particularly those capable of operating on LNG. Europe and Japan account for 6% and 3% of the alternative fuel capable orderbook in CGT, respectively.

The type of alternative fuel capable ships produced varies also significantly by region and country. Korea, China, and Japan lead in the construction of vessels capable of operating on LPG, methanol, ammonia, and ethane. China and Korea are also at the forefront of building vessels designed for LNG or biofuels; however, production in these segments is more widely distributed across multiple countries and shipbuilders.

Figure 2.4. China and Korea are leading the alternative fuel orderbook, as of September 2024



Note: Data based on September 2024, global orderbook in CGT.

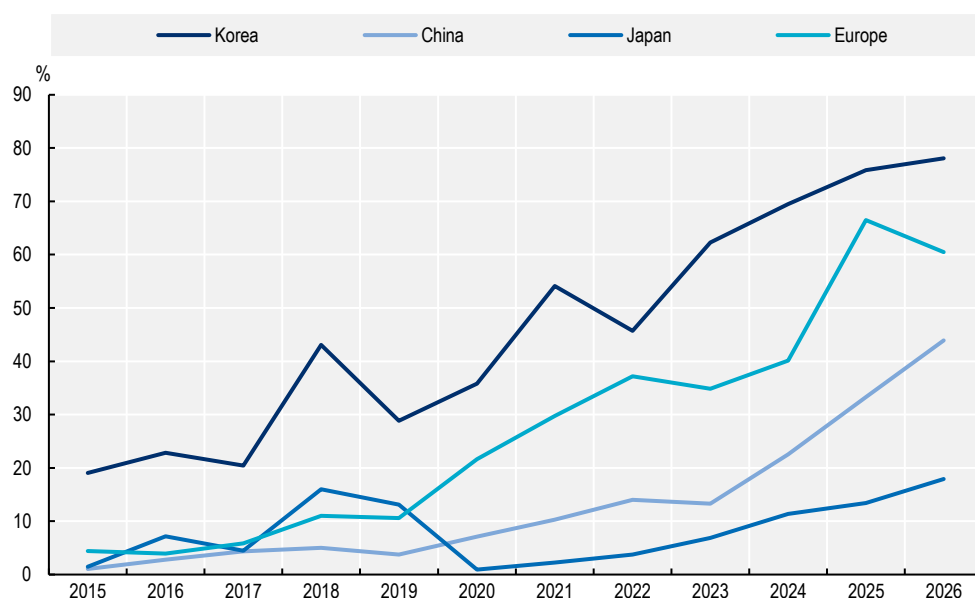
Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Another indicator for assessing major shipbuilding economies' capacity for low/zero-emission shipbuilding is their specialisation in building alternative fuel capable ships. This can be measured by the percentage share of alternative fuel capable vessels in total shipbuilding by each country or region per year.

Korea shows the highest degree of specialisation in low/zero-emission capable shipbuilding, with 69% of its ship deliveries in 2024 being alternative fuel capable. This figure is expected to rise to nearly 80% by 2026, primarily driven by LNG capable vessels, along with a growing number of methanol capable ships. Europe and China have also seen significant increases in their production of alternative fuel capable vessels. Europe expanded from 4.4% in 2015 to 40% in 2024, while China grew from 1% to 22.5% over the same period, reflecting a broader industry shift toward low/zero-emission shipping technologies. Japan exhibits slower growth in alternative fuel capable shipbuilding activity, remaining below 10% throughout the studied period. This slower adoption is largely linked to the type of vessels Japan primarily produces, such as bulk carriers, which have seen a slower uptake of alternative fuels compared to vessel types like containerships, cruise ships, or carriers designed to transport alternative energy sources, like LNG carriers.

China, Korea and Japan, as dominant global shipbuilders, are well-positioned to benefit economically from the transition to zero-emission shipping, according to a study by the International Council of Clean Transportation (ICCT). The paper examines the shipbuilding market, government support, and potential revenue gains from alternative fuel capable shipbuilding. Since zero-emission capable vessels are more expensive than conventional vessels, early movers could gain a competitive advantage. If all bulk carriers, tankers, and container ships built in 2030 were zero-emission capable vessels, global additional revenues could reach between USD 6.9 and 36 billion, increasing shipbuilder propulsion revenue by 86% to 452% (Cho, Ünalán and Meng, 2024^[24]).

Figure 2.5. Share of alternative fuel capable vessels in shipbuilders' total production (2015-2026)

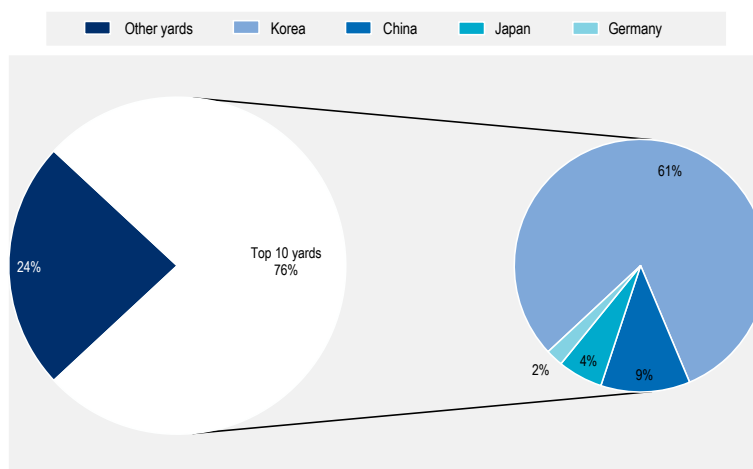


Note: Europe is EU27 as well as Norway and United Kingdom, studied period includes 10 years of completions and 2 years of orderbook, completions in CGT.

Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Alternative fuel capable vessel construction shows very high levels of concentration, with the top 10 yards globally accounting for 76% of production in the past 10 years. These yards are typically characterised by their large scale, advanced technological capabilities, and extensive experience in complex vessel construction. Primarily based in Korea, China and Japan, these companies benefit from considerable investment in research and development (R&D), allowing them to advance the adoption of new fuel technologies such as LNG, methanol, and ammonia. Korean shipbuilders, such as Hyundai Heavy Industries and Samsung Heavy Industries, are prime examples, recognised for their ability to construct large, high-value ships like LNG carriers and ultra-large containerships, which are increasingly being equipped with alternative fuel capabilities. These yards often have strong relationships with marine engine manufacturers and other suppliers, facilitating the integration of cutting-edge fuel systems.

Figure 2.6. High level of concentration in yards constructing alternative fuel capable ships (2015-2024)



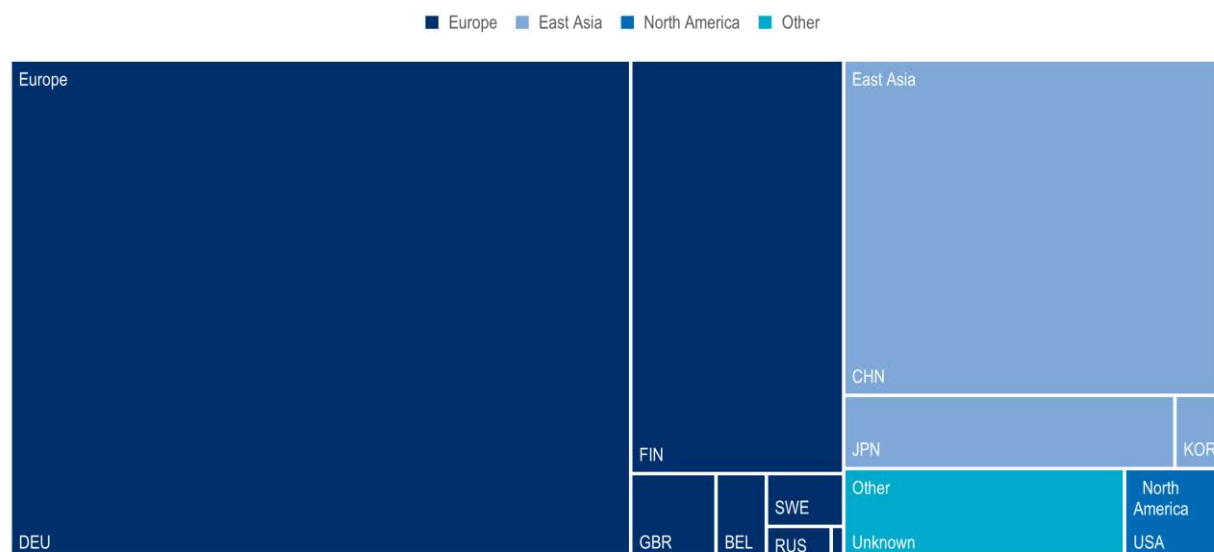
Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Beyond ship construction, marine engine development is a critical factor in the transition to alternative fuels, determining the feasibility and scalability of new fuel types across vessel segments. Advancements in engine technology enable compatibility with low/zero-emission fuels, while dual-fuel and retrofit solutions help bridge the transition from conventional fuels. The development and deployment of these technologies vary by region, making the distribution of marine engine manufacturing capabilities an interesting indicator of technological leadership and industrial readiness for alternative fuel adoption.

Marine engine design for alternative fuel capable vessels is dominated by European designers.

German marine engine designers, primarily M.A.N. Energy Solutions, account for over half (51%) of all engines in the global fleet and orderbook of alternative fuel capable ships worldwide (in terms of numbers of vessels). Following Germany, Chinese marine engine designers account for 21% of engines for alternative fuel capable vessels, reflecting its expanding role in marine technology and upstream equipment sectors. Finland ranks third with 14.5%, while Japanese engine designers make up 4%. In North America, engine design is also noteworthy, although smaller in scale. The US represents 1.5% of global engines for alternative fuel capable vessels, reflecting its presence in the alternative fuel marine technology space. This regional variation in engine production highlights the differing levels of specialisation and market focus across the global marine equipment industry.

Figure 2.7. Engine designers for alternative fuel capable vessels in the fleet and orderbook (as of September 2024)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

2.2.2. Deep dive on methanol and ammonia

Methanol and ammonia have been selected for a closer examination of their respective orderbooks due to their distinct growth trajectories and strategic importance in the future of maritime decarbonisation.

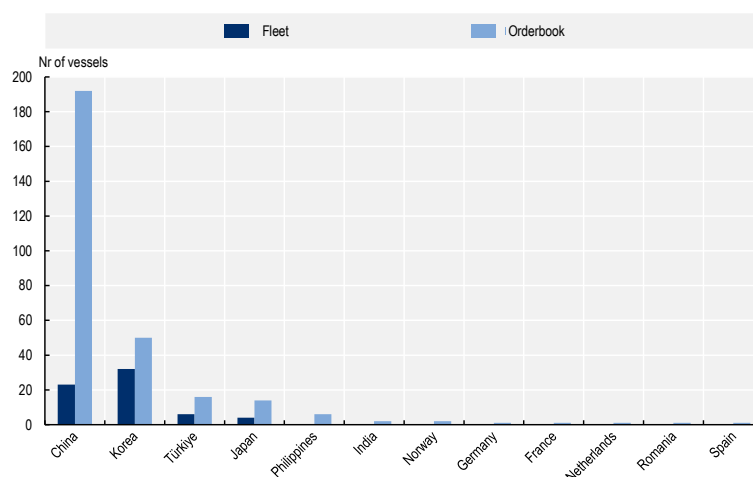
Methanol is currently experiencing the most rapid increase in ship orders, with the share of methanol capable vessels increasing from 0.12% in the global fleet to 9.33% in the orderbook by September 2024, highlighting its rising prominence as an alternative fuel (Clarksons Research, 2024^[11]). Green methanol can be classified into bio-methanol, derived from biomass or biogas, and e-methanol, produced by combining green hydrogen with CO₂ from direct air capture or biogenic sources. Both have minimal lifecycle emissions, making them viable for net-zero shipping. Further, methanol's chemical properties make it an attractive fuel alternative. Despite its lower energy density, it shares combustion characteristics with heavy fuel oil, simplifying storage and handling compared to ammonia or LNG. Additionally, methanol's 'drop-in' capability allows modern dual-fuel marine engines to operate on both methanol and heavy fuel oil (Methanol Institute, 2023^[25]).

Ammonia, on the other hand, is being prioritised for its long-term potential, with the International Energy Agency (IEA) projecting that it will account for 44% of final energy consumption in the shipping sector by 2050 (International Energy Agency, 2023^[26]). Ammonia can be produced entirely carbon-free using renewable energy. Its higher energy density compared to hydrogen and its ability to be stored in liquid form at manageable temperatures make it a more viable long-term solution for deep-sea shipping. However, the widespread use of ammonia as a marine fuel presents safety challenges. Due to its toxicity and corrosiveness, increased handling frequency will require strict safety measures and operational protocols to minimise risks during storage, bunkering and ship operations (Global Centre for Maritime Decarbonisation, 2024^[27]).

Given the significant increase in methanol capable vessels in the orderbook, it is relevant to assess the differences between the current maritime fleet and orderbook projections up to 2028. The economies showing the highest orderbook growth relative to their current fleet of completed vessels are of particular interest. As can be seen in Figure 2.8, this comparative assessment underscores the remarkable increase

in methanol capable orders, as the orderbook projections indicate a substantial increase, reaching up to 2000% of the present completed fleet in China. Notably, the methanol capable vessel segment has not only seen orderbook expansion but also the diversification in the countries engaged in constructing these vessels, with Türkiye, Philippines, India, Norway, Germany, France, the Netherlands, Romania and Spain having received methanol capable ship orders. China and Korea maintain their prominence as major suppliers and China is projected to surpass Korea in terms of the highest number of orders.

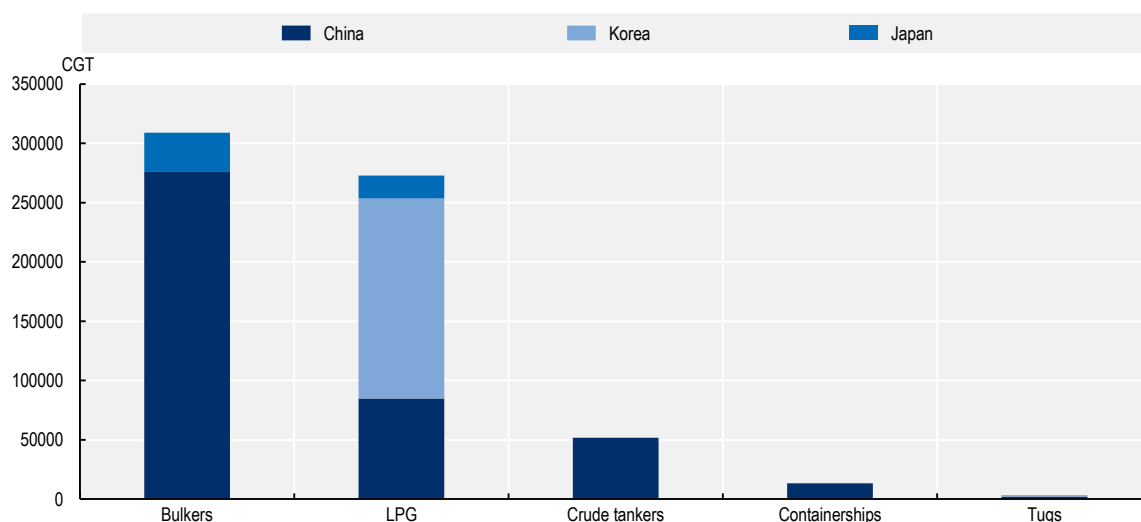
Figure 2.8. Methanol capable fleet vs orderbook by builder country



Note: Data as of September 2024.

Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

The orderbook of ammonia capable vessels in CGT is concentrated in China, Korea and Japan, which show clear differences in the scale and diversification of production (Figure 2.9). China has the largest orderbook for ammonia capable vessels, with approximately 400,000 CGT. This reflects a highly diversified portfolio, spanning bulk carriers, LPG carriers, crude tankers, containerships, and tugs. This broad mix of vessel types underscores China's strategic approach to establishing a strong presence across multiple segments of ammonia capable shipping. Korea's ammonia capable vessel production is more concentrated, with a total of about 150,000 CGT, the majority of which are LPG carriers. Japan, with around 50,000 CGT, primarily focused on bulk carriers and a smaller proportion dedicated to LPG carriers. The limited diversification in ammonia capable vessel production in Korea and Japan suggests a narrower focus, likely driven by their traditional shipbuilding capacities in these specific vessel types.

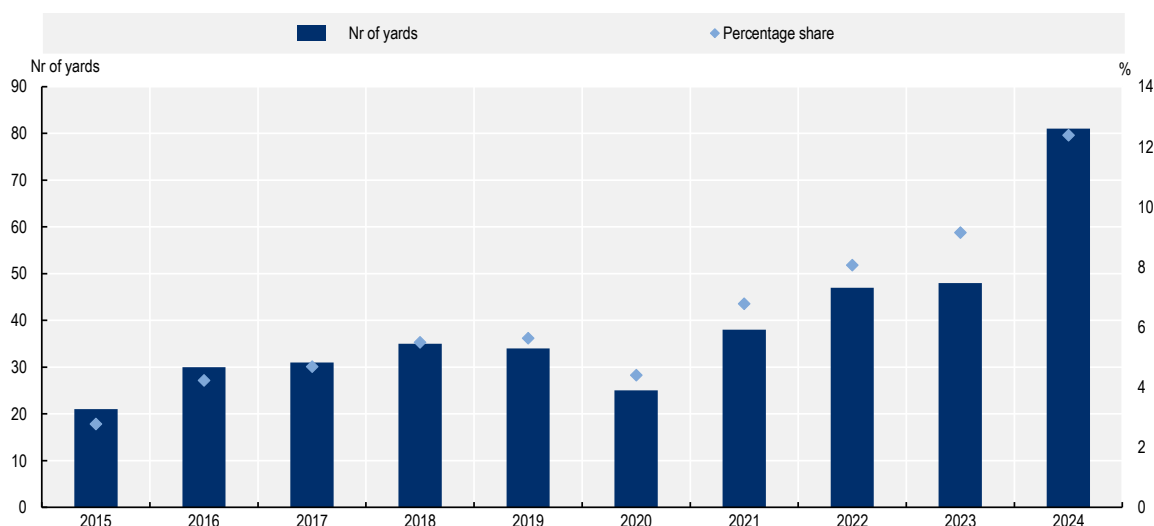
Figure 2.9. Orderbook ammonia capable vessels in selected countries (2024-2027)

Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

2.3. Active yards

Assessing active shipyards provides valuable insights into the annual capacity of shipyards to construct alternative fuel capable vessels (OECD, 2023^[28]). Examining the number of yards that have produced at least one alternative fuel capable vessel each year over the past decade, the analysis further stratifies the data by vessel type, builder country, and alternative fuel type. This approach enables a clear comparison between shipyards constructing alternative fuel capable vessels and the overall industry activity, which includes all shipyards building ships, whether conventional fuel capable or not. The ratio of “alternative-to-total active yards” highlights the proportion of active yards contributing to alternative fuel vessel construction relative to the total number of active yards. The “built date” of a ship, as recorded in Clarksons World Fleet Register (WFR), is used to track annual output, including vessels delivered in 2024, providing an assessment of the shipbuilding sector’s capacity to support decarbonisation efforts.

Figure 2.10. Yards constructing alternative fuel capable vessels make up 13% of all active yards in 2024

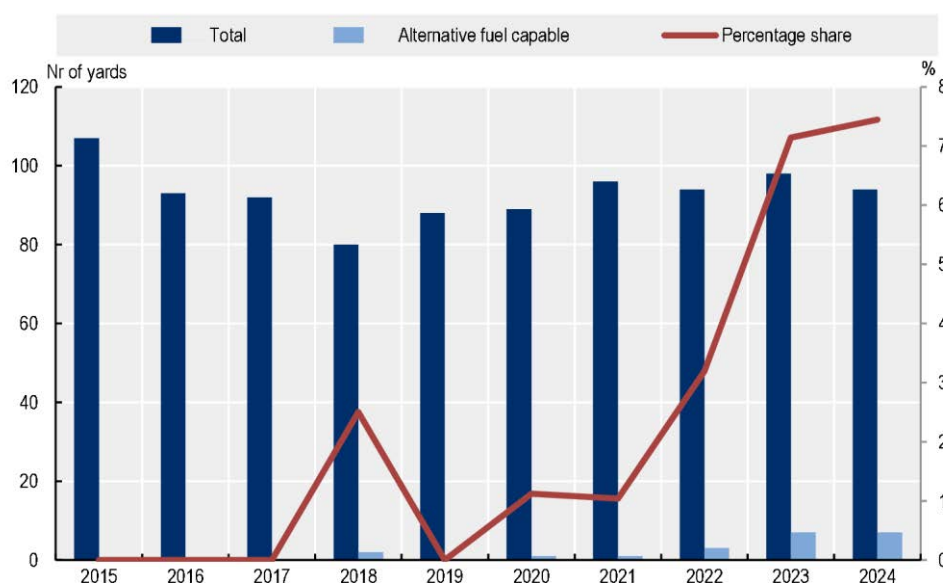


Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

To better understand shipbuilding capacity for different vessel types, it is of interest to analyse active yards constructing different vessel and fuel types separately.

For bulk carriers, the number of shipyards constructing alternative fuel capable vessels remains relatively modest. In 2024, only around 7.5% of shipyards involved in bulker construction—equivalent to 8 out of 94 active yards—were identified as building ships capable of operating on alternative fuels. This indicates that, while the capacity for constructing such vessels exists, it is still highly concentrated among a few shipyards. However, there has been a notable expansion in this area over the past five years (see Figure 2.11). A closer look at the data reveals a discernible shift in the type of alternative fuels being adopted for bulkers. The initial surge in the construction of alternative fuel capable bulkers between 2017 and 2019 was predominantly driven by the introduction of biofuel capable ships. Subsequently, the continued increase in the number of shipyards venturing into the construction of alternative fuel capable bulkers has been primarily associated with LNG.

Figure 2.11. Active yards constructing alternative fuel capable bulkers

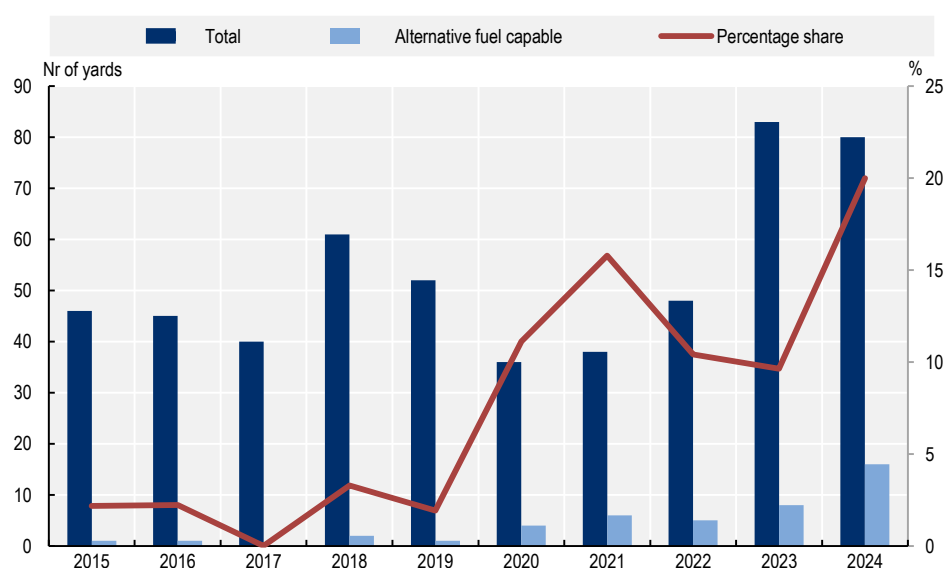


Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Active yards constructing containerships show a high share of alternative fuel capable production.

Notably, 20% of shipyards involved in containership construction are producing vessels capable of operating on alternative fuels in 2024. This growing trend begins to gain momentum in 2020. In line with the overall number of yards constructing containerships, the total and relative number of yards building alternative-fuel capable vessels has experienced fluctuations in studied period but has seen a general increase since 2015 (Figure 2.12). This growing trend is primarily attributable to two factors: a substantial rise in the construction of LNG capable containerships and an increase in the construction of methanol capable vessels. The latter see a particular surge in 2024, with five shipyards actively involved in constructing methanol fuel capable containerships.

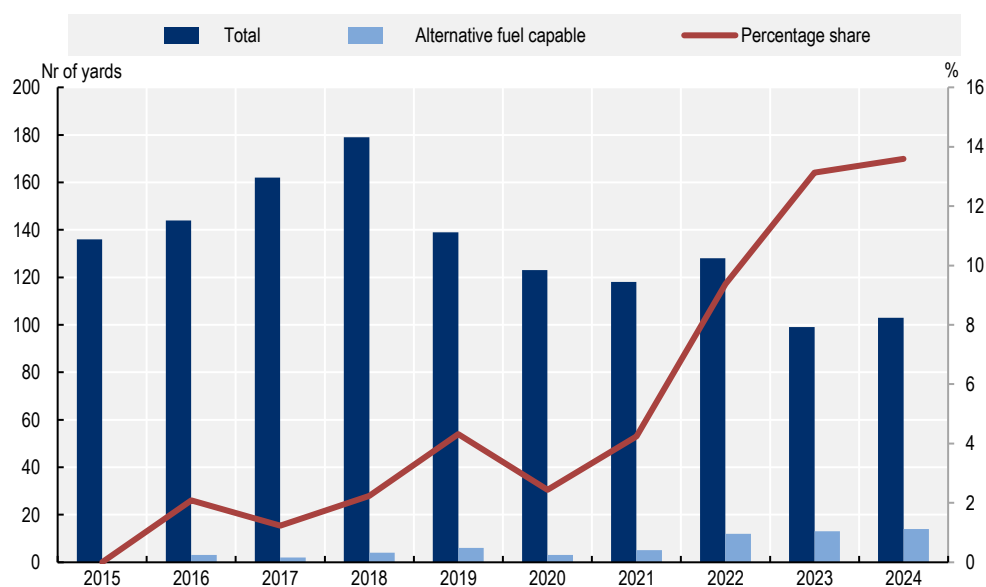
Figure 2.12. Active yards constructing alternative fuel capable containerships



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

In the tanker segment, the proportion of shipyards engaged in building alternative fuel capable vessels remains relatively low. Yards having constructed at least one alternative fuel capable ship represent 14% of the total active yards, indicating a lower rate of construction compared to other ship types. Interestingly, while the total number of yards producing tankers has seen a decline—from a peak of 179 in 2018 to 103 by 2024—the count of shipyards constructing alternative fuel capable tankers has experienced a gradual but consistent rise. This number grew from just four active yards in 2018 to 14 by 2024 (see Figure 2.13).

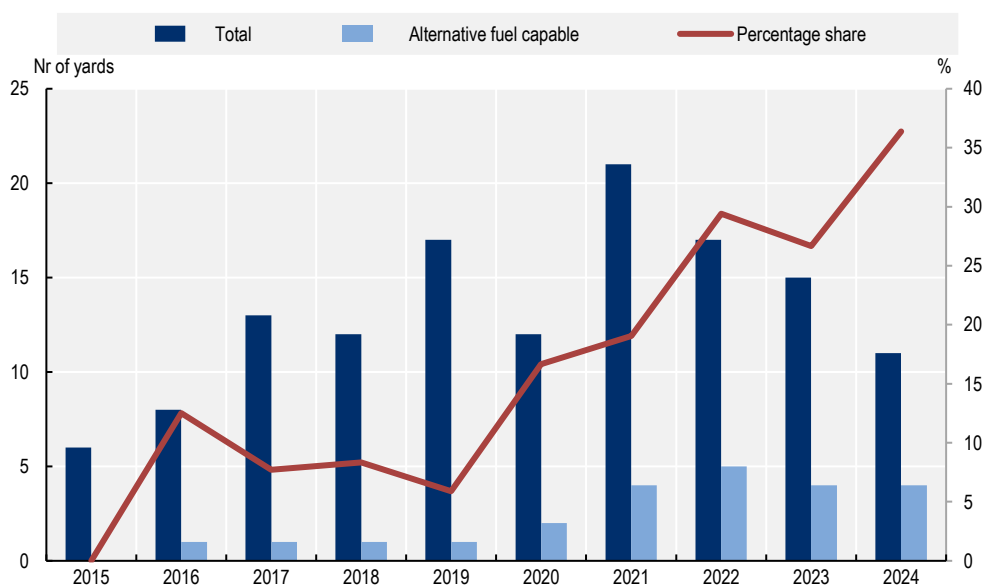
Figure 2.13. Active yards constructing alternative fuel capable tankers



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

For cruise ships, the percentage of shipyards engaged in building alternative fuel capable ships has consistently been the highest among all ship categories. In 2024, this figure reached a peak of 36%, underscoring the significant share of the shipyards capable of constructing low-carbon cruise vessels. This trend can be largely attributed to a decrease in the number of shipyards exclusively manufacturing conventionally fuelled ships (Figure 2.14). This increase in the number of shipyards building alternative fuel capable cruise ships became particularly pronounced after 2019. As in other segments of ship construction, the increase in yards specialising in alternative fuel technologies for cruise ships linked to the rise in the construction of LNG capable vessels. LNG, a transition fuel, highlights the importance of the maritime industry's ongoing efforts to adapt and prepare for a future that increasingly relies on alternative fuel sources (DNV, 2023^[29]).

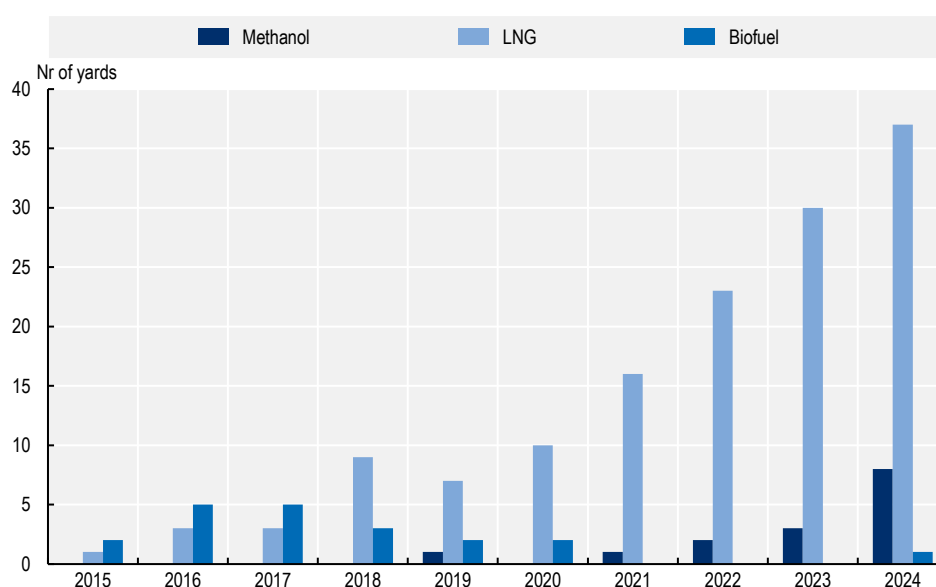
Figure 2.14. Active yards constructing alternative fuel capable cruise ships



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Analysis by fuel type confirms LNG predominant position, with most shipyards focusing on constructing LNG capable vessels (37 yards). Concurrently, there is a noticeable increase in the number of shipyards expanding into the construction of methanol capable ships (8 yards), reflecting a diversification in fuel alternatives being pursued (Figure 2.15). Yards constructing biofuel capable ships present a contrasting picture. After peaking in 2017, constructing biofuel-powered vessels has seen a decline. This shift can partially be attributed to the growing prominence of other alternative fuels, such as LNG, which has seen a rapid increase in adoption since 2019. Shipyards that previously focused on building biofuel capable tankers and containerships up until 2020 might have shifted towards LNG (and methanol), driven by the growing industry uptake of these fuels. The apparent decrease in biofuel capable ship construction, particularly among tankers and containerships, might also reflect a selection bias towards ship types that traditionally have not favoured biofuels as much. For instance, trends in active yards might be different for ferries, where biofuel capable vessels could represent a larger percentage.

Figure 2.15. Trends in active yards by alternative fuel capable ship type



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

2.4. Price difference in vessel types

Determining the price premium for alternative fuel capable vessels offers insight into the additional capital required to adopt zero/low-carbon maritime technologies. It can therefore help assess the economic viability of alternative fuel options, enabling stakeholders to forecast long-term operational costs. By identifying the financial disparity between conventional and alternative fuel capable vessels, it can also help inform the development of targeted policy incentives, aiming to narrow the cost gap and accelerate the maritime industry's shift towards zero/low-carbon technologies and practices.

To identify the price difference for alternative fuel capable vessels, the analysis employs a matching method to identify comparable vessels based on year of contract, country of origin, and size—with a degree of tolerance for variations. Due to data limitations, the analysis is focused on the pricing premium for LNG capable and methanol capable ships compared to conventionally fuelled vessels. The final output contextualises the price premia by three comparison pairs: 1) LNG capable and conventional fuel vessels, 2) methanol capable and conventional fuel vessels, and 3) LNG capable and methanol capable vessels. Data on prices for LNG capable vessels was available from 2019 to 2024, and for methanol capable vessels from 2022 to 2024. Price data were sourced from Clarksons WFR and the author's data collection.

Table 2.1 provides an overview of results. While no ship segment yielded a match for every year within the study period, most contained multiple data points. For bulk carriers, containerships, pure car carriers, and various types of tankers, it was possible to generate estimates within specific builder countries. When this was not feasible, cross-country comparisons were explicitly stated. However, several limitations should be considered. In most categories, the data is based on fewer than five price observations per fuel type, meaning that in a given year, builder country, and ship size, fewer than five individual price points were available for both alternative and conventional fuel vessels. This limited sample size reduces the robustness of direct comparisons and may introduce variability in the estimates. Additionally, differences in market conditions, regulatory frameworks, and contract structures across countries and ship types further constrain the generalisability of the findings.

Analysis in prices indicates a noticeable price premium for LNG capable (on average 12.8%) and methanol capable ships (11.4%) across vessel types. The extent of this premium varied significantly by ship type, with bulkers showing the highest price premium for both LNG and methanol capable ships. This observation aligns with previous findings, such as the DNV GL estimates, which suggest the construction costs of new LNG-powered ships are approximately 10-20% higher than those of conventional ships (DNV GL, 2020^[30]). Similarly, A.P. Moller – Maersk A/S, one of the largest shipping companies in the world, has indicated that for methanol capable vessels, this capability adds approximately 10-15% to the total CAPEX of each vessel (MAERSK, 2021^[31]). For certain ship categories such as containers and bulk carriers, the premium appeared to be on a downward trend, despite year-on-year price fluctuations being a common occurrence across all types. Although the higher CAPEX can pose a financial burden for shipowners, it creates additional revenue streams for shipbuilders and shipbuilding economies. Since vessels designed for alternative fuels are more costly than conventional ships, this shift presents a significant economic opportunity— particularly for early adopters, who can establish a competitive edge in the market.

Interestingly, the analysis revealed that several vessel types (bulkers, containerships, crude tankers, pure car carriers) exhibit a lower price premium for methanol capable ships than for LNG capable counterparts. Further, the price comparison between methanol and LNG indicates that methanol capable ships are more cost-effective for pure car carriers, crude tankers and containerships (with a price difference of -2.6%, -4.3% and -1.5%, respectively). As with other price premia calculations, the analysis is limited due to the low number of individual prices available for comparison.

Table 2.1. Percentage price difference for selected vessel types

	LNG capable vs. conventional fuel (%)	Methanol capable vs. conventional fuel (%)	Methanol capable vs. LNG capable (%)
Bulker	22.4**	13.3*	N.A.
Container	15.2	13.0	-1.5*
Product tanker	12.3*	12.7*	N.A.
Chemical tanker	3.0*	12.3*	N.A.
Crude tanker	9.4*	4.6*	-4.3*
Pure Car Carrier	11.4*	6.6*	-2.6*

Note: *based on <5 prices per fuel capability **compares cross-country data

Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

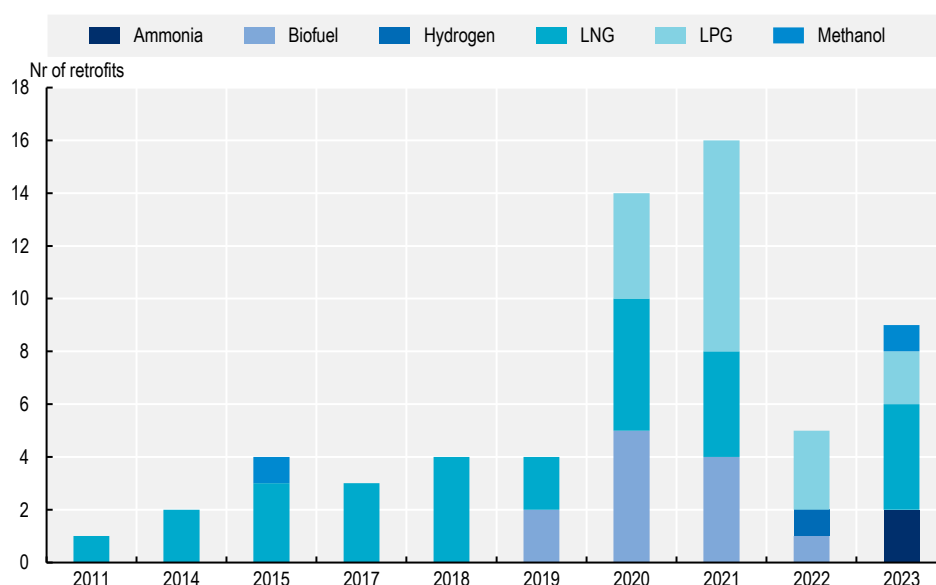
2.5. Alternative fuel conversions

An alternative fuel conversion is defined as the retrofit of a fuel engine, enabling it to operate using a chosen alternative fuel. This process involves adapting or upgrading existing engine components and fuel systems to ensure compatibility with fuels that offer environmental benefits over conventional options, such as reduced greenhouse gas emissions. The integration of comprehensive fuel systems into existing vessels is complicated by the need for larger fuel tanks to accommodate fuels with lower energy density, which must be placed without affecting the ship's structural integrity or cargo capacity. Additionally, retrofitting ships requires finding space for fuel preparation equipment near the engine room, installing costly and larger double-walled fuel piping with minimal disruption to the ship's structure, and enhancing safety measures for venting, purging, ventilation and fire and gas leak detection (Lloyd's Register, 2023^[32]). The alternative fuels included in this analysis are LNG, LPG, methanol, biofuel, ammonia and hydrogen.

For alternative fuel conversions, LNG appears as the predominant choice, accounting for 28 of the 64 total conversions identified, as of February 2024. This is followed by conversions to LPG and biofuels, with 19 and 12 conversions respectively. The data also reveals a nascent adoption of ammonia and methanol as alternative fuels (2 conversions each), and a single case of hydrogen fuel conversion (Clarksons Research, 2024^[11]). Looking to annual trends, fuel conversion activities fluctuated between 1 and 4 occurrences per year between 2011 to 2019. A notable increase was observed in 2020, with the count increasing to 14 conversions, and peaking at 16 in 2021. This uptick can be principally attributed to an increased frequency of LPG conversions. The first ammonia conversions were recorded in 2023 but they were not applied to oceangoing vessels (Figure 2.16).

The increase in conversion activities, particularly from 2020 onwards, appears to align with the growing presence of alternative fuels (beyond LNG) in the orderbook. While this trend highlights the shipbuilding industry's growing contribution to support GHG emissions reduction from shipping and the evolving regulatory environment supporting these shifts, alternative fuel conversions still represent a small fraction of total conversions and a relatively small number compared to increase of alternative fuel newbuilds. Although alternative fuel conversions are expected to increase in the future as the industry looks to replace and retrofit the existing fleet, concerns remain about potential bottlenecks in the conversion process, which could hinder progress toward net-zero goals.

Figure 2.16. Trends in alternative fuel conversions by fuel type (2011-2023)

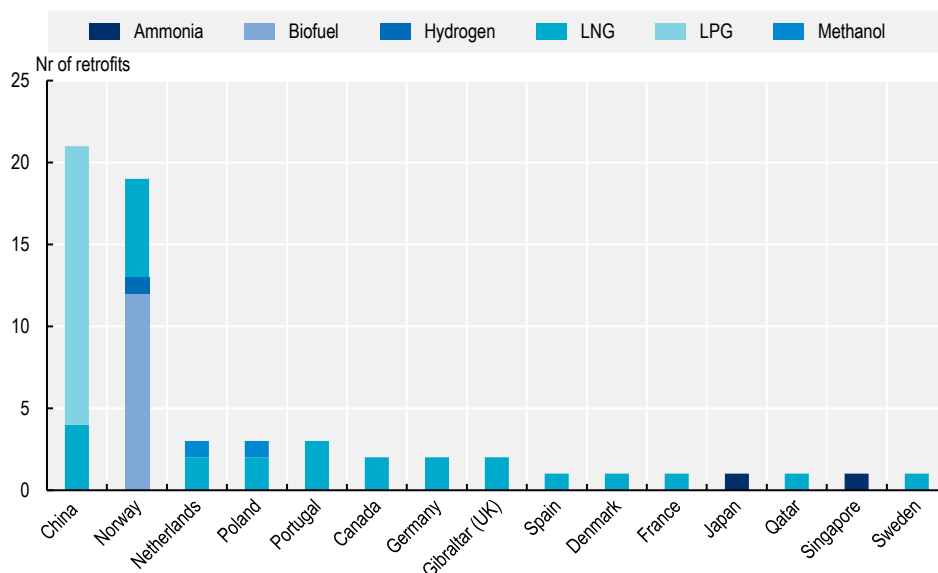


Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Examining the geographical distribution of retrofits, China emerges as the leader in cumulative alternative fuel conversions (21 total), with Norway following just after, and other European shipbuilding economies such as the Netherlands, Poland, and Portugal also making significant contributions. In China, the focus of conversions predominantly lies on LPG, alongside some LNG projects. Norway demonstrates the most comprehensive range of fuel conversions, including biofuels, LNG and hydrogen. Both the Netherlands and Poland have carried out LNG and methanol conversions (Figure 2.17). It is noteworthy that several countries (and territories) are actively participating in the fuel conversion market despite not having a large ship construction industry, such as Qatar, which had 134 vessels in the orderbook from 2011 to 2023. This seems to suggest that retrofitting activities are not necessarily contingent on a country's

shipbuilding capacity. The main Shipbuilding Committee members involved in fuel conversions are Norway, the Netherlands, Poland, Portugal, Denmark, France, Japan and Sweden.

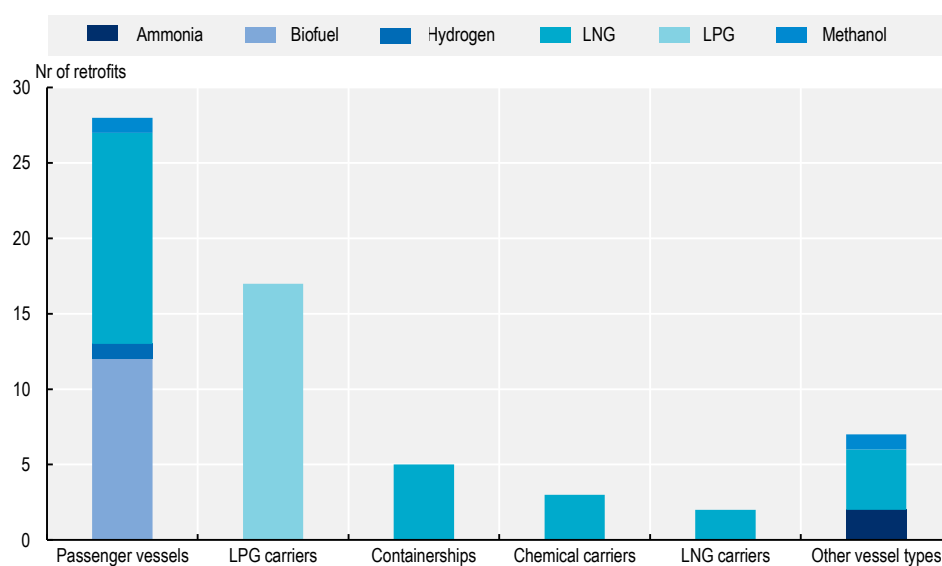
Figure 2.17. Total alternative fuel conversions per country and territory (2011-2023)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

The predominant vessel type for alternative fuel conversions, with the exceptions of LPG and ammonia, are passenger vessels. These are the only type of vessels for which the data indicates biofuel or hydrogen conversions. For containers, chemical carriers and LNG carriers, the conversions recorded are exclusively for LNG. LPG conversions have been documented solely for LPG carriers. Meanwhile, ammonia conversions have been uniquely undertaken on a tug and a platform supply vessel (Figure 2.18).

Figure 2.18. Total alternative fuel conversions by vessel type (2011-2023)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

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Notes

¹ Alternative fuel capable vessels are defined as dual fuel vessels designed to run on fuel oil and an alternative fuel (e.g., LNG, methanol, ammonia, etc). Alternative fuel ready vessels are designed to be more easily retrofitted to future alternative fuel use but are currently not capable of running on the designated alternative fuel. Vessels can be both capable of running on one alternative fuel and ready to run on another, e.g. LPG capable vessels that are ammonia ready.

3

Technology developments in energy efficiency, digitalisation and low-carbon innovation

Energy-saving technologies (ESTs), including new digital solutions, offer a cost-effective and reliable pathway to reducing emissions, while also reducing the overall demand for carbon-neutral fuels. These technologies provide immediate benefits by lowering fuel consumption, making the current business case for their adoption stronger, especially when compared to the higher costs of alternative fuels. At the same time, ongoing innovation in carbon-neutral fuel technologies and ESTs plays a crucial role in advancing new solutions and improving their accessibility and commercial viability. This chapter gives an overview of EST uptake in newbuilding and retrofits, key developments for emerging digital technologies in shipping and shipbuilding as well as innovation trends in the industry, by assessing patenting activity in low/zero-emission technologies.

Key messages

This chapter examines the evolving landscape of newbuilding and retrofitting of energy saving technologies, digital technologies and trends in climate change mitigation technology patenting for maritime and waterways transport. Several key conclusions can be drawn from the analysis.

- **Energy saving Technologies (ESTs) are increasingly adopted by the global fleet** (29% of tonnage), with opportunities for further retrofitting work. Japan and China are main suppliers in ESTs to the global fleet, with Chinese shipbuilders expected to see substantial growth in their supply based on the current orderbook.
- **EST retrofits have increased but show strong geographical concentration:** there is a clear upward trend in both the retrofitting of EST, with China emerging as a leading player in both the retrofitting of propeller and hull ESTs. A significant proportion of retrofit activities, particularly wind EST and carbon capture and storage retrofits, are also being carried out by shipyards in Europe. For retrofits of ESTs, bulk carriers and containerships are most retrofitted across different technology types (making up 32% and 29%, respectively).
- **Vessels are at similar age at the time of retrofit but undergo different retrofitting lengths:** the average age of vessels being retrofitted with ESTs is around 10 years, pointing towards a trend of relatively younger vessels undergoing EST and alternative fuel retrofits. This pattern could imply an industry perspective that retrofitting newer vessels serves as a strategic approach to ensure their commercial viability and regulatory compliance in the mid- to long term. Fuel conversions generally require more time to complete compared to other retrofit types (+45 days), potentially indicating a risk of bottlenecks in retrofitting capacity as demand increases.
- **Digitalisation is slowly increasing and plays a key role for data collection and monitoring, reporting, and verification (MRV):** while digitalisation in shipping has the potential to enhance operational efficiency and reduce emissions and customer demand for vessels integrated with advanced digital technologies is growing, the maritime industry's uptake of 'smart' technology is slowed by higher costs and lower developmental stages of new technologies compared to other industries.
- **The global patenting activity in low/zero-emission maritime technologies has decreased:** despite the growing pressure to decarbonise the maritime sector, the share of low/zero-emission innovation in maritime technologies peaked between 2010 and 2015. The EU and Japan have consistently shown the most patent filings in this field, but China has rapidly increased its patenting activity in the past five years, surpassing other key innovating countries.

3.1. Energy-saving technologies

With the limited supply of alternative fuels and gradual transition to a low/zero-emission fleet, complementary solutions are essential to meeting the shipping industry's emissions reduction targets.

Across the fleet, ESTs are being increasingly adopted by shipowners to reduce emissions and comply with the International Maritime Organisation (IMO)'s tightening energy efficiency and carbon intensity regulations. The use of ESTs offers several key advantages. They reduce energy costs and contribute to the reduction of cumulative greenhouse gas (GHG) emissions over time, providing

significant long-term environmental benefits. They also decrease the shipping industry's dependency on renewable energy sources, which may face supply limitations as demand increases, and, by improving energy efficiency onboard, these technologies can significantly lower the overall cost of the transition to alternative fuels. In fact, estimates suggest that investments in onboard energy efficiency could reduce the demand for investment in alternative fuel production by up to 10 times, making the transition more economically viable for the industry (Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping, 2022^[1]).

Table 3.1 provides an overview of selected technology types, their estimated cost as well as emissions reduction potential based on IMO GreenVoyage2050's Energy Efficiency Technologies Information Portal (IMO Green Voyage 2050, 2024^[2]). Complementary information on onboard carbon capture is based on DNV's latest estimates (DNV, 2024^[3]).

Table 3.1. Overview of selected ESTs

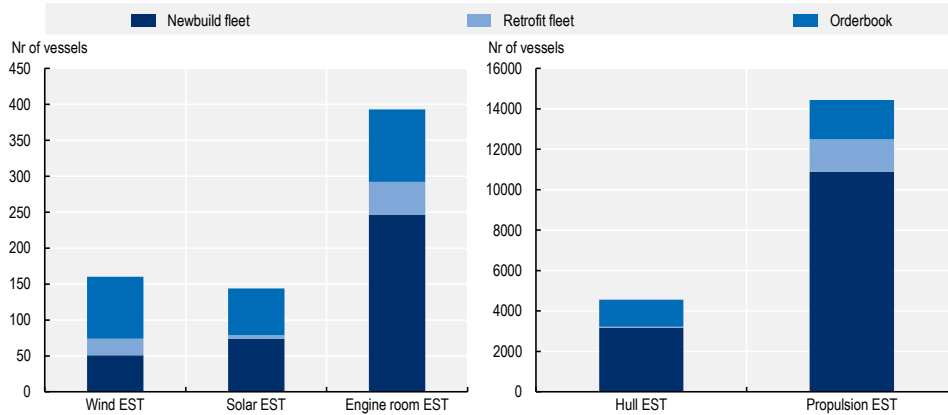
	Propeller EST	Hull EST	Solar EST	Wind EST			CCS
	Propeller modifications	Hull modifications	Solar panels	Flettner rotors	Fixed sails or wings	Wind kites	Onboard carbon capture (OOC)
Type of Technology	Includes propeller duct, pre-swirl stator, rudder bulb, wake equalising duct and propeller boss cap fin, for vessels with changed operational profiles and slow steaming, applicable to all vessel types and ages, particularly large vessels	Includes air lubrication system, bow enhancement, hull fin, suitable for LNG carriers, bulk carriers, and container ships, especially in off-design conditions	Suitable for vessels in sunny areas and requires significant deck space	Spinning vertical cylinders to create lift and propulsion. Suitable for tankers, some bulk carriers, general cargo carriers, RoRos	Harnesses wind power with flexible, rigid, or turbosails. Best suited for ships with ample space, not recommended for container ships	Generates a pulling force from wind power, reducing engine power needed for propulsion	Captures and stores CO2 emissions from ship operations, including post-combustion systems that clean exhaust gases and store CO2 onboard, and pre-combustion systems separating carbon to produce H2
Expected Cost	USD 400,000 - 500,000	Fixed cost of USD 100,000 plus USD 250,000 - USD 700,000 material costs	USD 420,000 - 510,000 for a 150 kW system	USD 400,000 to USD 950,000 per rotor	USD 170,000 to USD 300,000 per mast	USD 280,000 for a 160m ² kite to USD 3,420,000 for a 5,000m ² kite	Initial setup is complex and costly, influenced by space requirements, and the need for CO2 storage infrastructure
Reduction Potential	2% - 5% on main engine fuel consumption	Fuel savings of 3% - 5%	0.5% - 2% in auxiliary engine fuel consumption	3% - 15% on main engine fuel consumption	1% - 10% on main engine fuel consumption	1% - 5% on main engine fuel consumption	10 - 90%

Source: IMO GreenVoyage2050 (2024), Energy Efficiency Technologies Information Portal, <https://greenvoyage2050.imo.org/technology-groups/#EnergyConsumers> (Accessed on 30 September 2024).

In the global fleet, as of October 2024, ESTs were installed on over 9,397 ships, representing around **35% of the fleet's tonnage**. Propulsion-related ESTs, such as propeller ducts, pre-swirl stators, rudder bulbs, wake equalising ducts, and propeller boss cap fins, have seen the most widespread adoption, with 14,435 vessels either already fitted, retrofitted, or set to be fitted in the orderbook. Hull-related ESTs, including air lubrication systems, bow enhancements, and hull fins, have been implemented on 4,554 vessels. Although wind, solar, and engine room ESTs have seen slower uptake, they are now gaining traction, with 160 wind ESTs, 144 solar ESTs, and 393 engine room ESTs installed across the fleet and orderbook. Notably, wind and propulsion-related ESTs have a high proportion of retrofits compared to

newbuild installations, indicating the growing importance of retrofitting as a strategy for improving energy efficiency in the existing fleet (Clarksons Research, 2024_[4]).

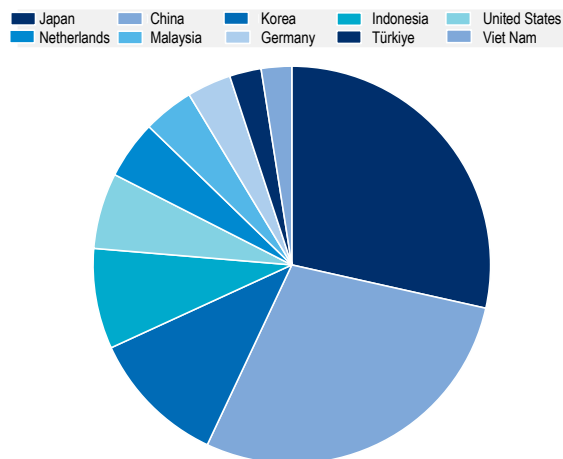
Figure 3.1. Total number of vessels fitted with ESTs (as of September 2024)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Japan and China emerge as the main suppliers of ESTs to the global fleet, making up around 28% respectively. Korea and Indonesia are also notable suppliers. The market for equipment suppliers is diverse, with numerous countries from different regions of the world (Asia, Europe and North America) providing around or below 5% of ESTs within the global fleet (Figure 3.2).

Figure 3.2. Global share in ESTs in the fleet by builder country (as of September 2024)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024)

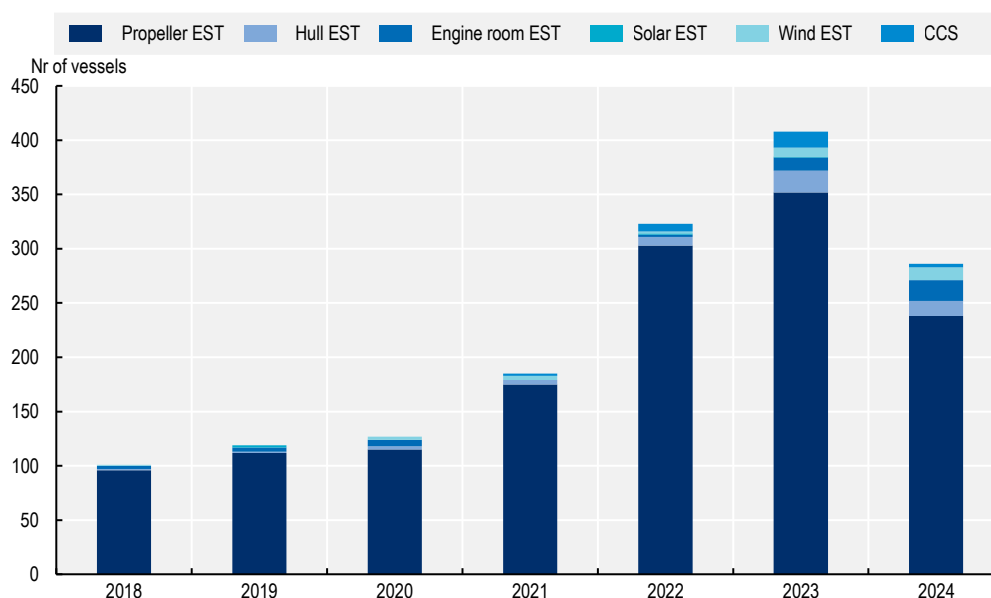
3.1.1. Deep dive on EST retrofits

The average age of the global fleet is increasing, currently standing at 13 years on a gross tonnage (GT) weighted basis, up from a low of 9.7 years in 2013. Further, with 34% of GT now over 15 years old and increasingly stringent decarbonisation measures being implemented, many vessels face potential

regulatory compliance challenges. According to Clarksons Research, under the Carbon Intensity Indicator (CII), it is estimated that approximately 45% of the existing tanker, bulk carrier, and container fleets could receive a D or E rating by 2026 unless modifications are made to their speed or specifications (Clarksons Research, 2024^[5]). This context makes retrofitting ESTs an increasingly attractive option to extend the operational life of older vessels and improve their energy efficiency to meet regulatory requirements. Figure 3.3 illustrates the trend in retrofitting vessels with ESTs from 2018 to 2024.

Total retrofits increased from 101 in 2018 to 408 in 2023, with propeller ESTs being the most widely adopted, growing from 96 vessels in 2018 to 352 in 2023. Hull and engine room ESTs have also seen increased uptake, while wind and solar technologies have experienced slower adoption. The following section will explore key retrofits in more detail, breaking them down by country and vessel type to better understand which vessels are being retrofitted and which countries have the greatest capacity for retrofits.

Figure 3.3. Retrofits of selected EST technologies (as of Sep 2024)



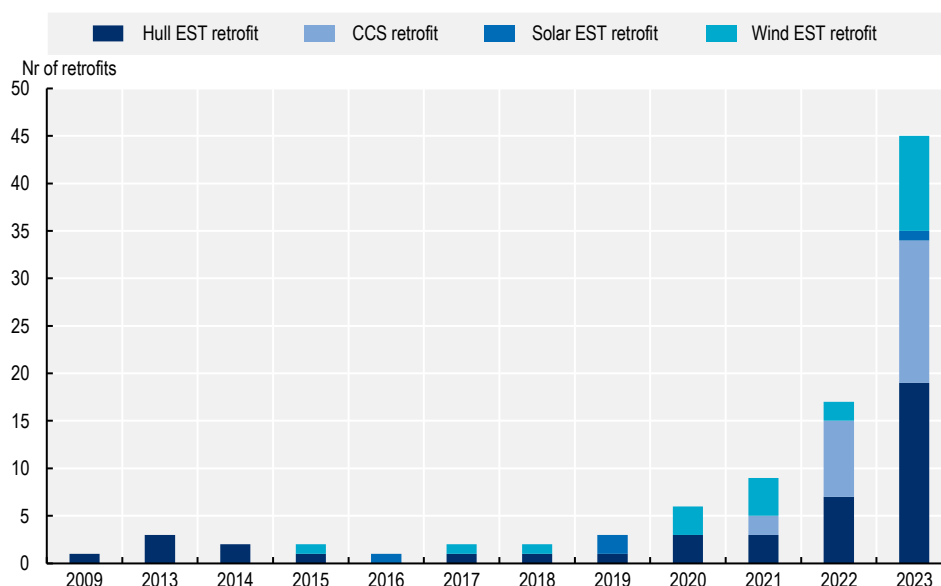
Note: Data for 2024 is only available until Sep 2024.

Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Hull, CCS, solar and wind EST retrofits

Within the selected group of EST retrofits, representing technologies that saw limited uptake before 2020, hull modifications are the most prevalent form, accounting for 51 of the documented retrofits. This is followed by onboard CCS with a total of 26 retrofits, wind energy-saving technologies with 23, and solar power implementations, which are the least common with only 4 cases. Throughout the period studied, retrofits in selected ESTs were relatively rare until a notable increase in adoption in 2020 and a significant surge in 2023. This upswing is primarily attributed to increases in hull modifications as well as CCS and wind energy technologies retrofits. The first instances of CCS retrofitting emerged in 2021, marking it as the most recent technology to be adopted (Figure 3.4). The significant increase in retrofits in 2023 may reflect ship owners' reaction to and anticipation of stricter regulations by the IMO, such as the introduction of the CII in 2023.

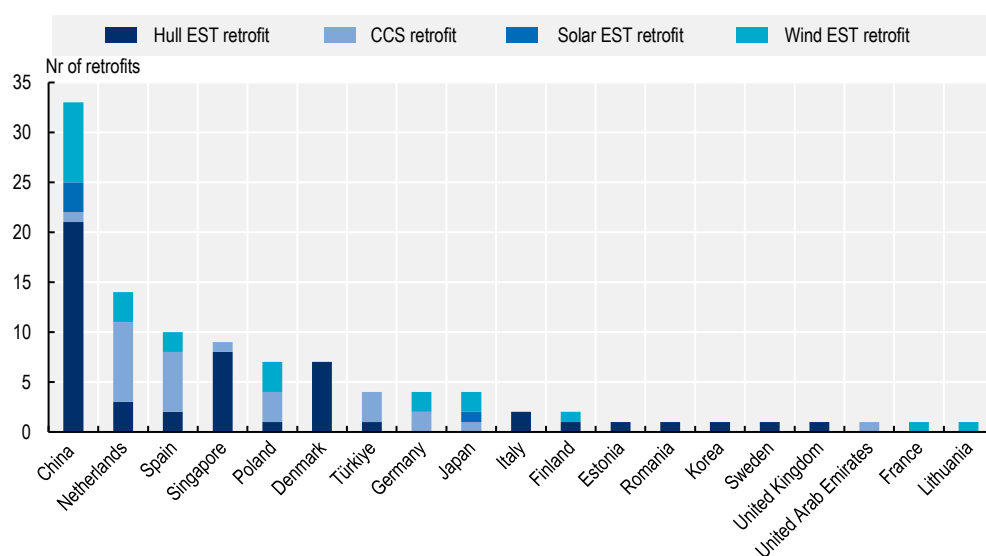
Figure 3.4. Trends in selected EST retrofits by technology (2009-2023)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

China has undertaken the most retrofits in hull modifications (21 out of 51 total), **wind energy** (8 retrofits out of 23), **and solar energy** (3 retrofits out of 4). Other countries actively involved in EST retrofits include European economies such as the Netherlands, Spain, Poland, Denmark, Türkiye, Germany, Italy and Finland. Outside Europe, Singapore, Japan, Korea and the United Arab Emirates have also undertaken EST retrofits. For CCS retrofits, the Netherlands has completed the most, followed by Spain (Figure 3.5).

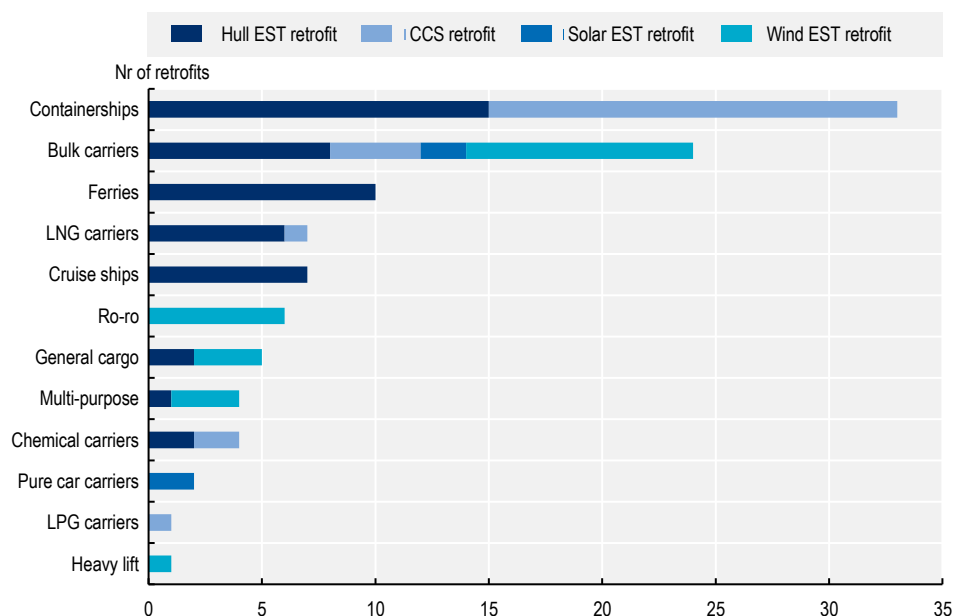
Figure 3.5. Total retrofits of selected EST by country and territory



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Containerships are the primary vessel type undergoing both hull EST retrofits and CCS retrofits. Hull EST retrofits are also notably prevalent in bulk carriers, ferries, LNG carriers and cruise ships, representing a significant portion of the total retrofits. In the case of CCS retrofits, bulk carriers, LNG carriers, chemical carriers, and LPG carriers are the vessel types where these retrofits are commonly found. Solar EST retrofits are implemented in bulk carriers and pure car carriers, whereas wind EST retrofits are most frequently applied to bulk carriers and Ro-Ro vessels, with presence also in general cargo, multi-purpose, and heavy lift vessels (Figure 3.6).

Figure 3.6. Total retrofits of selected technologies by vessel types (as of Sep 2024)



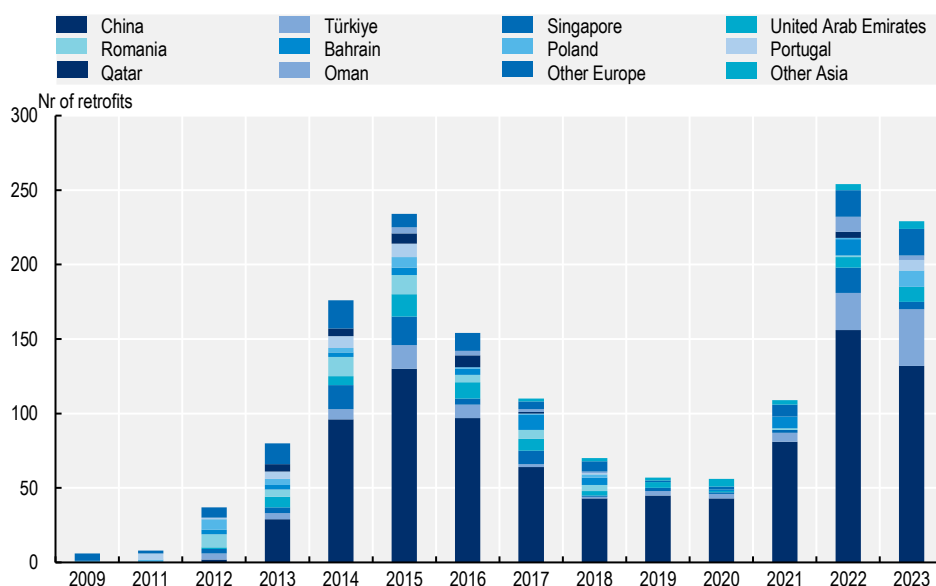
Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Propeller EST retrofits

Propeller EST retrofits are analysed separately due to their more frequent occurrence compared to other EST retrofits, with a total of 1,596 retrofits observed during the period analysed (2009 to 2023).

Notable peaks in propeller retrofits occurred in 2015, with 233 retrofits, and in 2022, with 254 retrofits. This may be attributed to ship owners reacting to elevated fuel costs in 2008, 2011-2014 and again in 2022, as well as the introduction of IMO regulation on ship energy efficiency in 2016 and 2023. Older ships were designed for higher engine loads, not maximum fuel efficiency at reduced speeds. Given the surge in fuel prices, the inefficiency of propulsion systems operating under ‘part load’ conditions could risk undermining competitiveness, motivating propulsion EST retrofits (Wärtsilä, 2021^[6]). China leads in the total number of these retrofits, with Türkiye, Singapore and Middle Eastern countries like the UAE, Bahrain, Qatar and Oman also showing significant activity. In Europe, Romania, Poland and Portugal are particularly active in undertaking propeller retrofits (Figure 3.7).

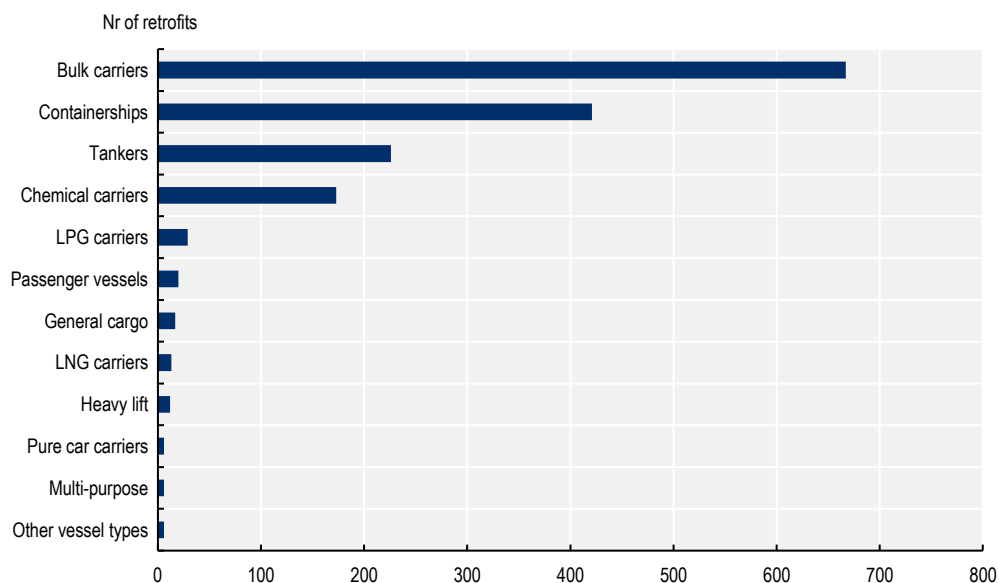
Figure 3.7. Propeller EST retrofits by country (2009-2023)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Propeller retrofits are most undertaken for large commercial vessels, like bulk carriers, container ships, tankers and chemical carriers.

Figure 3.8. Total propeller EST retrofits by vessel type (as of September 2024)



Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

Vessel age and length of retrofits

Table 3.2 indicates that the median age of vessels undergoing EST retrofits as well as alternative fuel conversions (which were analysed separately in Chapter 2).

Vessels are retrofitted for EST and alternative fuels at a similar age, hovering around the 10-year mark. This suggests that the ships being retrofitted are relatively young within their operational lifecycle. Such a trend may imply that it is commercially advantageous to retrofit these vessels rather than opting for demolition and acquiring new ships. It could reflect shipowners' strategic decision to invest in existing assets, enhancing their performance and extending their serviceability in response to evolving regulatory and market conditions.

In terms of the duration of these retrofits, fuel conversions are notably more time-consuming, with a median length of 67 days. This is significantly longer than the 22 days median duration for both EST and onboard CCS retrofits. The considerable additional time required for fuel conversions likely reflects the complexity and scale of the work involved in such retrofits and could substantiate potential bottlenecks in the future (Lloyd's Register, 2023^[7]).

Table 3.2. Median age of vessels and length of retrofit by activity type

	Fuel conversion	EST retrofit
Age of vessel (years)	9.5	10
Length of retrofit (days)	67	22

Note: The average retrofit duration varies by vessel type and fuel conversion complexity. For some fuel conversions, fewer than three data points are available, limiting the robustness of the estimates.

Source: OECD calculations based on data from World Fleet Register, Clarksons Research, <https://www.clarksons.net/wfr/fleet> (2024).

3.2. Digital technologies

Over the past decade, a set of digital technologies commonly referred to as Industry 4.0 has begun to reshape businesses across sectors. Despite the various definitions and terminology, the primary goal of these technologies is broadly acknowledged as the collection, exchange and processing of data to support autonomous decision-making systems. Notable examples of Industry 4.0 technologies include the Internet of Things (IoT), Big Data analytics, artificial intelligence (AI), robotics, cloud computing (CC), blockchain technology and additive manufacturing (Agarwala, Chhabra and Agarwala, 2021^[8]).

Digitalisation in the shipping industry encompasses the utilisation of data analytics for a diverse set of operations: from determining the best routes and sailing conditions and digital monitoring of ship health to the strategic scheduling of ship maintenance to help reduce machinery downtime and prevent ships from being out of service. Consequently, 'smart' vessels are expected to enhance operational efficiency and substantially reduce their GHG emissions (Bureau Veritas, 2022^[9]). Digitalisation also has great potential to optimise ship construction to better meet environmental regulation and customer demand. While the required technologies are often less mature and commercially viable in the maritime industry compared to other sectors, stakeholders see opportunities in adapting advanced digital technologies from other industries, potentially enhancing competitiveness and supporting decarbonisation efforts in the maritime sector (Sullivan et al., 2020^[10]).

Table 3.3 provides an overview of digital maritime technologies and their potential contributions to enhancing a vessel's operational efficiency and reducing GHG emissions across a ship's life cycle.

Table 3.3. Key digital maritime technologies and possible contribution to decarbonisation

Digital Technology	Description	Decarbonisation contribution
Big Data and AI	Use of sensors on the ship to transmit digital information to improve Machine Learning for a reliable Artificial Intelligence (AI)	Better analysis of various ship systems to optimise vessel and voyage operations and reduce fuel consumption
Digital twinning	Digital replica of the ship on shore permitting real-time monitoring of the ship and its machinery	Provision of predictive maintenance to machinery to ensure an efficient ship with reduced GHG emissions
Internet of Things (IoT)	The control of machinery remotely by using machine to machine communication	Ship operation safer, less maintenance and fuel consumption
Blockchain digital ledger	Efficient movement of cargo from one port to another on ships to ensure faster loading and unloading operations	Increased efficiency of the vessel and less time in harbour resulting in reduced GHG emissions
Additive manufacturing/ 3D printing	Manufacture of machinery parts on ship through digitised drawings of machinery	Reduction of spares on-board and easier repair of defective machinery for efficient operation

Source: (Lloyd's Register; Qinetiq; University of Strathclyde, 2015^[11]) (Ichimura et al., 2022^[12])

3.2.1. Digitalisation in shipping operations

Digital technologies have the potential to collectively enhance the operational efficiency of ships, optimise routing, and reduce fuel consumption. The volume of data associated with real-time tracking technologies has grown significantly in recent years, with nearly two-thirds of ships now employing digital equipment to enhance efficiency, improve commercial performance, navigation and support maritime security (Mirović, Miličević and Obradović, 2018^[13]). Notably, the integration of real-time tracking and automation tools has improved port logistics, with Just-in-Time (JIT) arrivals demonstrating fuel savings of up to 14.2% in the container sector (International Maritime Organization, 2022^[14]). Collaborations between ports, such as the Green and Digital Corridor initiative between Singapore and Rotterdam, further highlight the role of digitalisation in supporting decarbonisation efforts. As part of the Corridor, Singapore and Rotterdam have successfully trialled port-to-port data exchange, enabling vessel arrival and departure coordination, and are collaborating with industry partners on a proof of concept for monitoring, reporting, and verifying (MRV) GHG emissions along the route, aligned with global and regional standards (MPA Singapore, 2024^[15]). Digitalisation enhances transparency on vessel performance, providing data that can measure the impact of energy-saving measures and helping to design and operate the next generation of energy-efficient ships. Digital verification tools can further help create an infrastructure of trust in emissions reporting, supporting industry-wide collaboration and facilitating new contractual arrangements incentivising energy-efficiency measures (Wang et al., 2016^[16]).

Emerging technologies, such as digital twins allow the creation of virtual replicas of ships, enabling real-time monitoring of ship machinery and conditions. By predicting maintenance needs, digital twins reduce downtime and improve fuel efficiency, contributing to GHG emissions reduction. Studies indicate that leveraging AI and Big Data analytics for ship routing and operation can reduce fuel consumption by up to 10%, depending on factors like weather conditions and route optimisation (DNV, 2024^[3]).

3.2.2. Digitalisation in shipbuilding

Shipbuilding, a traditionally labour-intensive and slow-moving industry, has also begun to adopt digitalisation to meet growing environmental regulations and customer demands. The integration of digital technologies, such as computer-aided design (CAD), computer-aided manufacturing (CAM), additive manufacturing (3D printing), and advanced robotics, allows for 3D modelling and visualisation, enabling shipbuilders to collaborate more effectively on complex ship structures and improve precision. This in turn can help streamline the construction process, reduce material waste, and improve overall efficiency (Diaz et al., 2023^[17]). A case study of Korean shipyards found that automation and digital tools led to a 68% production automation rate, significantly reducing ship delivery times (LR, 2022^[18]). Advanced robotics, which perform tasks such as welding and assembly, can also help to alleviate labour shortages in the shipbuilding industry while reducing costs (Lloyd's List, 2024^[19]).

3.2.3. Remaining challenges to wider adoption

Despite the potential benefits of digitalisation in maritime operations and shipbuilding, there are significant challenges hindering its wider adoption.

One of the main barriers is the high initial cost of implementing digital technologies. For instance, as of 2024, the cost of industrial robotic systems used in ship construction ranges from USD 50,000 to USD 150,000, which can be prohibitive for adoption in small and medium-sized shipbuilders. In addition to initial investment costs, ongoing expenses for maintenance and integration also pose challenges, particularly for smaller enterprises (DNV, 2024^[3]).

Cybersecurity concerns also increase as digitalisation becomes more widespread. The use of data-sharing platforms and cloud-based systems exposes ships and shipyards to potential cyberattacks, underscoring the need for robust cybersecurity measures and better industry-wide coordination to protect critical infrastructure. Furthermore, there is a significant skills gap in the maritime workforce, as many workers lack the technical expertise to operate and maintain digital tools. Addressing this skills gap will require considerable investment in training programmes and increased collaboration among industry stakeholders (LR, 2022^[18]).

3.3. Low/zero-emission patenting in marine technologies

To meet global emissions reduction targets and mitigate climate change, the maritime industry must transition to low- and (near) zero-emission alternatives. Continued innovation plays an important role in driving down costs, overcoming technical challenges, and ensuring that new technologies can be scaled effectively to meet the global shipping industry's net-zero goals. The development and uptake of low/zero-emission technologies in maritime transport is expected to provide significant opportunities for shipbuilders and marine equipment suppliers to develop technologies that meet the demands of this rapidly evolving industry. The following section provides an overview of key trends in low/zero-emission patenting activity in maritime technologies, focusing on key innovation trends based on patent counts in climate change mitigation technologies. The analysis encompasses design advancements, energy-saving mechanisms and alternative fuel onboard technologies, with a specific focus on their impact on zero and low-carbon shipping.

3.3.1. Methodology

To analyse innovation trends, patents are used as a proxy for innovation activity. Patent data is extracted from the European Patent Office (EPO) Worldwide Patent Statistical Database (PATSTAT). The data is derived from patent applications under the Patent Cooperation Treaty (PCT) as of September 2024, with

a reference period spanning from 2008 up to 2022 (OECD, 2024^[20]). The PCT provides a legal infrastructure for a worldwide system of patent classification. Administered by the World Intellectual Property Organization (WPO), the PCT allows inventors to seek patent protection for an innovation in numerous countries simultaneously, rather than having to file multiple national or regional patent applications (WPO, 2022^[21]).

Low/zero-emission patents are identified via the Y02-tag, which EPO defines as “Technologies or applications for mitigation or adaptation to climate change”. Specific Y02 patent codes (Y02T70/00) pertaining to “Climate change mitigation technologies related to transportation: maritime or waterways transport” are used to extract relevant patents (European Patent Office, 2024^[22]). These patent codes can be further classified into subcategories that provide a granular view of innovations in maritime technology groups based on their areas of focus, thereby permitting a more detailed study of trends and patterns within the low/zero-emission maritime technology landscape (see Table 3.4).

Patent data is valuable for examining innovation trends as it is standardised, making it easy to compare across different economies and time periods. Nevertheless, it also faces some challenges in measurement stemming from the collection and structuring of data. Box 3.1. provides an overview of limitations to patent data that should be taken into consideration when analysing results and are pertinent to policy implications derived from the findings. A more detailed explanation of how to report, measure and analyse data on innovation can be found in the OECD/ Eurostat Oslo Manual (OECD/ Eurostat, 2018^[23]).

Table 3.4. List of patent codes

Patent code	Description
Y02T70/10	Measures concerning the design or construction of watercraft hulls
Y02T70/50	Measures to reduce greenhouse gas emissions related to the propulsion system
Y02T70/5218	Less carbon-intensive fuels, e.g., natural gas, biofuels
Y02T70/5236	Renewable or hybrid-electric solutions

Source: (European Patent Office, 2024^[22])

Box 3.1. Challenges to using patent data as a proxy for innovation

Inventors from several countries

The increasing trend of collaborative inventions involving multiple inventors from different countries raises the issue of potential overcounting. To tackle this problem and ensure accurate counting of transnational inventions, “fractional patent counts” are employed. This method assigns a fraction of the patent to each participating country instead of a whole count, thereby preventing double counting.

Home advantage bias

When patent metrics are derived solely from the data of a single national patent office, there is an inherent home advantage bias. This means that local applicants are more likely to file a larger number of patents within their own country (or region) relative to the size of their inventive activities, compared to international applicants. Using patent data registered under the PCT can help reduce this bias by providing a unified international patent filing system.

Scope of patent systems

As patent systems evolve and corporate approaches towards safeguarding innovations shift, a rise in patent filings does not automatically indicate an increase in new inventions, it could also signify a broader adoption of intellectual property rights (IPRs), particularly patents, by companies.

Patent quality

Another limitation to utilising patent data is that it cannot provide information on the “importance” of patents/ inventions. OECD research suggests including patent citations (for instance, the frequency with which a patent is cited in subsequent filings) to better evaluate patents by their “quality”.

Source: (OECD/ Eurostat, 2018^[23])

3.3.2. Discussion of findings

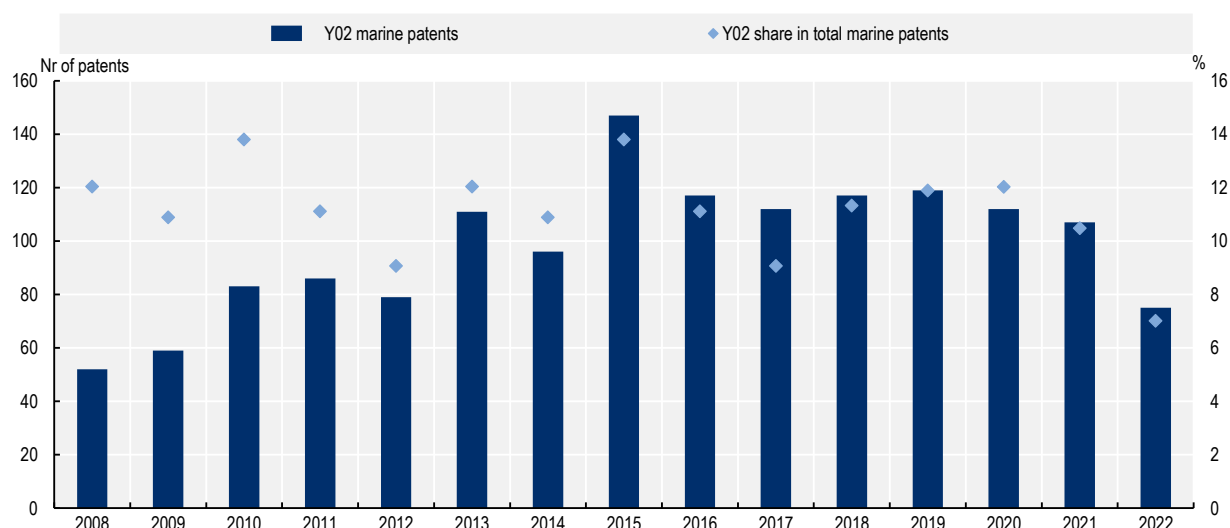
To gain insights into the main trends in innovation in low/zero-emission maritime technologies, it is useful to examine the shifts in patenting activity throughout the analysed period. Figure 3.9 delineates the trajectory of yearly low/zero-emission patents in maritime technologies from 2008 to 2022.

Both total patenting activity in low/zero-emission maritime technologies and its relative share has decreased. Examining the total number of Y02-tagged maritime technology patents, an increase is evident from 2008, peaking in 2015 with 114 patents globally. Patent counts start to taper off after 2015, culminating in a notable reduction by 2022. A similar trend is visible for the relative share of Y02-tagged patents in total maritime technology patents, which peaks in 2010 and 2015. This metric helps determine not only if low/zero-emission inventions have risen, but also if they have grown in relation to wider patenting activity in the maritime sector. After 2020, the share of low/zero-emission inventions shows a clear downward trajectory, reaching the lowest level over the studied period of around 7% by 2022. Despite the global increase in policies targeting low/zero-emission technology developments for the transition to net zero, Figure 3.9 shows that low/zero-emission maritime innovation activity in 2022 does not substantially surpass its level from over 15 years prior.

The peak in the early 2010s for low/zero-emission innovations and the subsequent drop in “climate change mitigation” inventions mirror broader technological trends (OECD, 2023^[24]). OECD analysis shows that after a surge from 2004 to 2011, low/zero-emission technologies’ share of global patents begins

to decrease around 2012, falling from 12.6% in 2011 to 9% in 2020. The period of relatively high share of low/zero-emission patents might be a result of surging fossil fuel prices in the late 2000s, indirectly putting an additional price on carbon, rather than an increased emphasis on environmental concerns (OECD, 2023^[25]).

Figure 3.9. Trends in number of low/zero-emission patents and percentage share in total maritime patenting (2008-2022)

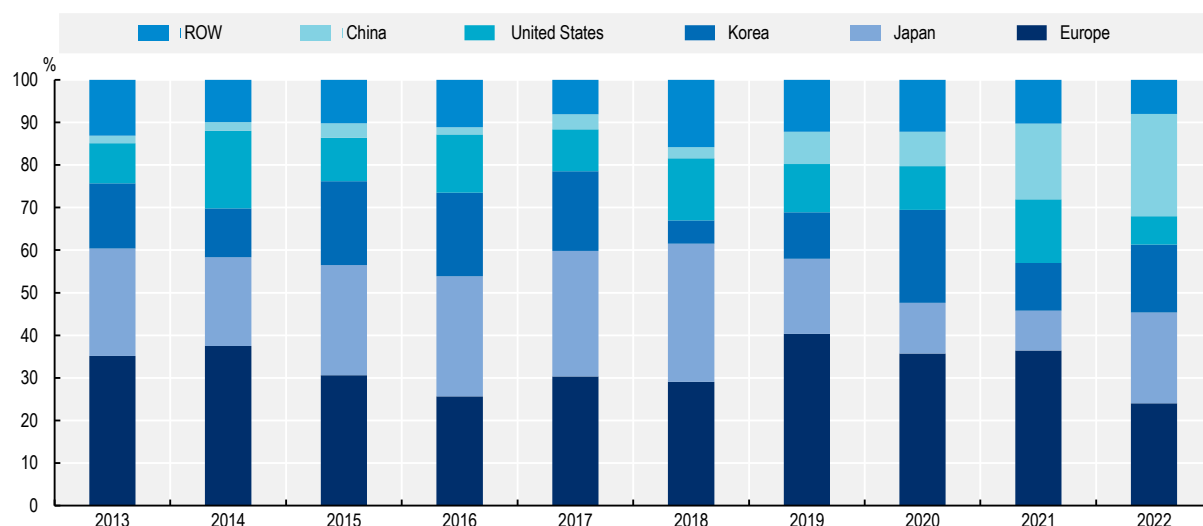


Source: OECD, STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, September 2024.

Patenting activity in climate change mitigation technologies related to maritime or waterways transport shows significant differences across shipbuilding economies. Figure 3.10 compares total low/zero-emission patent applications for a selection of economies taken from the total sample, which have exhibited consistently high levels of innovation over the studied period. Amongst high-level innovators, Europe (including EU27, Norway and UK) and Japan consistently maintain prominence as leading innovators for low/zero-emission maritime technologies. Japan's innovation activity shows a notable peak in 2015, registering 35 low/zero-emission patents out of a world total of 114 patents. Europe reaches its peak in 2019 with 48 low/zero-emission patents, including 40 patents by EU27.

The figure also includes data on China, the world's current largest shipbuilding economy. China has experienced a steadfast and considerable increase in its innovation activity trajectory. By 2021, its innovation activity in low/zero-emission maritime technology patents surpassed that of Japan, Korea and the United States and equalled the activity of all European shipbuilding countries, with 12 patents total. This ascent underscores China's increased focus and strategic direction towards "green" maritime solutions. It is primarily linked to a rapid increase in patents on measures concerning design or construction of watercraft hulls (patent code: Y02T70/10).

Figure 3.10. Share in low/zero-emission patenting in maritime technologies of selected economies (2013-2022)



Note: Europe is EU27 as well as Norway and United Kingdom (reflecting SEA Europe members).

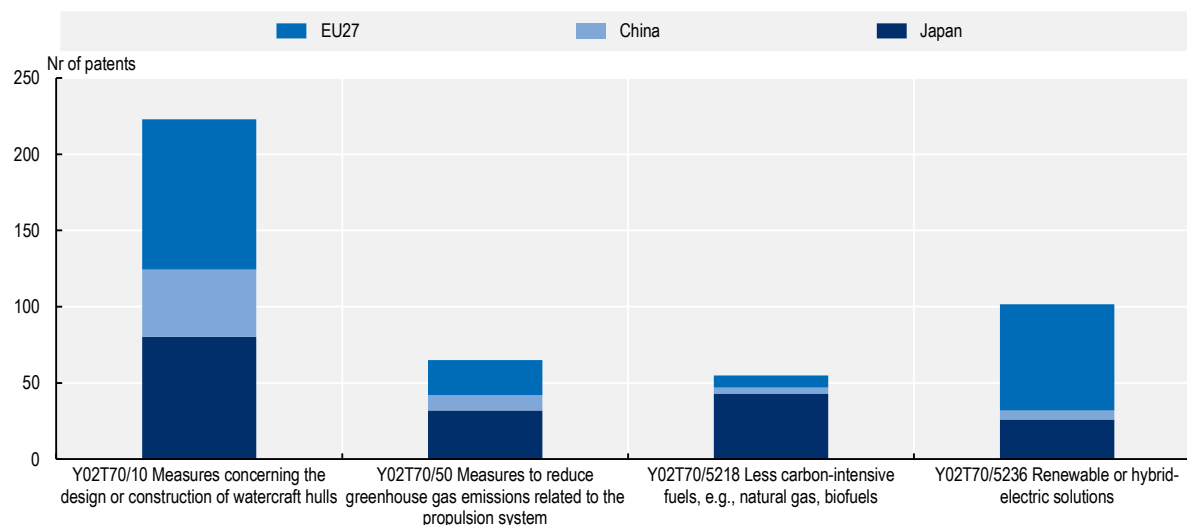
Source: OECD, STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, September 2024.

A significant share of clean technology development is localised within a select group of economies, revealing a marked specialisation across these countries (and regions) in their areas of focus and technical expertise. Looking to the EU27, Japan and China, the leading innovators in low/zero-emission maritime technologies over the studied period, distinct patterns in their respective capacity for climate change mitigation technology development and prioritised technology clusters become apparent (Figure 3.11).

In Japan, low/zero-emission innovation activity is notably concentrated in the design or construction of watercraft hulls (patent code: Y02T70/10), with a total of 80 patents over the studied period, making up almost 20% of world-wide low-carbon patents in this technology field. Less carbon-intensive fuels (Y02T70/5218) and measures to reduce greenhouse gas emissions related to propulsion (Y02T70/50) also show high levels of patenting, with 42 and 32 patents respectively. Similar to Japan, China shows the highest innovation activity in measures concerning the design or construction of watercraft hulls, with a total of 44 patents.

EU member states—unlike most shipbuilding economies—show high levels of patenting activity in renewable or hybrid-electric solutions (Y02T70/5236). This category has consistently shown high innovation levels and can be linked to the use of hybrid solutions for vessel types predominantly produced in the EU27 (e.g., ferries). Design or construction of watercraft hulls (Y02T70/10) is another domain where the EU27 has demonstrated significant activity, with a total of 99 patents. In terms of measures for reducing greenhouse gas emissions related to the propulsion system (Y02T70/50) and less carbon-intensive fuels (Y02T70/5218), patenting activity is relatively low over the studied period.

Figure 3.11. Main patent codes for low/zero-emission maritime patenting in selected economies (2013-2022)



Source: OECD, STI Micro-data Lab: Intellectual Property Database, <http://oe.cd/ipstats>, September 2024.

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4 Shipbuilders' decarbonisation strategy

This chapter provides an assessment of the capabilities of selected shipbuilders to develop and construct low/zero-emission vessels and support the increased adoption of new (including digital) technologies. The methodology employed aims to evaluate the extent of alignment between the existing manufacturing practices and low/zero-emission technology development and digitalisation strategies of major shipbuilding companies with the decarbonisation targets established by the IMO, which aims for net-zero shipping “by and around, i.e. close to, 2050”.

Key messages

This chapter provides an analysis of the decarbonisation targets and strategies of major shipbuilding companies in their respective regions, yielding key insights:

- **Not all major shipbuilding companies have set net-zero emissions targets:** while most large shipbuilders in the company mapping report net-zero targets for their operations, including Scope 3 emissions, not all explicitly align with net-zero ambitions (7 companies do not have a net-zero by 2050 target). Companies in Korea and Europe generally adopt stricter targets and strategies, reflecting regional variations in net-zero commitment.
- **Company strategies emphasise the construction of low/zero-emission vessels:** most major shipbuilding companies are actively supporting the adoption of low or zero-emission technologies and the integration of alternative-fuel capable vessels into the global fleet. They frequently participate in demonstration projects showcasing new technologies, underscoring their role in advancing maritime sustainability.
- **Digital and ‘Just Transition’ aspects are included in company strategies:** several companies emphasise the importance of digitalisation (for both marine technologies during ship operations and ship construction) as well as a ‘Just Transition’, including the development of green and smart skills.
- **ESG reporting has increased but access to green finance instruments remains limited:** there is a widespread commitment to adhering to ESG reporting standards (10 companies adhere to standardised ESG reporting method) and some companies utilise green finance instruments, such as sustainability-backed bonds or loans. However, this practice remains limited to a few frontrunners (3 companies reported).
- **Company size matters for decarbonisation ambition:** larger companies tend to display more ambitious targets and strategies for decarbonisation and digitalisation. Those with ambitious targets often achieve notable results in decarbonisation, digital transition, and green finance indicators. This suggests that ‘First Movers’ might be setting a benchmark for the industry, highlighting the importance of ensuring that the rest of the sector follows suit in the transition to net-zero.

4.1. Methodology to assess company strategies

Shipbuilders are adopting company strategies across several targets— from reducing GHG emissions in the manufacturing process to supporting the integration of alternative fuels and energy efficiency in vessels and advancing digitalisation. Studying these targets and strategies provides insight into the industry’s priorities and the extent to which they align with national and international decarbonisation, digitalisation and resilience decarbonisation goals. Furthermore, the analysis seeks to provide an overview of the shipbuilding industry’s landscape, focusing on key aspects such as net-zero and automation objectives, strategic roadmaps, technological developments, data availability, green financing strategies and research and development (R&D) investments in digital and low/zero-emission technologies. It is based on an examination of company reports and initiatives pertinent to decarbonisation and digitalisation efforts within the sector.

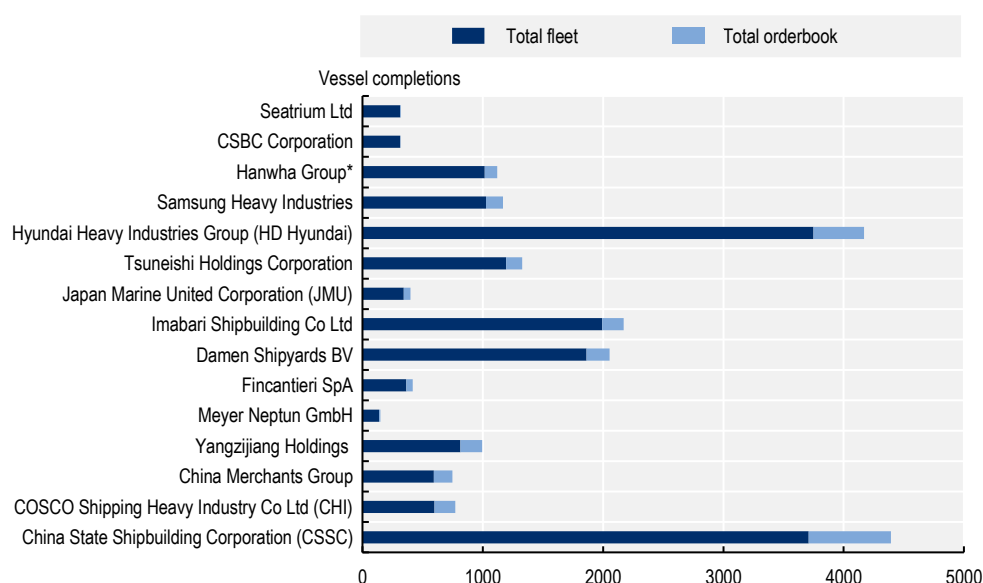
The analysis focuses on prominent shipbuilders across various regions, specifically targeting the major shipbuilding companies of China, Korea, Japan and the European Union. Shipbuilding companies were selected according to the following criteria: fleet size, orderbook, revenue and employee count. Relevant data were collected from Clarksons Shipping Intelligence Network (SIN), Clarksons WFR and ORBIS. The top three shipbuilder groups from each region were selected based on their contribution to global ship production and the availability of sustainability reports at the group level. Several large shipbuilders per region had to be excluded from the analysis due to a lack of available data on their decarbonisation/digitalisation efforts.

Table 4.1 provides a list of the selected shipbuilders. For China, four shipbuilders have been selected to reflect the importance of Chinese shipbuilding in the current orderbook. For Korea, Japan and Europe three shipbuilding companies are selected each. For the Rest of World (ROW), two companies are selected. Figure 4.1 shows their total ship completions and CGT in the global fleet and current orderbook, as an estimate of capacity.

Table 4.1. List of selected companies by region

China	Korea	Japan	European Union	ROW
China State Shipbuilding Corp (CSSC)	Hyundai Heavy Industries Group (HD Hyundai)	Imabari Shipbuilding Co Ltd	Meyer Neptun GmbH	CSBC Corporation
Yangzijiang Shipbuilding Holdings	Samsung Heavy Industries	Japan Maritime United (JMU) Corporation	Fincantieri SpA (*including VARD)	Seatrium Ltd
China Merchants Group	Hanwha Group	Tsuneishi Holdings Corporation	Damen Shipyards BV	
COSCO Shipping Heavy Industry Co Ltd (CHI)				

Figure 4.1. Total fleet and orderbook of selected companies



Note: *Data on vessel completions for the fleet by Hanwha Group is incomplete as the company was previously Daewoo Shipbuilding & Marine Engineering Co., Ltd. (DSME), and historical data from that period has not been fully integrated.

Source: Authors' elaborations based on Shipping Intelligence Network, Clarkson Research, <https://sin.clarksons.net/>, as of January 2024.

4.1.1. Criteria for assessment

The research methodology employs a set of indicators informed by disclosure framework assessment methodologies from entities like the Science Based Targets initiative (SBTi) sector (Science Based Targets, 2021^[1]), as well as previous analysis carried by the OECD on firms' decarbonisation efforts in the steel sector (OECD, 2024^[2]). These indicators are designed to determine each shipbuilding company's alignment with a net-zero 2050 target, covering aspects such as emission reduction targets, green vessel technology targets, decarbonisation strategies, digitalisation strategies, participation in voluntary initiatives, data disclosure practices, investment in R&D for decarbonisation, green financing, and climate-related risk assessments.

Table 4.2 gives an overview of the key indicators assessed as well as the assessment criteria. Information is gathered from annual reports, company websites and newsroom articles, focusing on the shipbuilders' efforts towards decarbonisation during the vessel operation and construction phase. The analysis does not cover strategies that extend beyond the shipbuilder company, such as the development of green ports or alternative fuel supply infrastructure.

Table 4.2. Overview of indicators and assessment criteria

Indicator	Assessment criteria
Emission reduction target	- Does the company have a GHG emissions reduction target? - What type of target (net-zero by 2050)? - What are the medium-term and long-term targets up to 2050? - Scope of emissions covered (scope 1,2,3) - Alignment with a 1.5°C trajectory based on scenario analysis
Green vessel/ technology target	- Does the company have targets for development of low- or zero-emission vessel technology? - Data on (near) zero-emission vessel completions - Specific targets for (near) zero-emission technology development
Decarbonisation strategy	- Presence of a decarbonisation strategy or plan to achieve targets - Type of technologies and alternative fuels promoted - Inclusion of carbon credits or offsets - Investments in R&D for digital, low-, and zero-emission technologies - Coverage of workers, skills or 'Just Transition' aspects
Digitalisation strategy	- Presence of a digitalisation strategy or plan - Type of digital technologies promoted - Coverage of worker and skill-related aspects
ESG reporting and green finance	- Company ESG reporting - Compliance with standardised ESG methodology - Compliance with standardised Co2 methodology - Compliance with the Task Force on Climate-related Financial Disclosures (TCFD) - Compliance with Science Based Targets indicators (SBTi) - Issuance of green finance instruments
Voluntary initiatives	- Participation in voluntary decarbonisation/digitalisation initiatives - Scope of the initiative and stakeholders involved

4.2. Findings on company strategies

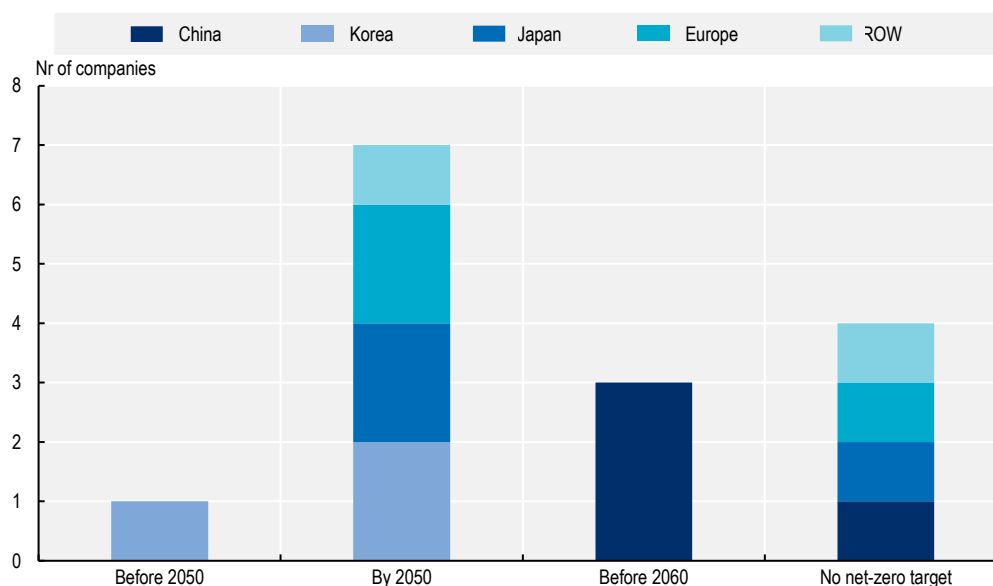
The following section gives an overview of the key findings from the selected shipbuilding companies. It includes findings on 1) targets and GHG emissions, 2) company strategies (including decarbonisation, digitalisation and 'Just Transition' strategies), and 3) ESG reporting and voluntary initiatives.

4.2.1. Targets and GHG emissions

Most companies included in the mapping have committed to achieving net-zero emissions by 2050. Notably, Hanwha stands out for setting a target of reaching net-zero by 2045. Conversely, Chinese firms

deviate from this trend, aligning instead with their national government's objective of attaining carbon neutrality before 2060. Four companies have not specified a clear net-zero target date. Korean shipbuilding companies, overall, demonstrate the most stringent approach to their decarbonisation commitments, uniformly aligning with the net-zero by 2050 goal (Figure 4.2).

Figure 4.2. Selected companies' net-zero emission targets



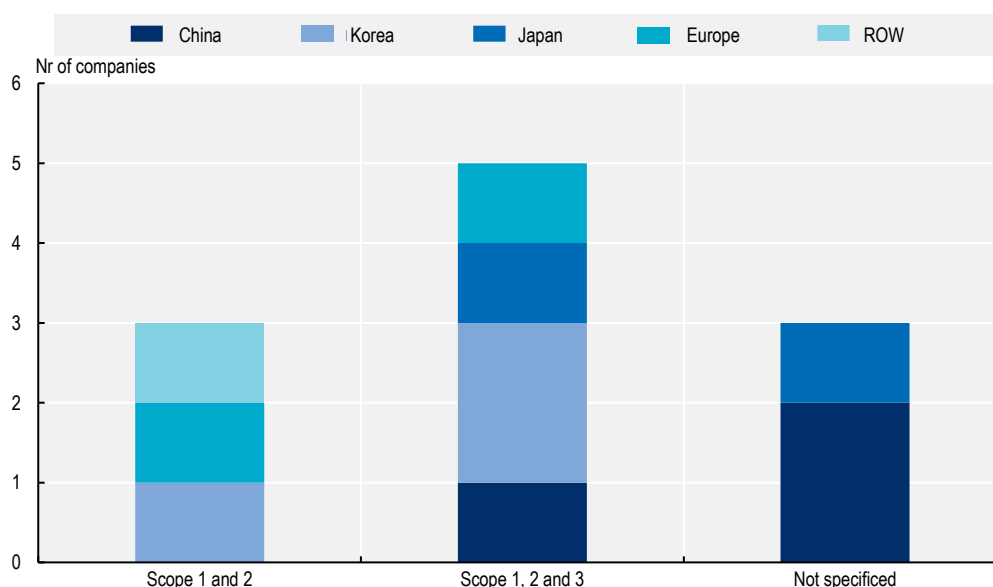
Note: "No net-zero target" is defined as no explicit target to reach net-zero emissions by a specified date, the company may include other targets, e.g., on energy usage or relative emissions reductions.

Among the companies with established net-zero targets, an analysis of their emissions scope reveals varying degrees of inclusion. As a reminder:

- Scope 1 emissions are direct emissions from company-owned and controlled resources.
- Scope 2 emissions cover indirect emissions from the generation of purchased electricity, steam, heating and cooling consumed by the reporting company.
- Scope 3 emissions include all other indirect emissions that occur in a company's value chain, including both upstream and downstream emissions (Science Based Targets, 2023^[3]).

Most shipbuilding companies have committed to addressing Scope 1, 2, and 3 emissions, showing a comprehensive approach to decarbonisation that encompasses direct and indirect emissions throughout their value chain. Some companies have a narrower focus, primarily on Scope 1 and 2, which pertains to the direct emissions from their operations and the indirect emissions from the energy they procure. Three companies have not explicitly stated which scope their net-zero targets encompass, leaving ambiguity regarding the breadth of their emissions reduction commitments (Figure 4.3).

Figure 4.3. Scope of emissions under net-zero targets



4.2.2. Strategies by target area

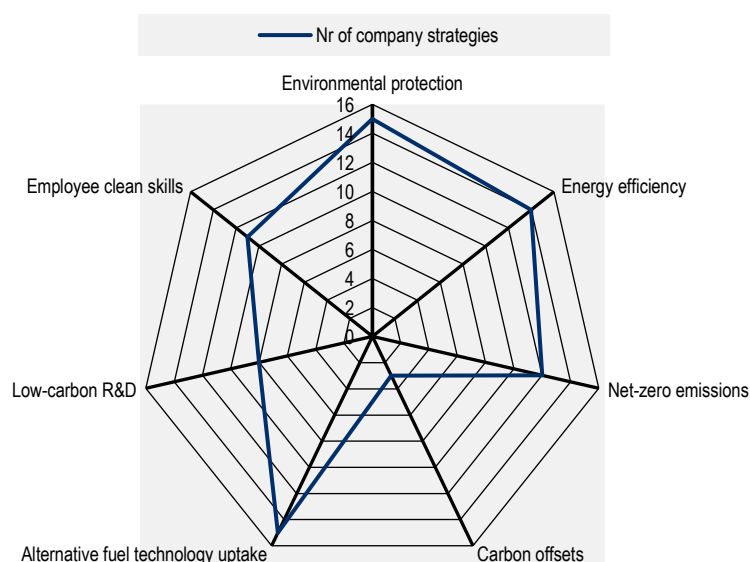
In analysing major shipbuilders' decarbonisation strategies, the focus is on the existence and specifics of decarbonisation plans, the promotion of clean technologies and alternative fuels, and the role of carbon credits or offsets. Additionally, the strategies are studied for how they address digitalisation and incorporate considerations for workforce transition and skills development in support of industry transformation.

Decarbonisation strategies focus on shipbuilders' impact on the environment, energy efficiency and new technology uptake. As seen in Figure 4.4, company strategies show a strong focus on environmental protection (covered by 15 companies). Energy efficiency and savings are also prominently featured, along with the shift towards renewable energy sources and the development of green technologies (15 companies). In contrast, only three shipbuilding companies mention the use of carbon offsets as a strategy for balancing emissions. Exceptions include Samsung Heavy Industries, which is implementing an internal carbon pricing scheme. Moreover, while not as frequently highlighted, green R&D is explicitly addressed by 8 shipbuilding companies, with reports including annual R&D expenditure and descriptions of specific research programmes.

Leading shipbuilders' strategies point to increasing R&D for green and smart solutions. Key technologies highlighted in company reporting to enhance energy efficiency and reduce emissions are improved hull designs, propellers and ducts, as well as technologies for alternative fuels like LNG, ammonia, methanol and hydrogen. The integration of digital technologies is also evident, for example developments for Maritime Autonomous Surface Ships (MASS) or monitoring vessel performance in actual sea conditions.

To support these innovations, dedicated research institutes within the industry, such as Hyundai Maritime Research Institute, as well as national innovation platforms are being utilised. Collaborations with research institutes and universities are further employed to drive the development of energy-efficient and emission-reducing solutions. Chinese shipbuilder CSSC stands out with 63 national innovation platforms, collaborating closely with public institutions. Most shipbuilding companies in the studied sample commit to a series of low/zero-carbon projects over the next decade.

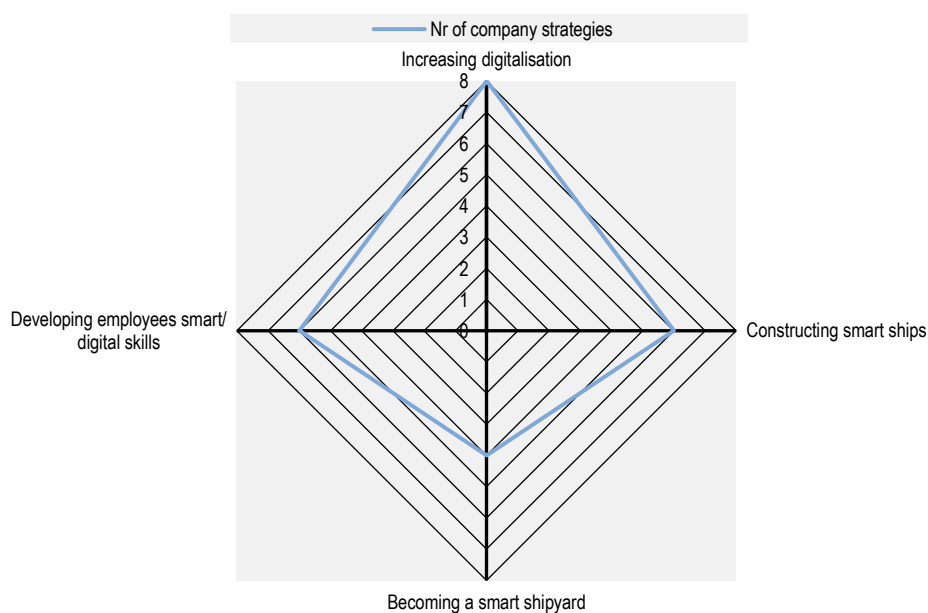
Figure 4.4. Companies' decarbonisation strategies by target areas



Note: "Low-carbon" refers to low/zero-emission technologies, these can be technologies related to alternative low/zero-emission fuels or ESTs.

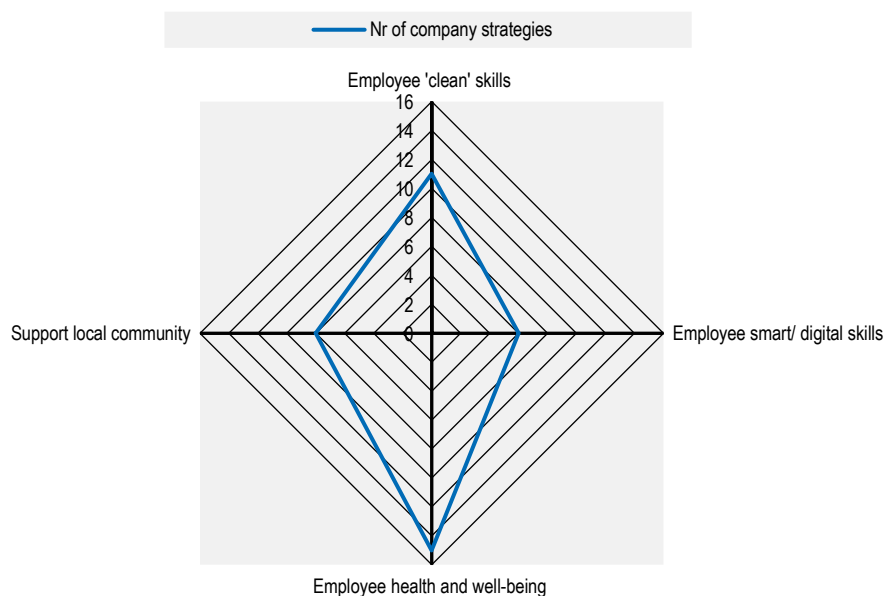
Support for digitalisation is highlighted across major shipbuilders. Examining digitalisation efforts more closely, most companies make a general reference to "increasing digitalisation" or developing "digital solutions". A significant focus is put on developing smart solutions and constructing smart ships, indicating a move towards integrating advanced technologies in shipbuilding. Four companies explicitly highlight their strategy to transform their facilities into smart shipyards, which involves adopting digital tools and systems to enhance efficiency and productivity. Six companies have further adopted initiatives to develop employees' skills for an increasingly digital work environment (Figure 4.5).

Figure 4.5. Companies' digitalisation strategies by target areas



Shipbuilders emphasise employee health, well-being, and workforce development in their reporting. Supporting a just transition— that is, shifting the industry towards new practices equitably, while safeguarding employment, fostering skill development, and providing social protection for workers and communities— is a target most companies highlight in their reporting and information sharing. Emphasis on employee health and well-being is a common thread. Ten companies specifically underscore the development of “clean” skills associated with low/zero-emission solutions, though there is room for more concrete strategies from others. Additionally, a handful of companies spotlight their support for the broader local community (Figure 4.6). These efforts align with the principles outlined in the 2023 IMO GHG Strategy calls for a “just and equitable transition”, including upskilling and supporting seafarers and providing assistance to Least Developed Countries (LDCs) and Small Island Developing States (SIDS) to ensure a fair shift toward net-zero shipping globally (International Maritime Organization, 2023^[4]).

Figure 4.6. Companies’ Just Transition strategies by target areas



4.2.3. ESG reporting and voluntary initiatives

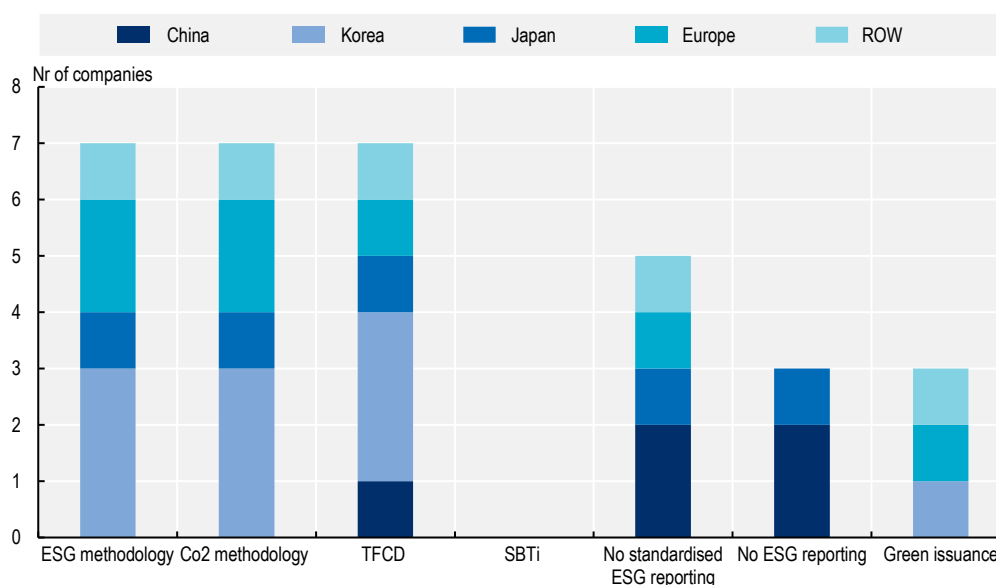
Access to finance is crucial for investments in new vessel construction and industry transformation, necessitating increased compliance with ESG (Environmental, Social, Governance) indicators. Half the companies in the mapping adhere to standardised, international ESG reporting methodologies, such as the Global Reporting Initiative (GRI). Additionally, seven companies utilise standardised Co2 management methodologies, including certifications like the Co2 Reduction Management scheme aligned with the ISO 14001 “Environmental Management” standard and the ISO 14051 “Material Flow Cost Accounting” standard (see Figure 4.7).

Several companies follow the Task Force on Climate-related Financial Disclosures (TCFD) guidelines, reflecting a broad commitment to transparency in climate-related financial risk reporting. However, none of the companies surveyed have adopted the SBTi, a framework encouraging organisations to set emissions reduction targets in line with climate science. Five companies lack adherence to a standardised

ESG reporting method, and three do not provide information on ESG indicators, highlighting areas for improvement in ESG transparency and accountability.

Most leading shipbuilders have not issued green bonds or loans (or do not report it), with the exception of Fincantieri SpA, Hyundai Heavy Industries and Seatrrium Ltd. Fincantieri SpA secured three major sustainability-linked loans: a EUR 300 million construction loan in 2021 with Intesa Sanpaolo and CDP for a cruise ship, a EUR 500 million revolving credit in 2022 with Intesa Sanpaolo, and a EUR 700 million guarantees facility in 2023 with BNP Paribas (Fincantieri, 2023^[5]). Hyundai Heavy Industries reported a USD 300 million Senior Unsecured Green Bond in 2023, backed by the Korean Development Bank (KDB). Seatrrium Ltd. issued two sustainability backed USD 500 million loans in 2021 and 2023, reported on the Singapore Exchange (SGX) (Clarksons Shipping Intelligence Network, 2024^[6]).

Figure 4.7. ESG reporting by companies

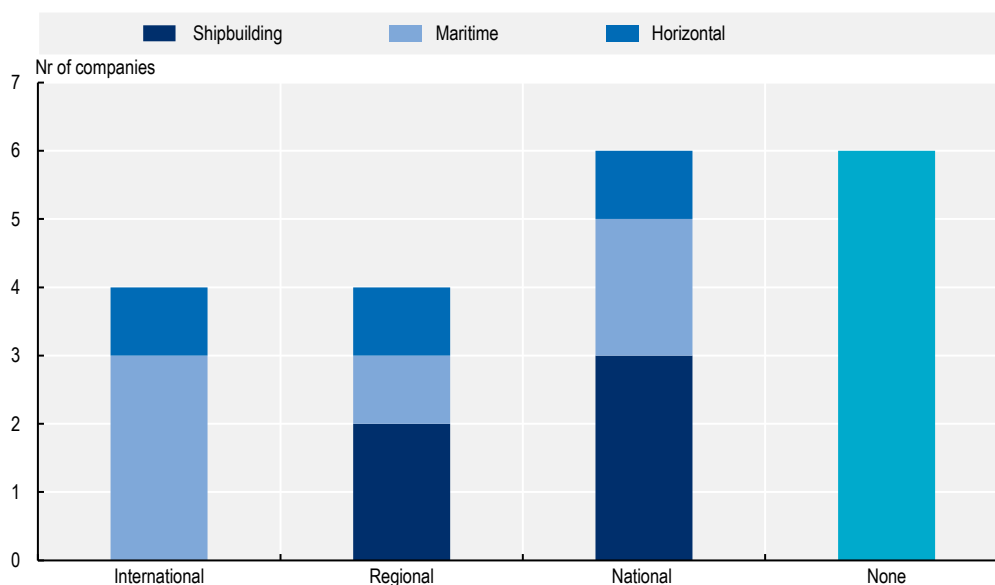


Actors across the maritime industry are pursuing decarbonisation targets through voluntary initiatives and public-private partnerships. They include a variety of decarbonisation initiatives linked to fostering demand and supply of low-carbon fuel and technology alternatives, such as voluntary commitments to stricter operational standards for ship owners and operators or innovation networks to improve the technology and commercial readiness of low-carbon technologies. Further, they can finance initiatives that favour 'climate-aligned' companies, such as the Poseidon Principles for financial institutions (Poseidon Principles, n.d.^[7]), and that can help incentivise shipping companies to invest in low-carbon technologies and practices to reduce their emissions.

Most major shipbuilding companies engage in voluntary collaborations with other businesses, governments and academic institutions at various levels—international, regional and national (Figure 4.8). Internationally, examples include the Getting to Zero Coalition and the UN Global Compact, while regionally, the focus is primarily on European Union initiatives, such as Zero-Emission Waterborne Transport (ZEWI). Nationally, collaborations often occur among shipbuilders from the same country or with governmental bodies. It is noteworthy that shipbuilders' participation in international efforts tends to be directed towards broader maritime sector concerns or cross-industry issues, whereas regional and national involvements often zoom in on shipbuilding-specific or technology-driven agendas. Fincantieri

SpA shows a particularly diverse mix of initiatives, predominantly within EU programmes like the ZEWT and various national projects. Six shipbuilders do not engage in any reported voluntary initiatives.

Figure 4.8. Breakdown of voluntary initiatives by sector and jurisdiction



Note: It is possible for one company to be included several times in a jurisdiction, e.g. if company has national initiatives relating both to shipbuilding and horizontal.

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5

Alternative fuel supply and infrastructure

Alternative fuel capable ships are only as “clean” as the fuel they run on. Establishing robust supply chains for low/zero-emission fuels— from production to storage and bunkering at ports— is essential for decarbonising shipping. However, building the necessary infrastructure, ensuring safe fuel handling, and securing the required investment remain critical bottlenecks, and addressing these challenges will require substantial efforts from the shipping industry, ports, energy suppliers, governments, and financial stakeholders. This chapter examines developments in the production and bunkering of selected alternative fuels, with a particular focus on methanol and ammonia. It also maps offtake and supply agreements already established between shipowners, ports, and energy suppliers to help create demand and ensure a reliable fuel supply for the sector.

Key messages

This chapter analyses current trends and forecasts in alternative marine fuel production and infrastructure, with the following main conclusions:

- **LNG and biofuel are currently the most widely used alternative fuels, but forecasts point to future ammonia uptake:** Shipping's LNG and biofuel consumption has been on the rise in recent years, with LNG consumption reaching approximately 11 million tonnes in 2022, accounting for about 5% of total marine fuel usage. By 2030, other alternative fuels, particularly ammonia and methanol, are projected to grow, representing 6% and 1% of total energy consumption by the shipping industry, respectively.
- **There is a general increase in alternative fuel port infrastructure despite strong differences by region and fuel type:** There is a noticeable global rise in alternative fuel bunkering and storage, especially for LNG, demonstrating progress in marine industry's shift towards alternative fuels. For methanol and ammonia port bunkering remains limited and shows uneven readiness across the globe.
- **A mapping of offtake demand from the maritime sector in OECD countries shows that methanol is the most common focus of current supply agreements:** out of 104 offtake/supply agreements studied between ship owners, ports and/or energy suppliers, 44% focus on methanol. Europe is the leading region for offtake agreements, with Denmark, Norway, and France being home to most buyers, while Finland, Norway, and the Netherlands are key locations for suppliers.

5.1. Challenges to alternative fuel supply and production

The availability of low/zero-emission fuels is a primary concern for the shipping industry as it targets maritime decarbonisation. The IMO's current GHG Strategy represents the internationally guiding regulation for a net-zero maritime industry "by and around" 2050, as well as its medium-term target to "increase uptake of zero or near-zero GHG emission technologies, fuels and/or energy sources to represent at least 5%, striving for 10%, of the energy used by international shipping by 2030" (International Maritime Organization, 2023^[1]). The incorporation of low/zero-emission solutions on board of vessels is inextricably linked to the concurrent development of onshore facilities capable of supplying and handling emergent fuel types (DNV, 2023^[2]). Industry experts have highlighted that establishing a robust supply for alternative fuels is likely to represent a significant challenge, potentially acting as a critical bottleneck for maritime decarbonisation that will necessitate substantial investment to overcome (Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping, 2022^[3]). There are several challenges to alternative fuel production and bunkering that limit maritime decarbonisation efforts.

A major obstacle is the lack of infrastructure for carbon-neutral fuels like e-methanol and e-ammonia, which currently do not have the necessary systems in place for widespread production, distribution, and supply. Additionally, there is uncertainty over the long-term fuel solution, making it difficult for stakeholders to commit to specific technologies. Regional disparities in alternative fuel supply chains further complicate adoption, as bunkering options beyond LNG remain limited, creating fuel supply risks.

The cost gap also poses a significant barrier, with alternative fuels costing up to 300% more than conventional fuels. Current projections suggest that, without strong regulatory support, there is no viable business case for these fuels, even in the long term (Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping, 2024^[4]). This is also reflected in the significant investment needs around alternative fuel supply

infrastructure. A 2020 study conducted by the University Maritime Advisory Services and the Energy Transitions Commission estimated that achieving shipping decarbonisation through an ammonia-dominant approach by 2050 would require an investment of USD 1.2 trillion to USD 1.6 trillion. This translates to an annual expenditure of USD 60 billion to USD 80 billion between 2030 and 2050. Of the total investment, 87% would need to be allocated to fuel production, storage, and distribution (Global Maritime Forum, 2020^[5]). This raises further questions around the distribution of decarbonisation costs across the maritime value chain, which remains uneven.

Regulation is essential to bridging the cost gap and ensuring the economic viability of alternative fuels. This includes both regulatory measures at the IMO as well as national and regional legislation to develop alternative fuel supply chains and support technology advancements. Forward-looking green industrial strategies can position countries as leaders in alternative fuel production, while strong regulations— such as incentivising long-term offtake agreements— are key to drive investment in renewable fuels. Beyond decarbonisation, renewable fuel production can also stimulate economic growth, create jobs, enhance domestic value chains, and generate positive spillover effects across industries (Global Maritime Forum, 2024^[6]).

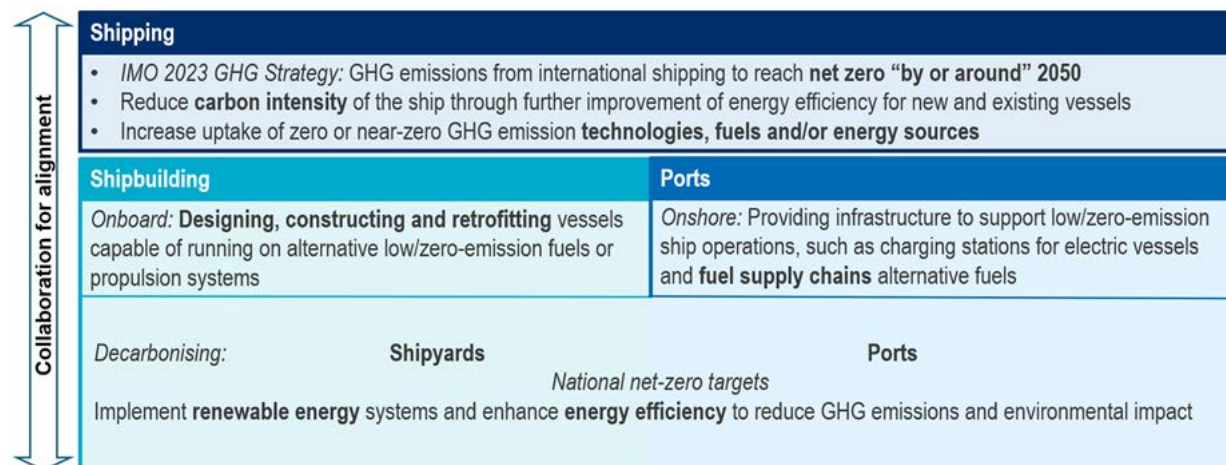
Regulatory measures and government support to establish alternative fuel supply chains are complemented by voluntary efforts from stakeholders throughout the supply chain, including cargo owners, financiers, and port authorities, who are seeking tangible commitments to emissions reduction (COZEV, 2023^[7]). In 2024, 50 maritime companies issued a “Call to Action” for faster adoption of zero-emission fuels, investment in low/zero-emission vessels, and global hydrogen infrastructure. Signatories pledged to use zero-emission services by 2040, while hydrogen producers committed to supplying a significant share of the 11mt green hydrogen needed for IMO’s 2030 emission targets, with 50% sourced from developing economies (RMI, 2024^[8]).

5.2. Interplay between onboard technology and onshore fuel supply

While alternative fuel supply is not within the direct purview of the shipbuilding sector, the establishment of these supply chains, alongside the infrastructure for alternative fuels, constitutes an area of strategic importance for maritime decarbonisation.

The interconnected and convergent pathways among shipbuilding processes, maritime operations and the provisioning of marine fuel supplies show the critical intersections that underpin the transition to a decarbonised maritime industry (UNCTAD, 2022^[9]). To achieve international shipping’s decarbonisation goals, as expressed in the IMO 2023 GHG Strategy, sector-wide maritime decarbonisation requires relates the application and management of new technologies and practices in vessel operations. This must be underpinned by low/zero-emission ship construction and circularity, reflecting the importance of efficient vessel design and the reusability of materials, including ship recycling marine fuel supply. At the same time, decarbonising ship operations requires alternative fuel production and bunkering at ports, which encompasses infrastructure and port measures (including storage and bunkering) to establish the supply chains of alternative fuel production and distribution.

Infographic 5.1. Interplay between shipping, shipbuilding and ports



The separate maritime value chains are similarly impacted by wider industry-efforts as well as regulation in support of maritime decarbonisation. As such, there is a symbiotic relationship between the utilisation of various fuel types, the assortment of available energy sources for maritime propulsion and the essential role of regulatory and financial frameworks in facilitating the maritime industry’s shift to alternative fuels and technologies (Ricardo & DNV, 2023^[10]).

5.3. Developments in alternative fuel production and bunkering

Historically, over 99% of the energy consumed in international shipping was made up of oil products (International Energy Agency, 2023^[11]). As of 2024, conventional fossil fuels continue to dominate the marine fuel market, with shipping consuming approximately 280 million tonnes of fuel annually. In 2023, the reported fuel oil consumption for ships with a gross tonnage of 5,000 or more trading internationally was 213 million tonnes, with nearly all of it being fossil fuels, including heavy fuel oil, light fuel oil, and diesel/gas oil (DNV, 2024^[12]).

5.3.1. Alternative fuel production

LNG and biofuels are the most widely used alternative fuels in shipping, with vessels capable of running on these fuels having experienced a significant uptake in the global fleet. LNG consumption was around 11 million tonnes in 2022, making up about 5% of total marine fuel usage. Biofuels, particularly biodiesel blends, represented over 7% of total bunker sales at the Port of Rotterdam and approximately 1% at the Port of Singapore, totalling an estimated 0.4 million tonnes of pure bio-based diesel in 2023, up from 0.3 million tonnes in 2022 (DNV, 2023^[13]).

Methanol’s chemical properties make it an attractive fuel alternative. Despite its lower energy density, it shares combustion characteristics with heavy fuel oil, simplifying storage and handling compared to ammonia or LNG. Additionally, methanol’s ‘drop-in’ capability allows modern dual-fuel marine engines to operate on both methanol and heavy fuel oil (Methanol Institute, 2023^[14]). Green methanol can be classified into bio-methanol, derived from biomass or biogas, and e-methanol, produced by combining green hydrogen with CO₂ from direct air capture or biogenic sources. Both have minimal lifecycle emissions, making them viable for net-zero shipping. The existing methanol capable tonnage in the global fleet incentivises increased production capacity for methanol. Based on methanol projects targeting shipping, the Global Maritime Forum estimates that around 3,5 million tonnes per years could be available

to shipping by 2030. The cost estimation of green methanol based on expected projects ranges between \$900 and \$2,500 per metric tonne VLSFO equivalent (Global Maritime Forum, 2024^[15]). A supply gap for green methanol could arise in 2030 in parts resulting from constraints on availability of biogenic CO₂ and high cost of transportation. This could lead to an increase in cost and in hindering the potential creation of production sites.

Ammonia is expected to see the highest uptake in the global fleet by 2050, with the IEA projecting that it will account for 44% of final energy consumption in the shipping sector by mid-century (International Energy Agency, 2023^[16]). Ammonia can be produced with zero carbon emissions when derived from renewable energy sources. Its greater energy density relative to hydrogen and its ability to be stored as a liquid under relatively manageable conditions make it a promising long-term option for deep-sea shipping. However, ammonia as a marine fuel comes with significant safety challenges due to its toxic and corrosive nature. The increased frequency of handling will therefore necessitate stringent safety measures and operational protocols to mitigate risks associated with storage, bunkering, and ship operations (Global Centre for Maritime Decarbonisation, 2024^[17]). Nearly 8 million tonnes of near-zero-emission ammonia production capacity is expected to come online by 2030, representing about 3% of total capacity from 2020. However, these projections are not specifically aimed at the shipping sector (International Energy Agency, 2021^[18]). According to the Global Maritime Forum (GMF) and Zero-Emission Shipping, announcements suggest approximately 32 million tonnes of green ammonia per year could be available for shipping by 2030. Cost estimates for ammonia by 2030 range from \$900 to \$2,700 per metric tonne, Very Low Sulphur Fuel Oil (VLSFO) equivalent, which could result in a substantial premium compared to conventional bunkers. Notably, US-produced green ammonia may achieve cost parity with VLSFO due to the tax credits included in the US Inflation Reduction Act (IRA) (Global Maritime Forum, 2024^[15]).

While there is still uncertainty over which fuel will be the dominant choice for shipping in the short, medium and long-term, IEA forecasts suggest that for 2030 international shipping activity will result in the following share of final energy consumption: 8% of share in energy consumption for biofuels, 6% for ammonia, 4% for hydrogen and 1% for methanol (International Energy Agency, 2023^[16]).

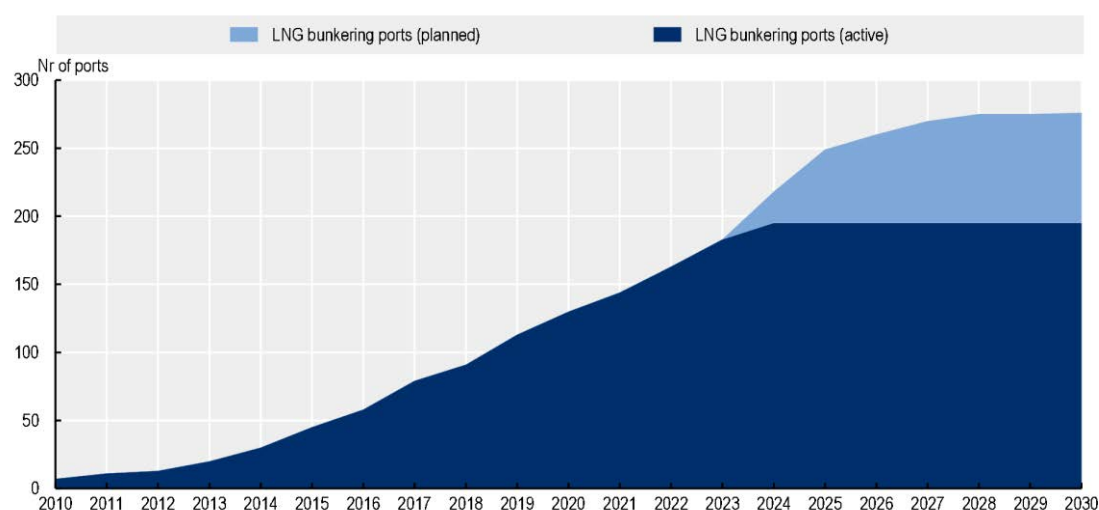
5.3.2. Storage and bunkering at ports

The availability of fuel supply and bunkering infrastructure at ports is essential to support the maritime sector's net-zero transition. In addition to assessing (future) production capacity, it is of interest to look to the existing supply chains for alternative fuels and projections of bunkering capacity across global maritime ports. Storage and bunkering are defined as follows:

- **Storage (active and potential):** port storage facilities worldwide for alternative fuels, highlighting their capacity and readiness to ensure consistent supply;
- **Bunkering (active and potential):** the operational and potential bunkering facilities for alternative fuels, which are critical touchpoints for fuelling vessels. Bunkering options include: 1) Truck to Ship (TTS), where fuel is transferred from a truck directly to the vessel, suitable for smaller operations; 2) Ship to Ship (STS), involving fuel transfer between vessels, often used offshore; and 3) Shore to Ship, where fuel is supplied from onshore facilities to the ship, enabling larger volume transfers (ALG, 2024^[19]).

The analysis maps estimates for current and projected bunkering from several sources. For methanol and ammonia bunkering, data from Clarksons World Fleet Register is used. The data is complemented by IEA data from the Hydrogen Production and Infrastructure Projects Database.

The number of ports globally offering LNG bunkering has doubled over the past five years, increasing from approximately 80 to 195 active LNG bunkering ports (and 81 planned facilities) in 2024. Projections suggest that this figure will surpass 250 by 2026 (see Figure 5.1).

Figure 5.1. Current and projected LNG bunkering in ports (2008-2030)

Note: Data as of October 2024.

Source: Clarksons Research (2024), World Fleet Register, <https://www.clarksons.net/wfr/fleet>.

For methanol, bunkering capacity stood at 12 active port facilities end of 2024, and 24 potential bunkering projects under development. Current bunkering infrastructure projects are mainly concentrated in Europe, East Asia and North America but potential bunkering points to an increase of bunkering for the rest of the world. For instance, a \$1.1 billion memorandum of understanding (MOU) for green bunkering in Egypt was signed by SCZONE and Scatec at COP28 (Clarksons Research, 2024^[20]).

For ammonia, potential bunkering projects could be identified primarily in East Asia and Europe. There is currently no ammonia bunkering for shipping in operation, but the world's first ship-to-ship transfer of ammonia using vessels at anchorage in a working port environment was carried out in September 2024 (Yara, 2024^[21]). A 2024 study by the Global Maritime Forum and RMI suggests that a global green ammonia trade could develop, with fuel transported from low-cost production regions like the US, South America, Australia and Sub-Saharan Africa to key bunkering hubs. While early-adopter ports should have sufficient supply by 2030, competition for the cheapest volumes may be intense, benefiting those who secure supply early (Global Maritime Forum; RMI, 2024^[22]).

Table 5.1. Number of ports with current and potential methanol and ammonia bunkering in different regions in 2024

	Methanol		Ammonia
	Bunkering	Bunkering (potential)	Bunkering (potential)
North America	2	2	5
East Asia	4	2	12
Middle East & North Africa	1	4	6
Europe	4	11	18
ROW	1	5	17

Note: Data as of October 2024.

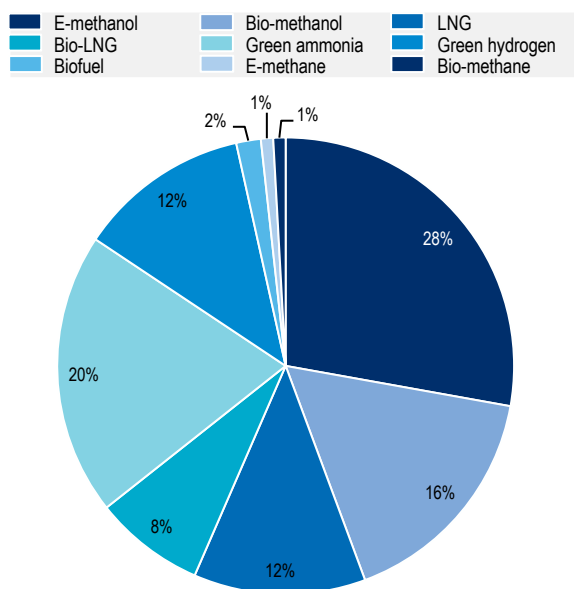
Source: Data from (Clarksons Research, 2024^[23]) and (International Energy Agency, 2024^[24]).

5.3.3. Offtake agreements for alternative marine fuels

Industry stakeholders have highlighted the importance of offtake or supply agreements between shipowners, ports, and energy producers to help aggregate fuel demand for clean fuels, thereby facilitating bankable investments in large-scale production capacity and fostering certainty for fuel buyers and suppliers (Mærsk Mc-Kinney Møller Center for Zero Carbon Shipping, 2022^[3]).

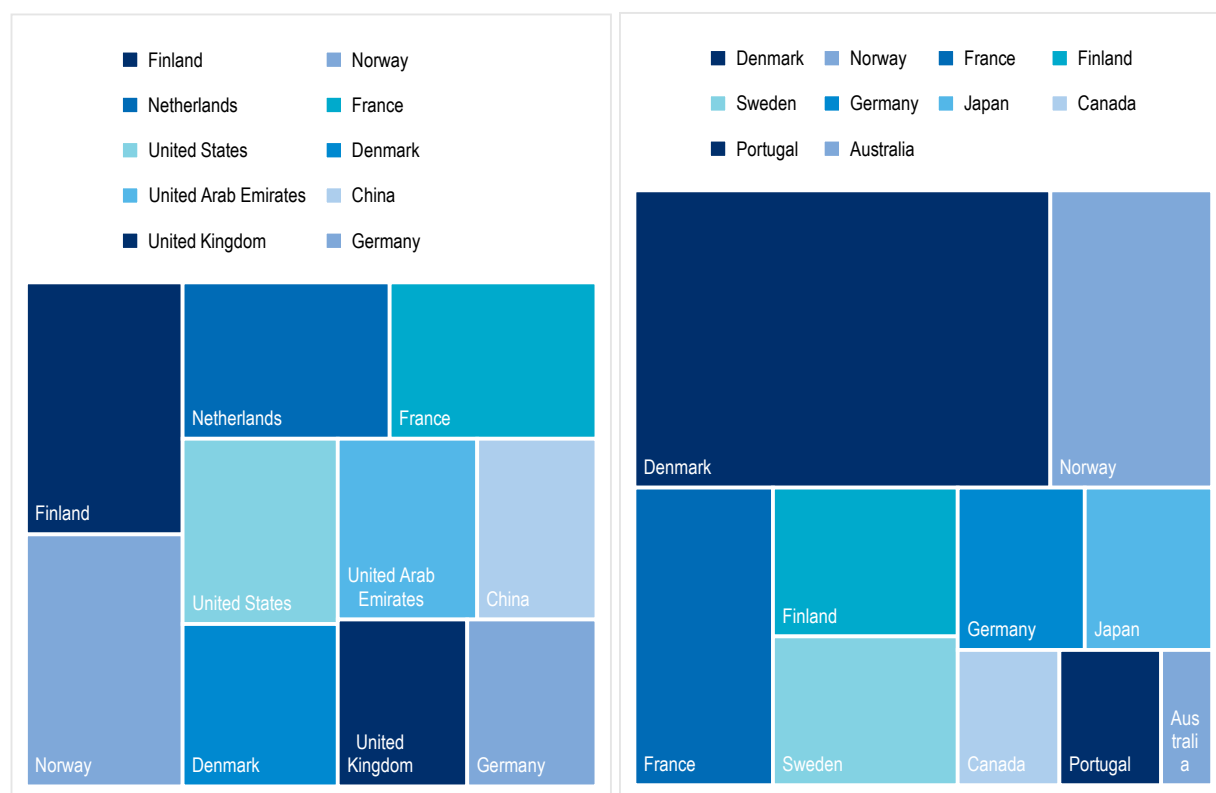
Which alternative fuel has attracted the most attention from maritime stakeholders in OECD countries? To help answer this question, a mapping of 104 offtake and supply agreements was conducted, focusing on demand for alternative fuels from maritime stakeholders in OECD countries. The analysis includes only agreements that explicitly involve maritime sector actors—such as shipowners or ports—and excludes broader energy deals not directly linked to maritime use. While the suppliers involved may originate from both OECD and non-OECD countries, the scope is limited to agreements reflecting demand from within the maritime sector in OECD countries. Of the agreements, 28% are dedicated to e-methanol and 16% to bio-methanol, indicating a clear preference for methanol-based solutions. LNG and bio-LNG together account for 20% of the agreements and represent the largest share of announced production capacity, suggesting strong near-term investment and scalability. Green ammonia is the subject of 20% of agreements, reflecting growing interest despite limited infrastructure readiness. Green hydrogen appears in 12% of cases, largely in early-stage projects.

Figure 5.2. Breakdown of alternative fuel offtake agreements by fuel type



Among the mapped offtake/ supply agreements, the suppliers of alternative fuels are predominantly located in Finland, Norway, the Netherlands, and France, with significant contributions—outside of Europe—also coming from the United States and the United Arab Emirates. On the demand side, Denmark leads in alternative fuel offtake agreements, largely driven by the shipping company A.P. Møller – Maersk A/S, which has entered into 15 identified agreements. Buyers in Norway, France, Sweden, Germany and Japan have also entered into several offtake/ supply agreements, reflecting a growing effort by the industry to ensure supply of alternative fuels for future ship operations (see Figure 5.3).

Figure 5.3. Main countries for fuel supply (left) and demand (right) in offtake agreements



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6

Mapping maritime decarbonisation policies

Regulatory certainty through a stable and predictable climate policy framework is key for both short- and long-term decarbonisation efforts of maritime stakeholders. This chapter maps and assesses the policy measures adopted by selected jurisdictions to support the decarbonisation of the maritime sector, focusing on shipping and shipbuilding. By providing a cross-country comparison of policies, it aims to enhance transparency and enable benchmarking of policy target areas and instruments across different regions. This is further complemented by in-depth analyses of policies in selected economies and international efforts, with a focus on regulation by the International Maritime Organisation (IMO) and green shipping corridors.

Key messages

This chapter maps and assesses policy measures adopted by selected economies to support the decarbonisation of the maritime sector, focusing on shipping and shipbuilding. The main conclusions from the policy mapping exercise are the following:

- **Almost 80% of countries have maritime decarbonisation strategies, with focus on the shipping sector:** decarbonisation strategies primarily target the shipping sector but 26% of measures include shipbuilding as a secondary target within broader maritime strategies. 5 out of 25 governments do not have a comprehensive strategy specifically tailored to the maritime sector, which can pose the risk of diluting policy coherence and reducing targeted implementation.
- **Implementation of mandatory decarbonisation measures remains limited:** most countries focus their policy efforts on developing comprehensive roadmaps and strategies, as well as diverse government support programmes aimed at advancing decarbonisation across various domains. While there are few mandatory measures across jurisdictions, the IMO is currently in the process of adopting legally binding mid-term measures.
- **Policies tend to focus on innovation, alternative fuel uptake and supporting infrastructure:** the most common policy target areas are fostering innovation and technology adoption within the sector. This encompasses both onboard technology enhancements and the development of onshore infrastructure, particularly regarding alternative fuel supply. This focus underscores policymakers' recognition of the interdependence between technology uptake in the global fleet and the requisite support infrastructure on land.
- **Policies on increasing access to finance are currently limited:** while some policymakers have adopted measures to facilitate finance and investment for maritime decarbonisation (including five taxonomies for green finance), the policy focus on finance is less than other areas, and could benefit from further efforts given the importance of mobilising capital for the transition.
- **Governments target policy support for digitalisation across the maritime sector:** a significant number of policies target digitalisation, covering aspects from operational technology in shipping to the integration of digital processes in shipbuilding. The policy support extends to the development and adoption of smart technologies, with a clear emphasis on fostering innovation within this domain.

6.1. Methodology for the policy mapping

Regulatory certainty and pathway clarity through a foreseeable and reliable climate change policy environment are necessary for long-term decarbonisation by shipping and shipbuilding companies. Climate and environmental regulation can delay ship orders when there is uncertainty on ambitions and instruments or accelerate ship replacement/ retrofitting and increase demand for alternative fuels and low/zero-emission technologies by imposing new energy efficiency or carbon intensity standards to comply with. Be they policy-pull (e.g., regulation and emissions standards) or technology-push (e.g., research and development (R&D) support), policy instruments can be powerful tools to help establish clear standards

for credible net-zero commitments and incentivise low/zero-emission technology innovation and uptake in the global fleet.

The following section provides an assessment the type of policy measures employed across selected economies to help decarbonise the maritime sector. The identification of relevant horizontal and sectoral policies is based on the Climate Actions and Policies Measurement Framework (CAPMF), developed by the OECD Environment Directorate (OECD, 2022^[1]). It has been adapted to specific characteristics of the shipbuilding industry, with a focus on what industry of the maritime sector is being targeted as well as policy instruments relevant to the studied sector. The policy mapping is based on the following criteria for inclusion:

- *Measures that are currently in place, set to be implemented in the next year or were concluded one-year prior;*
- *Decarbonisation measures that target the shipbuilding (incl. marine equipment) or the shipping sector;*
- *Horizontal measures that have an explicit decarbonisation target for shipbuilding or shipping;*
- *Measures to promote the development and adoption of zero- or low-carbon alternative fuels that have an explicit target for shipping.*

In other words, the policy mapping does not include horizontal policies if they do not include the shipbuilding or (more broadly) maritime sector as a target sector. Similarly, policies designed to promote the development and adoption of zero- or low-carbon alternative fuels are only included if they explicitly mention a shipping or maritime target. For instance, if a country is implementing a programme to facilitate the production of green ammonia or hydrogen, this programme is only considered in the policy analysis if it specifically mentions the utilisation of ammonia within the maritime sector as one of its objectives.

Information on policies was collected from many sources. These include targeted websites compiling information of climate change mitigation/ decarbonisation policies as well as EU and national government websites. Further, information from other resources available to the Secretariat, including the Shipbuilding Committee inventory list, policy and market developments in non-Shipbuilding Committee countries and policies detailed in Shipbuilding Committee peer reviews was included. Figure 6.1 provides an overview of the type of sources employed to access information and data on relevant policies. A total of 25 countries (+ EU and IMO) were included in the policy mapping exercises, including Shipbuilding Committee members and associates as well as selected other shipbuilding economies¹.

Table 6.1. List of sources for the policy mapping

	Source	More information
Global coverage	IMO Docs, MEPC 80/WP.12	https://webaccounts.imo.org/
	STIP Compass	https://stip.oecd.org/stip/
	OECD PINE Database	https://oecd-main.shinyapps.io/pinedatabase/
	IEA Policies Database	https://www.iea.org/policies
	Climate change laws of the world (LSE)	https://climate-laws.org/
	Clarksons World Fleet Register, Shipping Intelligence News	https://www.clarksons.net/wfr
Regional coverage	EU Official Journal	https://eur-lex.europa.eu/oj/direct-access.html
	Other EU websites	For example, - EU Innovation Fund Project Portfolio Dashboard: https://cinea.ec.europa.eu/programmes/innovation-fund/innovation-fund-project-portfolio-dashboard_en - CORDIS: https://cordis.europa.eu/projects
National coverage	Government and ministry official websites	For example, NEDO: https://www.nedo.go.jp/
	Shipbuilding Committee resources	Shipbuilding Committee Inventory, policy and market developments in non-Shipbuilding Committee economies, peer reviews

6.1.1. Overview of policy target areas

Delving deeper into the targets of maritime decarbonisation policy, the analysis encompasses a breakdown of detailed policy target areas. Within this framework, a multi-dimensional approach is adopted, dissecting the diverse aspects of policy intervention. These areas of policy concentration encompass *innovation*, highlighting the pivotal role of continued advancements and scaling up of new technologies to drive decarbonisation. For onboard technologies, the following target areas are defined: *energy efficiency* and *alternative fuel uptake*. Additionally, the mapping extends to *alternative fuel production and infrastructure*, to reflect the important infrastructure requirements needed for the transition to cleaner energy sources.

Human capital and labour considerations form another critical segment of the policy spectrum, acknowledging the significance of a skilled and adaptable workforce for industry transformation. *Access to finance and investment* dimensions are also included to explore the incentives and mechanisms that stimulate capital flow towards green maritime initiatives.

Digitalisation is included to emphasise the role of technological integration and digitalisation in bolstering efficiency and reducing emissions. Another possible policy target area is *data collection* to better understand emission trends in the maritime industry.

Finally, the mapping includes the importance of *international cooperation* as the global nature of the maritime industry necessitates collaborative efforts and harmonised policies to effectively combat GHG emissions. This comprehensive breakdown serves as a foundational framework for a detailed examination of policy instruments across these diverse target areas, improving understanding of the multifaceted landscape of maritime decarbonisation policies.

In the context of the *timeframe* under scrutiny, the policy measures incorporated within this mapping exercise encompass a spectrum of initiatives. These include policies that were implemented after the year of 2008 and that are in-force or were concluded a year prior to 2024. Furthermore, the mapping extends to policies that have been officially announced, although their implementation is expected in the coming years. By considering both implemented policies and those on the horizon, this comprehensive approach ensures a holistic representation of the evolving policy landscape that governs the maritime sector's decarbonisation efforts.

6.1.2. Overview of policy instruments

To enhance comprehension regarding the diverse array of policy measures implemented across shipbuilding economies to facilitate the decarbonisation of the maritime sector, the mapping exercise establishes clear definitions for a range of policy instruments. These instruments encompass both demand-pull and technology-push approaches that governments employ to steer the industry towards new practices. In Table 6.2, an overview of these policy instruments is presented, accompanied by descriptions of each policy's characteristics and its potential utility in driving the decarbonisation agenda within the maritime sector. This systematic delineation of policy instruments should serve as a resource for stakeholders seeking to navigate the complex landscape of policies influencing the sector's decarbonisation efforts.

Table 6.2. List and description of policy instruments

Policy type	Policy instrument	Description
Strategy	Strategy, agenda, plan Target setting	• A comprehensive and coordinated plan developed by a government to guide and facilitate the transition of the maritime industry or shipbuilding sector toward reduced carbon emissions and improved environmental performance. • It typically includes plans for policy measures, regulations, and initiatives designed to achieve specific decarbonisation goals, promote the adoption of clean technologies and enhance the sector's overall competitiveness. • It outlines short-term and long-term targets, identifies key stakeholders, allocates resources and establishes a timeline for implementation, monitoring and evaluation to ensure the successful transformation of the maritime or shipbuilding sector. It helps promote policy coherence and regulatory certainty.
	Command-and-control regulation	• Government-imposed regulation and requirements designed to govern specific aspects of the maritime sector. • Regulatory policies may include laws, and regulations that prescribe limits on emissions, energy efficiency standards, performance and/or design standards or other mandatory measures aimed at reducing GHG emissions of shipping.
	Carbon-pricing mechanism Carbon tax	• Putting a price on negative externalities of an economic activity, in this case greenhouse gas emissions. • It can take the form of carbon taxes ('Pigouvian taxes') or emissions trading systems (ETS), where emitters are either taxed based on the amount of carbon they release or are required to purchase emission allowances in a market. The aim is to incentivise businesses and individuals to reduce their GHG emissions by imposing a financial cost on such emissions. • Carbon pricing can generate government revenue, which can serve as a consistent funding source for low-carbon R&D and investment.
Regulation	Standards, guidelines and certification	• Establishment of standards or guidelines for industry, which are not mandatory but encouraged. • Defining different type of activities to support uptake of "green" activities.
	State grant and state provided loans Public-private partnerships Export or home credits Tax breaks	• A financial incentive provided by governments or related entities to support specific activities, industries or practices. • In the context of maritime decarbonisation, government support may be directed towards promoting the adoption of cleaner technologies, such as alternative fuels or energy-efficient equipment, by reducing the cost burden on businesses or individuals engaged in these activities. • Type of support can include direct government interventions, green innovation investment, other public investment, green government procurement, state grants, state provided loans, public private partnerships, export or home credits and tax breaks.

	Voluntary approach	• Non-mandatory initiatives/ agreements in which industry stakeholders, often in collaboration with governments, commit to specific actions. • Participation is not legally required, but organisations choose to engage in them to demonstrate their commitment to environmental responsibility. • In the maritime sector, voluntary approaches might involve industry participants voluntarily adopting emission reduction targets or best practices to reduce their environmental impact. Voluntary approaches without government involvement or support are not included in the policy mapping.
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Source: (OECD, 2011^[2]) (OECD, 2022^[1]) (OECD, 2022^[3])

6.2. Discussion of findings

This section provides a summary of both general findings and ‘deep dives’ of individual shipbuilding economies and green shipping corridors. It offers an overview of the key insights derived from the policy mapping exercise, encompassing the global trends in maritime decarbonisation policies and their implications. It aims to serve as a resource for policymakers, stakeholders and industry experts to better understand the evolving landscape of policies aimed at achieving decarbonisation in the maritime sector.

The mapping shows a strong policy focus on the shipping sector (Figure 6.1). Approximately 56% of the policies primarily target shipping. Shipping with a secondary shipbuilding focus—i.e., measures that focus on the shipping industry while also including strategies and objectives for shipbuilding—accounts for about 26% of the policies. The remaining 17% predominantly target the shipbuilding sector. This distribution of policies reflects the dominant direction of decarbonisation efforts, which mainly emphasise the transport and operational phases of vessels, that currently account for the majority of GHG emissions.

Figure 6.1. Relative share of policies focused on the shipping and shipbuilding sectors

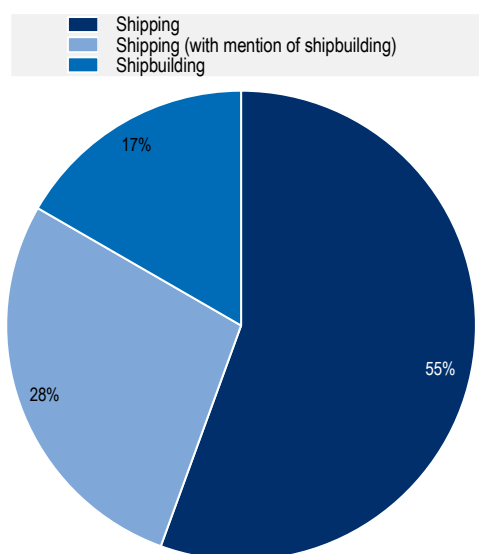
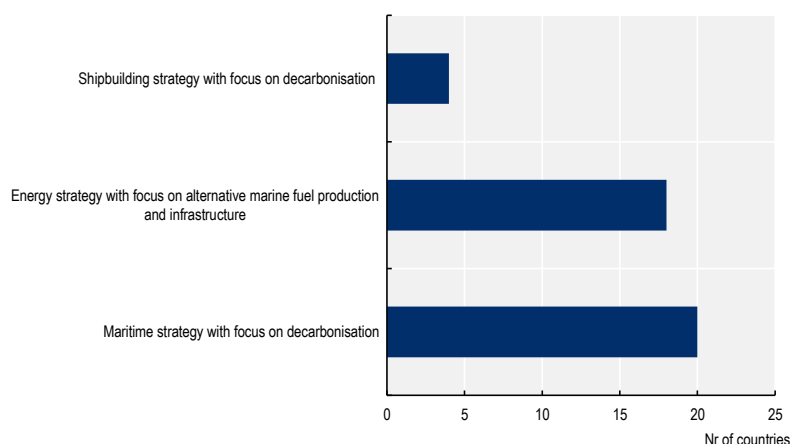


Figure 6.2 presents the types of decarbonisation strategies adopted by 25 countries, plus the EU.

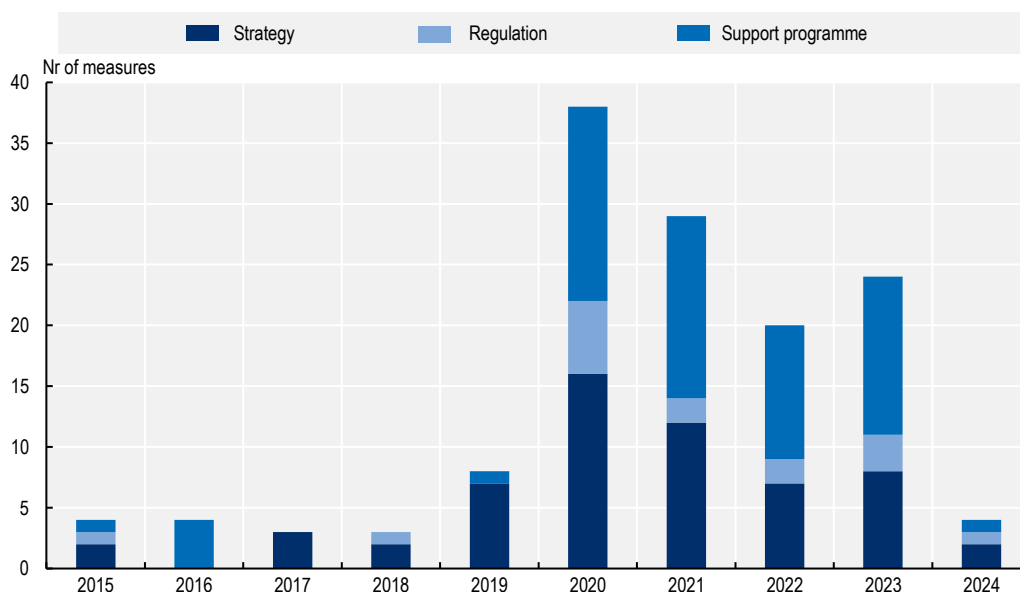
A total of 20 jurisdictions (77%) have implemented a maritime strategy with an emphasis on decarbonisation, which may include a broader transport decarbonisation strategy with a focus on the maritime sector or a dedicated maritime transport strategy (with specific net-zero targets). 11 of these strategies were adopted before the revised IMO GHG strategy and still reference the older IMO decarbonisation goals. Further, 17 jurisdictions (68%) have established an energy strategy centred on alternative marine fuel production and infrastructure. In contrast, only 4 countries (15%) have adopted a shipbuilding strategy focused on decarbonisation, highlighting that this area is currently less prioritised compared to broader maritime and energy initiatives.

Figure 6.2. Number of countries with selected decarbonisation strategies



To support their net-zero targets, many countries have introduced a growing range of policy measures and government support aimed at fostering the development and adoption of low/zero-emission maritime technologies and practices. Since 2020, there has been a notable rise in the number of such measures, reflecting stronger policy engagement with maritime decarbonisation. The uptake of regulatory instruments remains limited; instead, government roadmaps or strategies and support measures are more commonly employed. Beyond national initiatives, international collaborative efforts (often facilitated by the IMO) have emerged as another support tool. These partnerships reflect the inherently global nature of maritime transport and highlight the need for coordinated action to align decarbonisation pathways across jurisdictions.

Figure 6.3. Trends in introduction of strategies, regulation and support programmes (2015-2024)

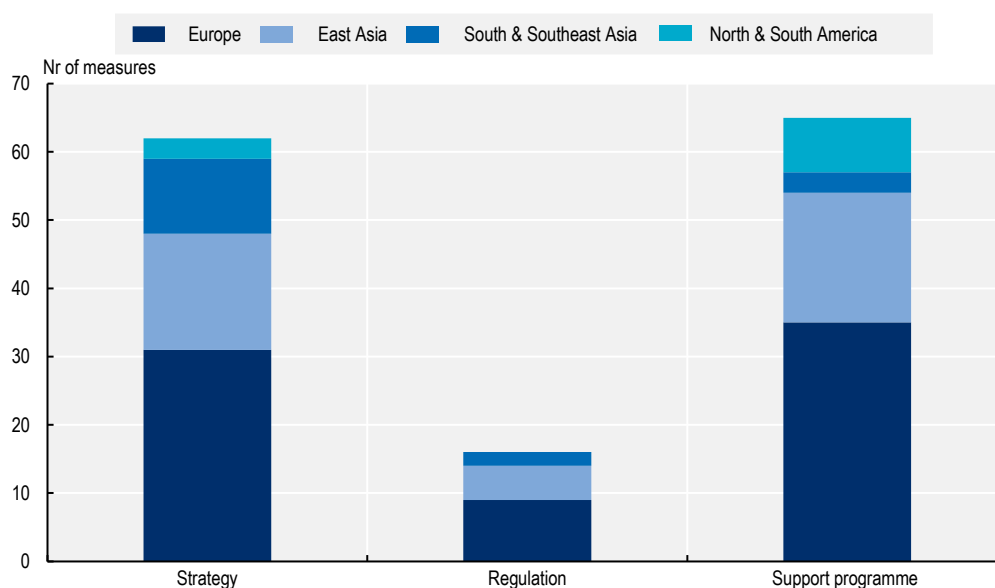


Note: The mapping captures measures introduced up to mid-2024. As a result, the 2024 data should be interpreted with caution, as it may not fully reflect the scope or momentum of recent policy activity.

Mandatory regulatory measures, including carbon pricing mechanisms, are rarely chosen as policy options for maritime decarbonisation (Figure 6.4). This relative unpopularity could be attributed to the

inherent cost implications associated with such regulatory and pricing mechanisms. These instruments create a shared economic burden for the entire sector, compelling a collective financial commitment to mitigate carbon-intensive practices and technologies across the industry. This stands in contrast to support programmes that facilitate the development and adoption of less carbon-intensive production and operational methods, all while remaining non-mandatory for economic stakeholders.

Figure 6.4. Breakdown of policy types by regions



6.2.1. Discussion of policy targets areas

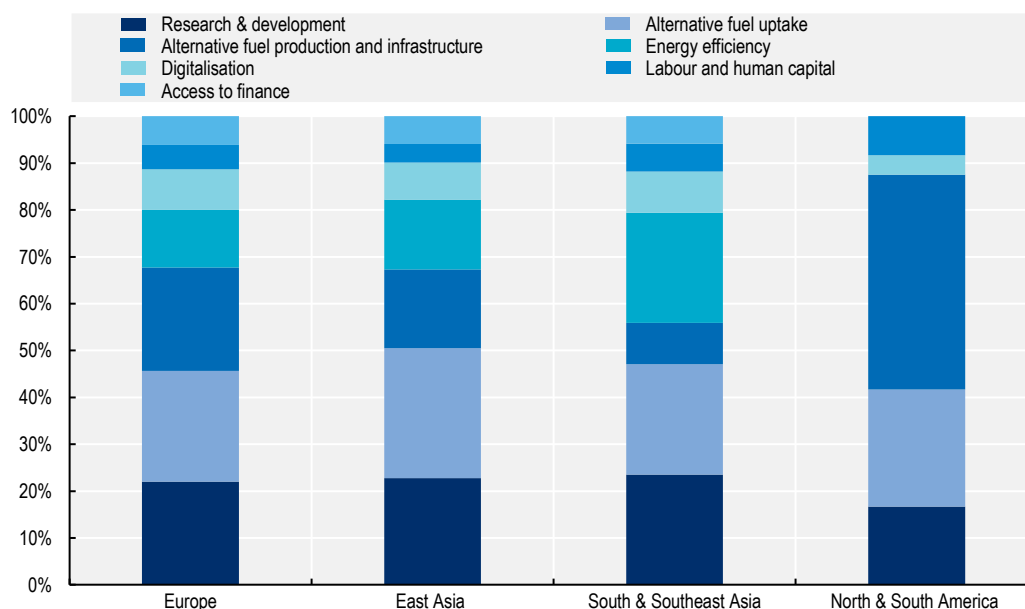
The policy target areas for maritime decarbonisation vary across regions, reflecting differing national priorities and circumstances (as seen in Figure 6.5). One of the most mentioned focus areas is the *uptake of alternative fuels*. This is particularly emphasised in Europe, East Asia, and Southeast Asia, where there is a significant commitment to advancing the adoption of green fuels for maritime transport. In the Americas, the focus leans more towards the *supply of alternative fuels*, including production, infrastructure and bunkering. This difference in policy emphasis is reflective of the fact that many countries in the Americas have a smaller contribution to global shipping and shipbuilding and instead prioritise ensuring a stable supply of alternative fuels for the shipping industry.

R&D for technology development is another critical area of focus for countries pursuing maritime decarbonisation. Across regions, there is a strong commitment to investing in R&D to advance technologies that support emissions reductions and overcoming the barriers to net-zero faced in the maritime sector. *Energy efficiency* as a target area receives greater emphasis in Southeast Asia and South Asia, where countries are implementing several measures to enhance operational efficiency and reduce energy consumption in maritime operations.

Access to finance is critical for achieving maritime decarbonisation, and there is growing recognition of the need for targeted financial frameworks. The development of finance taxonomies, such as the EU Taxonomy, K-Taxonomy, Singapore-Asia Taxonomy for Sustainable Finance, ASEAN Taxonomy, and China's Green Bond Endorsed Project Catalogue 2021 Edition, are supporting investment in green shipping initiatives. However, there remains a need for more financing options, particularly for the development of low-emission ships, to meet ambitious decarbonisation targets across the maritime sector.

Finally, in terms of *labour and human capital*, the European Union has introduced the most policy measures related to the maritime workforce in absolute terms. Labour-related policies are also relatively prominent in the Americas, as well as in South and Southeast Asia, with notable efforts in countries like India and Indonesia. These policies aim to ensure that the workforce is adequately prepared for the transition to greener maritime operations, incorporating training programs and measures to promote employment in new areas of the sector.

Figure 6.5. Relevance of selected policy target areas in different regions



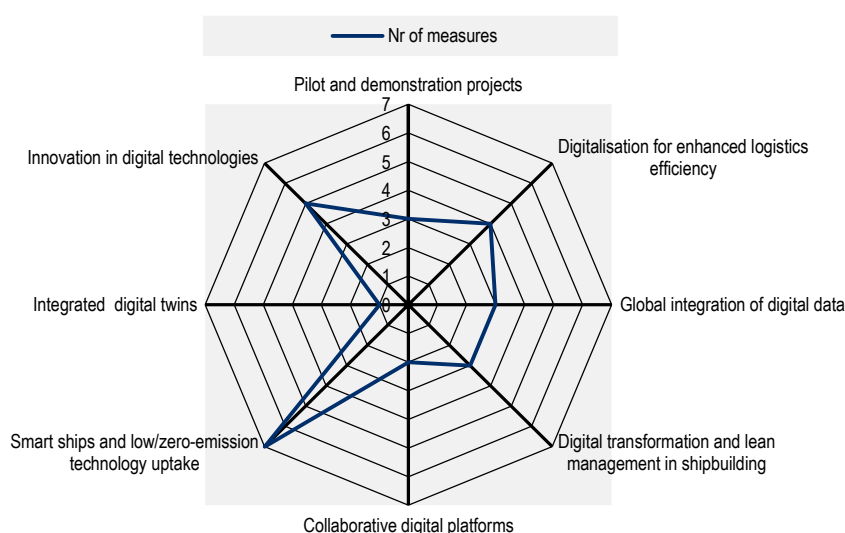
6.2.2. Deep dive on maritime digitalisation

Policies targeting digitalisation support both operational technology in shipping and the integration of digital processes in shipbuilding (see Figure 6.6). Generally, policies tend to include three target domains: 1) shipping operations enhanced by digital technologies, 2) the integration of digital solutions in shipbuilding processes and 3) the broader development of digital technologies in the maritime sector. Specifically, shipping-focused policies aim to boost the adoption of digital technologies among ship owners, enhancing the efficiency of maritime operations. They also underscore the role of digital technologies in optimising logistics and the importance of globally integrating digital data and enhanced data sharing during ship operation.

Policy targets on shipbuilding activities focus on leveraging digital transformation, artificial intelligence and automation to advance the competitiveness and efficiency of the shipbuilding and marine equipment industry. They highlight the integration of digital and lean management strategies to improve the quality and efficiency of ship assembly and construction. Additionally, they emphasise the development of collaborative digital platforms and management systems for shipyards.

Broader ship technology policy targets include supporting the development of digital solutions for enhanced energy efficiency and lower GHG emissions. Key initiatives include creating Integrated Green Vessel Digital Twins to improve vessels' lifecycle environmental performance and promoting digital innovation in performance monitoring to boost shipping efficiency.

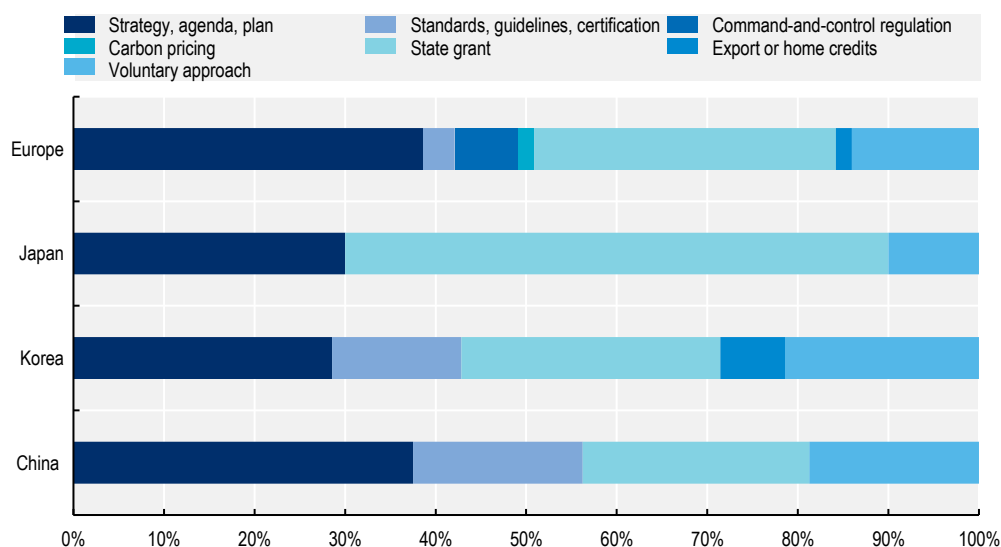
Figure 6.6. Breakdown of digitalisation policy measures by targets



6.2.3. Discussion of policy instruments

In the largest shipbuilding regions, a breakdown of specific policy instruments reveals a strong focus on strategies, agendas and plans as well as state grants and voluntary collaborations between industry and government to support them (Figure 6.7). The EU, Korea, and China have also established standards, guidelines, and certification processes, particularly aimed at facilitating financing for decarbonisation initiatives. The use of export and domestic credits to support decarbonisation efforts is especially prevalent in Korea. Notably, the EU distinguishes itself through its command-and-control regulations and carbon pricing mechanisms. For more detailed information on the policies of individual countries, please refer to the section on policy deep dives.

Figure 6.7. Relevance of selected policy instruments in different regions



6.3. Deep dive on selected policies

To better understand regional differences in policy mix and ambition, a more in-depth analysis of specific policies is merited. The following section provides ‘deep dives’ on selected policies in the EU, Korea, Japan and China, which have been selected due to their prominence in shipbuilding and marine equipment manufacturing industries, as well as their diverse and extensive policy portfolios aimed at decarbonising the maritime sector. It offers an in-depth analysis of these key regions’ unique policy landscapes, highlighting their respective strategies, priorities and initiatives in pursuit of a greener maritime industry. To set the stage for the analysis, it is helpful to begin with a brief overview of the IMO GHG strategy and IMO policy efforts, which serves as the foundational framework for advancing international shipping’s decarbonisation efforts.

6.3.1. International Maritime Organization (IMO)

The 2023 IMO GHG Strategy is a cornerstone in the global pursuit of decarbonisation within the maritime sector. Following revisions made during the Marine Environment Protection Committee (MEPC80) meeting in July 2023, this strategy stands as a key framework for shipping and shipbuilding economies, delineating a roadmap for international shipping’s future and fostering commitments for reducing greenhouse gas emissions. Central elements of the 2023 IMO GHG Strategy encompass a vision for the maritime industry’s trajectory towards more stringent decarbonisation targets, both in the mid- and long-term.

The strategy targets net-zero maritime GHG emissions “by or around, i.e. close to” 2050, aiming for a minimum reduction of 40% in GHG emissions per transport work by 2030. This target serves as a crucial marker for measuring progress in emissions reduction. Moreover, the strategy champions the adoption of zero or near-zero GHG emission technologies, fuels, and energy sources within international shipping. It sets an aspirational goal for these cleaner alternatives to comprise at least 5% (striving for 10%) of the energy used by the sector by 2030 (International Maritime Organization, 2023^[4]). Ongoing negotiations at the IMO are also focused on developing and implementing an IMO net-zero framework, with different possible components (International Maritime Organization, n.d.^[5]). If designed and implemented effectively, these measures could support market certainty to help unlock large-scale investment in low/zero-emission shipping.

Recognising the challenges in implementing such transformative changes, the strategy acknowledges barriers and lays out supportive measures. These measures encompass capacity building, technical cooperation, and R&D initiatives, all aimed at overcoming hurdles on the path to emissions reduction. Furthermore, the strategy’s significance extends to its commitment to align with global climate goals, particularly those outlined in the Paris Agreement. It emphasises efforts towards phasing out GHG emissions from shipping in accordance with the Agreement’s long-term temperature objectives, reinforcing the maritime sector’s role in addressing the broader climate crisis.

Under the International Convention for the Prevention of Pollution from Ships (MARPOL) Annex VI, the IMO has implemented obligatory measures aimed at enhancing the energy efficiency of vessels and curtailing their carbon intensity (International Maritime Organization, 2011^[6]). Since 2015, newly constructed ships have been subject to regulations concerning energy efficiency, requiring them to adhere to specified benchmarks on the *Energy Efficiency Design Index (EEDI)*. Moreover, the *Ship Energy Efficiency Management Plan (SEEMP)* presents a voluntary set of guidelines intended to enhance the operational efficiency of ships (International Maritime Organization, 2021^[7]). Since January 2023, the IMO has introduced ‘short-term measures’, which encompass the *Energy Efficiency Existing Ship Index (EEXI)*, the *Carbon Intensity Indicator (CII)*, and an improved version of SEEMP, all of which have entered into force.

6.3.2. EU

The EU is implementing comprehensive maritime decarbonisation policy mix, combining demand-pull, including mandatory fuel standards and carbon pricing mechanisms, and technology-push policy instruments.

As part of its “Fit for 55” legislative package, the EU has introduced various policies aimed at maritime decarbonisation. As of January 1, 2024, the **incorporation of maritime transport into the EU Emissions Trading System (ETS)** took effect, encompassing the entirety of emissions within EU and European Economic Area (EEA) ports and 50% of emissions from voyages to and from EU and ports, involving countries beyond the EU. Aligned with the “polluter-pays principle”, it is anticipated that proceeds from a minimum of 20 million ETS allowances, equivalent to approximately EUR 1.5 billion at current market rates, will be allocated to maritime initiatives under the ETS Innovation Fund through 2030. Eligible projects encompass enhancements in the energy efficiency of vessels and ports, pioneering technologies and infrastructure, alternative fuels, and zero-emission propulsion systems (European Commission, 2023^[8]).

Under the FuelEU Maritime Regulation, the EU introduced in 2025 an ambitious framework for progressively reducing the greenhouse gas intensity of ship fuels. The regulation sets a target of achieving a 75% reduction in greenhouse gas intensity by 2050, compared to levels observed in 2008. An important feature of FuelEU Maritime is the possibility to pool compliance across several ships from the same or different companies. Additionally, the regulation establishes a comprehensive system for monitoring and reporting emissions originating from ships, alongside incentives designed to promote the adoption of alternative fuels within the maritime sector. These incentives encompass the development of dedicated infrastructure for alternative fuels, investment in research and development (R&D) efforts for emerging technologies, and financial support for pilot projects (Council of the European Union, 2023^[9]).

EU binding regulations are often accompanied by technology-focused measures at the national level by member states, such as the Green Ship of the Future (GSF) programme in Denmark or the MARITIME.zero GHG programme in Germany. Further, government support is available through *Horizon Europe* and the *EU Innovation Fund* (European Commission, 2023^[10]).

6.3.3. Korea

Korea has actively implemented several overarching maritime decarbonisation strategies and initiatives targeting the shipbuilding sector. One noteworthy example to highlight its holistic approach to maritime decarbonisation is the “*1st National Plan for the Development and Popularization of Green Ship*”, which was introduced in 2020. As part of this initiative, the Korean government has allocated a budget of USD 870 million for innovative research and development programmes aimed at advancing green vessels over the period from 2022 to 2031. The primary objectives of Korea’s Greenship-K Promotion Strategy for 2030 encompass the advancement of carbon-free technologies related to hydrogen and ammonia, the creation of energy-efficient solutions with reduced carbon footprints and the domestic development of essential technologies for Liquefied Natural Gas (LNG), electrification and hybridisation within the maritime industry (KOMSA, 2021^[11]).

The Korean government has also taken notable steps towards shipping and transport decarbonisation by launching initiatives like the “*2023 National Action Plan Toward Green Shipping by 2050*” and establishing the “*Green & Smart Transportation Partnership*”. The Korean government’s approach has centred on crafting policies that directly address industry and industry clusters, with a particular focus on fostering the adoption of low-carbon and digital solutions.

6.3.4. Japan

As a prominent player in both shipbuilding and shipping, Japan has strategically designed initiatives that encompass both sectors. One such initiative is the “*Roadmap to Zero Emission from International Shipping*”, a collaborative effort spearheaded by the Ministry of Land, Infrastructure, Transport and Tourism (MLIT) in partnership with maritime industries, research institutions and public entities. This roadmap, formulated in March 2020, outlines Japan’s commitment to achieving zero emissions in international shipping. It not only sets the vision but also involves substantial research, development, and demonstrations of zero-emission technologies while actively participating in the formulation of relevant regulations within the IMO. The ultimate goal is to introduce the inaugural “*Zero Emission Ship*” by 2028, marking a significant milestone in their green shipping journey (MLIT, 2020^[12]).

Further bolstering these efforts is the New Energy and Industrial Technology Development Organization of Japan (NEDO), which has undertaken a substantial USD 280 million funding project dedicated to the development of next-generation ships. This sizeable investment aligns with Japan’s target to lead in innovative shipbuilding practices, aligning with their broader focus on shipbuilding as well (NEDO, 2023^[13]).

Japan also extends support to the commercial transport sector’s energy efficiency demonstration projects and accelerating zero-emission ship finance through a programme jointly managed by the Development Bank of Japan Inc. (DBJ) and ClassNK, a leading ship classification society. Under this programme, ClassNK assesses ships using a comprehensive scoring model developed by DBJ, evaluating factors such as decarbonisation, environmental performance and innovation. DBJ, in turn, provides investment and financing support, exemplifying Japan’s multi-faceted approach decarbonisation across the maritime and commercial transport sectors (Development Bank of Japan, 2022^[14]).

6.3.5. China

China has progressively advanced its maritime decarbonisation agenda through a combination of broad national strategies and targeted shipping and shipbuilding sector initiatives. Apart from broader plans that encompass green shipping targets, such as the *14th Five-Year Plan (2021-2025) for National Economic and Social Development and Vision 2035 of the People’s Republic of China*, and the *Action Plan for Carbon Dioxide Peaking before 2030 (‘1+N’)*, China has undertaken targeted initiatives focused on the transport sector. These initiatives include the *14th Five-Year Plan for the Development of Modern Comprehensive Transport System* (in 2021) and the *14th Five-Year Plan for the Development of Green Transport* (in 2021) as well as the *Action Plan for Large-scale Transportation Equipment Renewal* (in 2024). Strategies and plans by the Chinese central government are complemented by region initiatives, such as the implementation rules for the subsidy funds for port and shipping in the field of green transportation construction by the Shenzhen Transportation Special Fund or measures for the management of special support funds for transportation energy conservation and emission reduction in Shanghai.

In general, China’s existing policies designed to decarbonise its maritime sector primarily emphasise the development and adoption of clean energy, promoting cooperation between the shipping and port industries and establishing governance mechanisms to address shipping decarbonisation issues (GIZ, 2022^[15]). Before 2024, there was a lack of policy directives explicitly targeting shipping or shipbuilding decarbonisation, with relevant regulations spread across various documents and lacking clear national objectives and pathways (Yang, Cheng and Liu, 2023^[16]). However, in 2024, China introduced the “*Action Plan for Green Development in the Shipbuilding Industry (2024-2030)*” marking a significant shift towards a more focused policy on maritime decarbonisation and shipbuilding’s contribution.

China has introduced regulation to better assess GHG emissions from the shipping sector, with the “*Management Measures for Ships’ Energy Consumption Data and Carbon Intensity*” by the Maritime Safety Administration of China in November 2022. This specialised regulation focuses on collecting energy

consumption data from Chinese and foreign ships using Chinese ports, facilitating the assessment of Chinese ships' carbon intensity levels. The implementation of this regulation holds significant potential for promoting shipping decarbonisation from a technical management perspective (Clarksons Research, 2024^[17]). Furthermore, in March 2023, the China Association of the National Shipbuilding Industry (CANSI) endorsed new guidelines for the calculation and disclosure of carbon emissions of Chinese shipyards and ship repair facilities. These guidelines provide instructions on computing both direct and indirect carbon emissions, tailored to the unique attributes of these sectors.

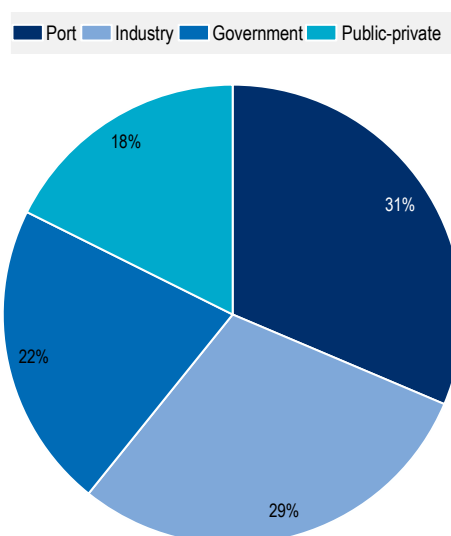
Most of China's existing shipping decarbonisation policies are non-mandatory, characterised by their advisory nature. This predominantly stems from the fact that policies formulated by the central government and its subsidiary departments often serve as guiding principles. However, beyond these guiding policies, the Chinese central government possesses the authority to enact mandatory policy instruments in the form of regulations and laws. Mandatory policies carry greater authoritative and binding force compared to their non-mandatory counterparts (Yang, Cheng and Liu, 2023^[16]).

6.4. Deep dive on green shipping corridors

Green shipping corridors represent key efforts to leverage international cooperation for maritime decarbonisation. The Green Shipping Corridor initiative was launched during COP26 in 2021 as part of the Clydebank Declaration, which saw 27 signatories commit to developing zero-emission shipping routes (Department for Transport, 2023^[18]). These routes, known as green shipping corridors, are designed to achieve zero GHG emissions between two or more ports through the adoption of alternative fuels, green vessels, and new, including digital, technologies ahead of regulatory requirements.

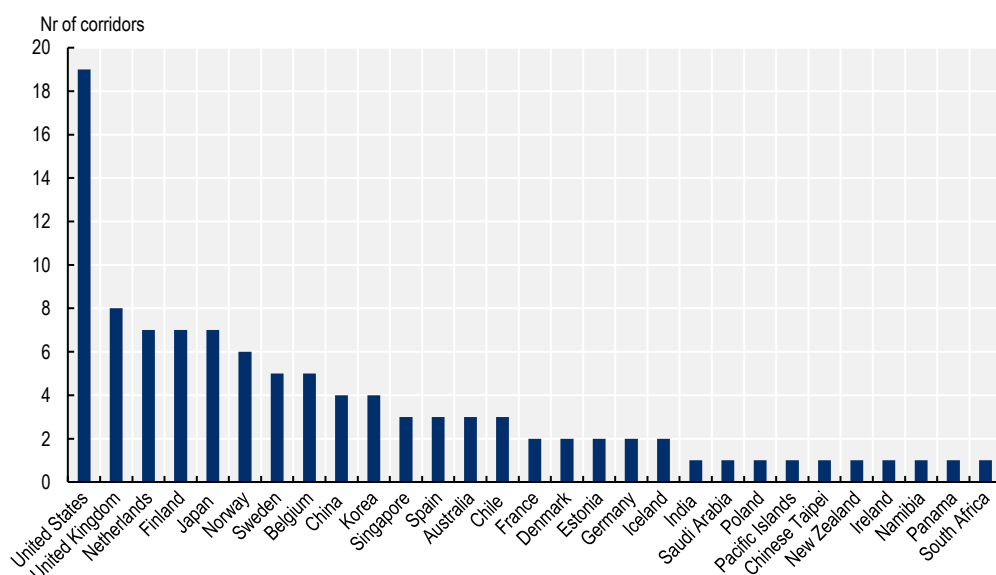
Green shipping corridors involve a diverse set of stakeholders, including governments, port authorities, shipping companies, and energy providers. The projects focus on areas such as developing alternative fuel supply chains, supporting the uptake of low/zero-emission technologies, and establishing safety regulations for alternative fuels. Although specific goals vary across projects, all initiatives share the aim of advancing decarbonisation in the maritime sector. The feasibility of these initiatives depends on factors like proximity to alternative fuel supply hubs, developed infrastructure, and favourable economic conditions. As can be seen in Figure 6.8, while the Clydebank Declaration involved national governments, bottom-up initiatives taken by the industry—such as ports, shipping companies, shipbuilders, and energy suppliers—constitute a significant portion of announced projects, with ports often acting as the main leaders.

Figure 6.8. Breakdown of green shipping corridors by stakeholder leadership



By mapping announcements globally, a total of 51 green shipping corridors could be identified. Figure 6.9. shows the number of green shipping corridors by country as of October 2024. The United States leads with 19 green shipping corridors, followed by the United Kingdom, the Netherlands, and Finland, each with 7 corridors. Japan, Norway, and Sweden also show notable involvement, with 6 and 5 corridors respectively. Several other countries, including China, Korea, and Singapore, are involved in 3 to 4 corridors each.

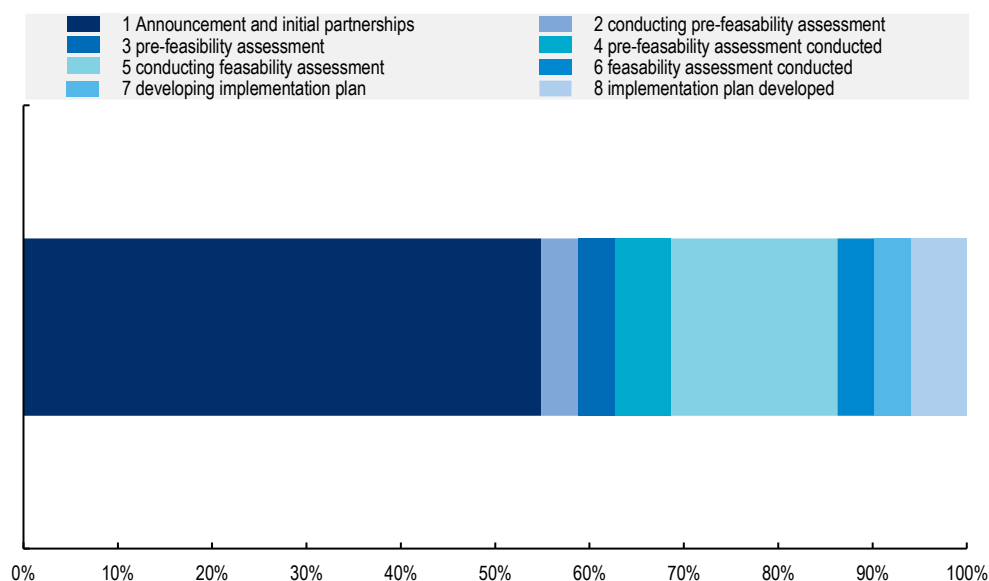
Figure 6.9. Number of green shipping corridors by country (as of October 2024)



The IMO identifies eight phases for the development of green shipping corridors, ranging from agreement and partnerships to successful implementation (Global Maritime Forum, 2023^[19]). However, most initiatives are still in the early stages, with only three out of the 51 initiatives reaching the

implementation phase. The uncertainties surrounding timelines, emission reduction targets, and risks have led some critics to question whether green shipping corridors risk “greenwashing” maritime decarbonisation efforts or if they will indeed play a significant role in decarbonising the maritime industry (Lloyd's List, 2023^[20]).

Figure 6.10. Readiness of green shipping corridors (as of October 2024)



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Notes

¹ The policy mapping includes Argentina, Brazil, China, Croatia, Denmark, Finland, France, Germany, India, Indonesia, Italy, Japan, Korea, the Netherlands, Norway, the Philippines, Poland, Romania, Singapore, Sweden, Thailand, Türkiye, the United States, the United Kingdom, and Viet Nam.

The Role of Shipbuilding in Maritime Decarbonisation

Impacts of Technology Developments and Policy Measures

Maritime transport underpins global trade, moving around 80% of goods worldwide. At the same time, it accounts for nearly 3% of global greenhouse gas emissions. As momentum builds for cleaner shipping, the shipbuilding industry plays a critical role in designing, constructing, and retrofitting vessels that can operate on alternative fuels, advanced propulsion systems, and energy-efficient technologies. With the International Maritime Organisation's 2023 GHG strategy setting a clear course, the industry is accelerating the development of low- and zero-emission solutions. Yet, challenges remain— an aging fleet, diverse ship types, high fuel costs, and investment gaps all influence the pace of progress. This report examines how new ship designs, fuel technologies, and technology innovations are reshaping maritime transport and explores the key enablers driving this transformation forward.



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